

# Matched-Analyte Spectral Audit

Raman vs Ag-SERS — GAIRA V5 grounding corpus

*Do Raman and Ag-SERS observe the same vibrational modes for the same analyte, or fundamentally different spectra?*

|                         |                                                              |
|-------------------------|--------------------------------------------------------------|
| Matched analytes        | 51                                                           |
| Spectra compared        | 435 (180 Raman / 255 Ag-SERS)                                |
| Preprocessing (primary) | A2 ASLS + Savitzky-Golay + SNV (exact Stage B)               |
| Common grid             | 520-1750 cm <sup>-1</sup> @ 2 cm <sup>-1</sup> (616 bins)    |
| Peak-match tolerance    | ±12 cm <sup>-1</sup>                                         |
| Sensitivity pipeline    | A1 ASLS + Savitzky-Golay + L2                                |
| Status                  | READ-ONLY audit — no models fitted, no preprocessing changed |

## PRINCIPAL FINDING

Ag-SERS spectra of 51 chemically distinct metabolites are ~95% identical to one another (median between-analyte cosine 0.945; 95% of each spectrum's variance is the corpus-mean background). Analyte identity survives as a small residual — sufficient for 73% within-SERS identification, but too faint to support cross-modal correspondence with Raman. Stage A/B's weak-similarity conclusion is spectroscopically corroborated.

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# Executive summary

51 matched analytes · 435 spectra · 785 nm · read-only

## What was asked

Stage A and Stage B concluded that Raman and Ag-SERS observations of the same analytes are only weakly similar. This audit inspects the spectra themselves — using the frozen Stage B corpus and the unmodified Stage B SNV pipeline — to decide whether the two techniques observe fundamentally different spectra, or the same vibrational modes with different enhancement.

## What was found

1. The Stage A/B conclusion is spectroscopically corroborated. Matched pairs separate from mismatched pairs by only +0.03 in cosine, and no analyte shows band correspondence above a mismatched-analyte null.
2. The mechanism is background dominance, not different chemistry. Ag-SERS spectra of 51 chemically distinct metabolites are ~95% identical to each other (between-analyte cosine 0.945 versus 0.349 for Raman); 95% of each Ag-SERS spectrum's variance is explained by the single corpus-mean spectrum — the signature of a citrate/colloid surface background.
3. The Ag-SERS data are not junk. Analyte identity is present as a small residual and supports 73% leave-one-out 1-NN identification within Ag-SERS against 2% chance; replicates agree at 0.999 raw. The analyte signal exists but contributes only ~5% of the spectrum.
4. Apparent band correspondence is a peak-density artefact. Ag-SERS yields ~46 features spaced ~24  $\text{cm}^{-1}$  against ~12 Raman bands spaced ~75  $\text{cm}^{-1}$ , so a  $\pm 12 \text{ cm}^{-1}$  window almost always finds a hit: observed recall 0.80 versus 0.71 for a mismatched analyte and 0.60 for random peaks.
5. A real preprocessing defect exists but does not drive the result. SNV collapses Ag-SERS replicate reproducibility from 0.95 (L2) to 0.49; the negative result nonetheless reproduces under L2 and after removing the shared component.
6. Rigid alignment does not rescue agreement: median optimal shift 0  $\text{cm}^{-1}$ , median cosine gain +0.036. Intensity redistribution is severe (peak-rank  $p$  0.12).

## What it means

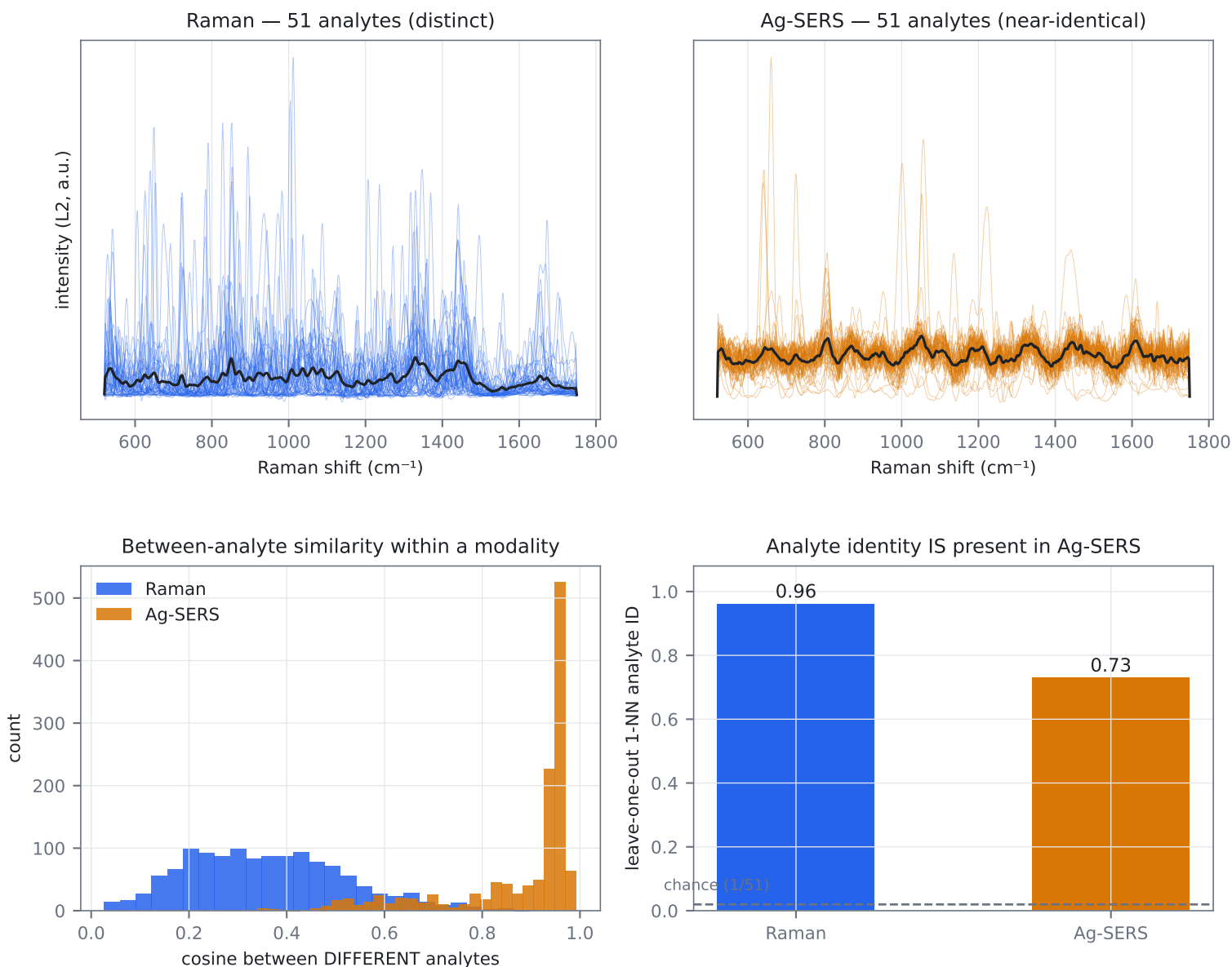
The cross-modal failure is a measurement-contrast problem in the Ag-colloid acquisition rather than evidence that Raman and Ag-SERS observe unrelated chemistry, and it is not a failure of representation learning. It also explains the Stage B encoder collapse: near-identical inputs produce near-identical embeddings. Progress requires Ag-SERS measurements in which the analyte, not the colloid, dominates the spectrum — higher effective surface coverage, blank-colloid difference spectra, or explicit background modelling — before any further representation work.

## Scope and limits

Read-only audit: no models were fitted, no preprocessing was altered, no GAIRA code or governing document was modified. All statistics are computed on the 51 matched analytes of the frozen Stage B corpus. The L2 pipeline appears only as a clearly-labelled sensitivity analysis.

# Decisive evidence 1 — the Ag-SERS arm is background-dominated

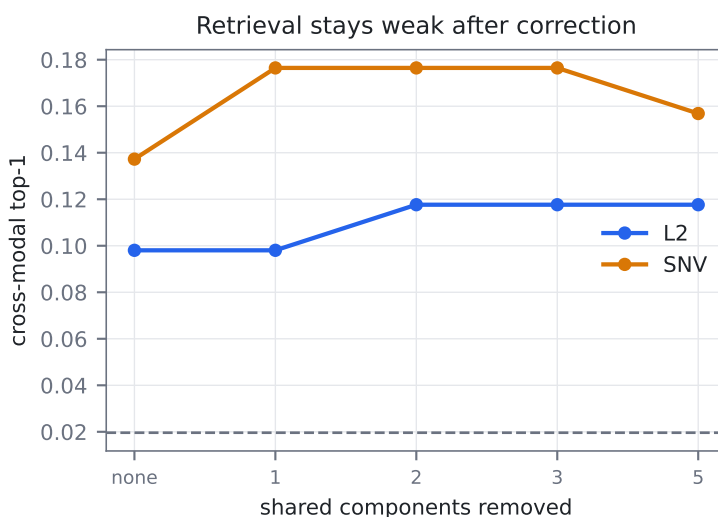
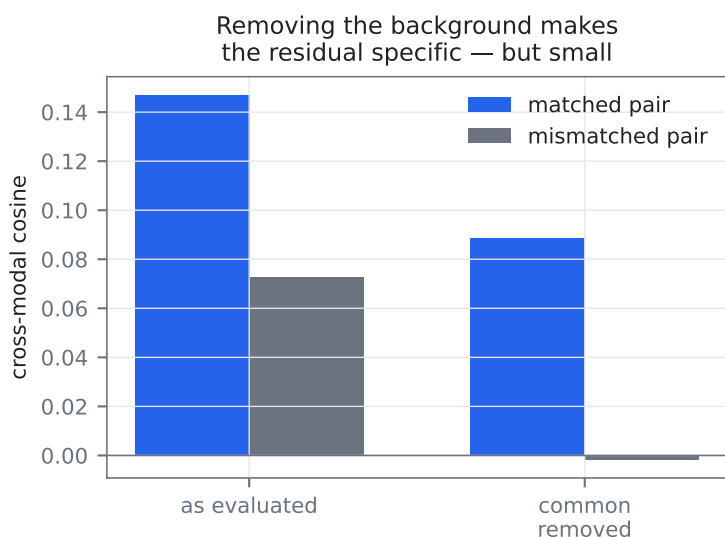
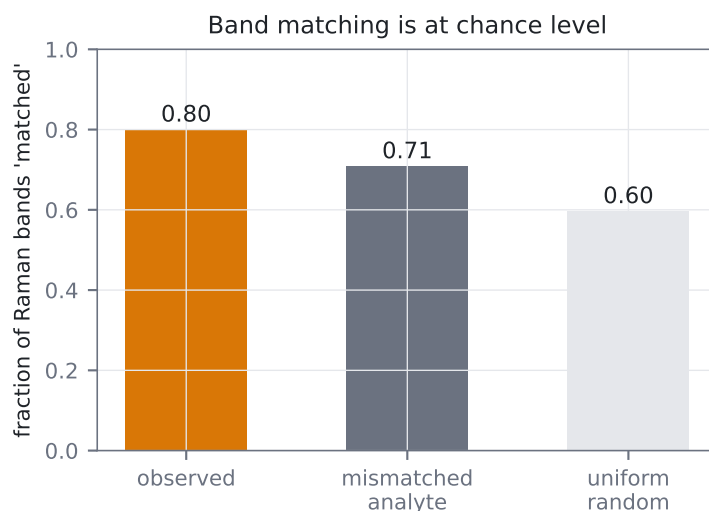
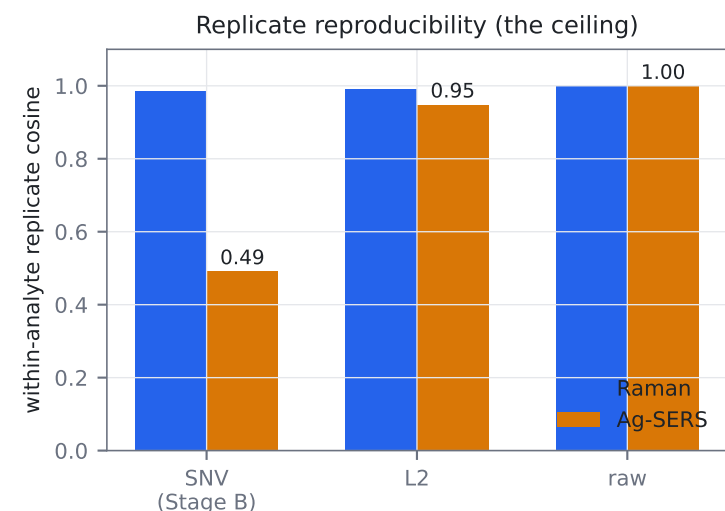
All 51 Ag-SERS mean spectra overlaid (L2) versus all 51 Raman mean spectra, same axes.



## Reading

Ag-SERS spectra of chemically unrelated metabolites are ~95% identical to each other (median between-analyte cosine 0.945 vs 0.349 for Raman), and 95% of each Ag-SERS spectrum's variance is captured by the single corpus-mean spectrum — the signature of a dominant shared colloid/surface background. Analyte-specific information nevertheless exists: leave-one-out 1-NN identification reaches 0.73 within Ag-SERS against 0.02 chance. The analyte therefore contributes a small, real residual on a large common component that has no Raman counterpart.

## Decisive evidence 2 — reproducibility, chance matching, background removal

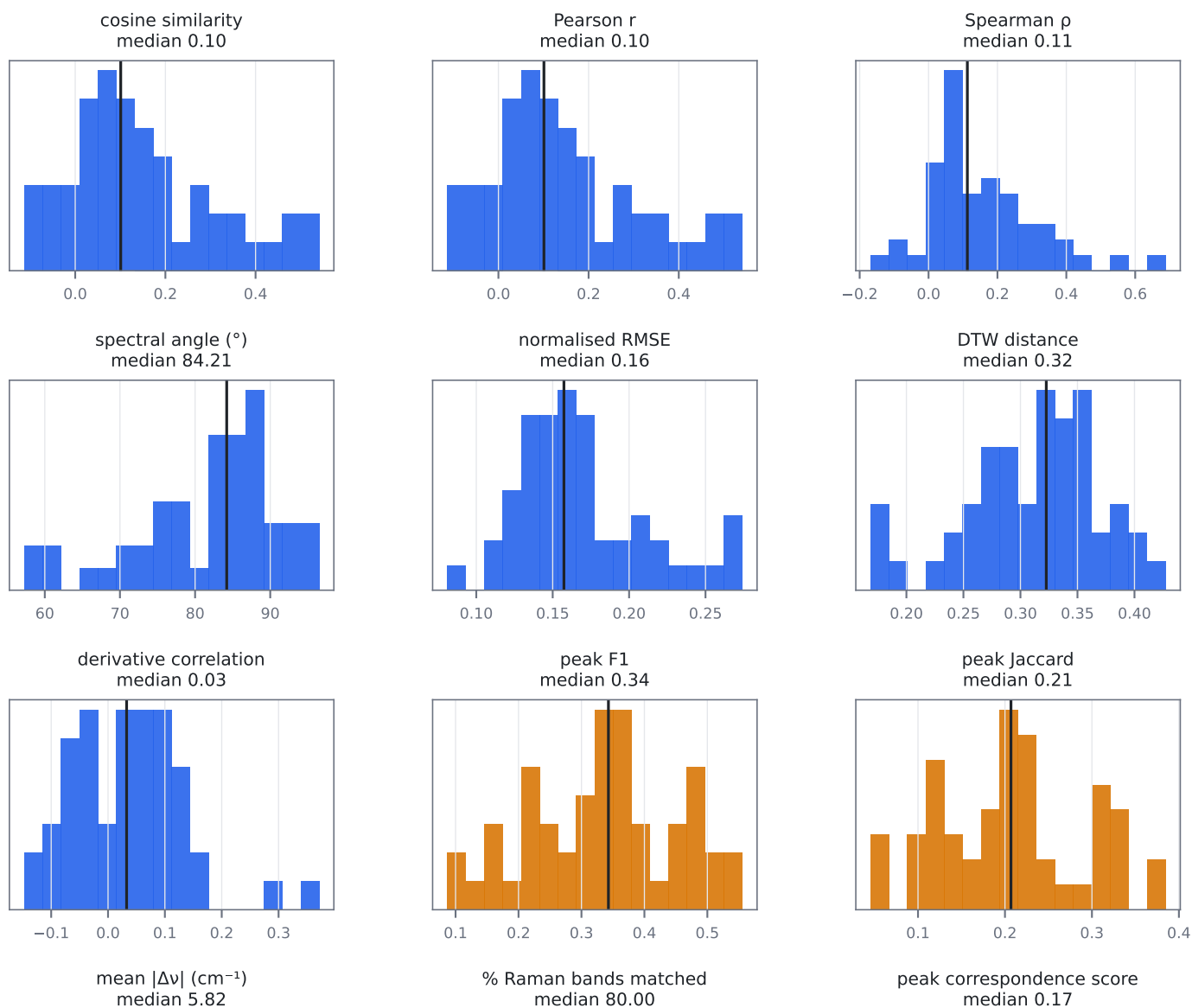


### Reading

SNV — the pipeline Stage A/B selected — collapses Ag-SERS replicate agreement from 0.95 (L2) and 0.999 (raw) to 0.49, because SNV mean-centres each spectrum so cosine becomes a correlation dominated by the low-amplitude residual. This is a genuine methodological defect, but it does not create the negative result: matching a Raman spectrum to a DIFFERENT analyte's Ag-SERS spectrum recovers 71% of bands versus 80% observed (0/51 analytes specific), and this holds identically under L2. Projecting out the shared component makes the surviving cross-modal similarity specific (mismatched falls to ~0) yet leaves top-1 retrieval at only 0.12–0.18 against 0.02 chance.

# Global statistics across 51 matched analytes (Part 9)

Primary pipeline: SNV, exactly as Stage B evaluated.



## Rankings

### Best full-profile agreement

urea (0.54), albumin (0.51), hypoxanthine (0.49), creatinine (0.47), xanthine (0.43), acetyl-coa (0.38), glycerol (0.35), triolein (0.35)

### Worst full-profile agreement

proline (-0.11), phenylalanine (-0.09), pyruvate (-0.09), fructose-6-phosphate (-0.05), ascorbate (-0.04), citrate (-0.03), uracil (-0.02), phosphate (-0.02)

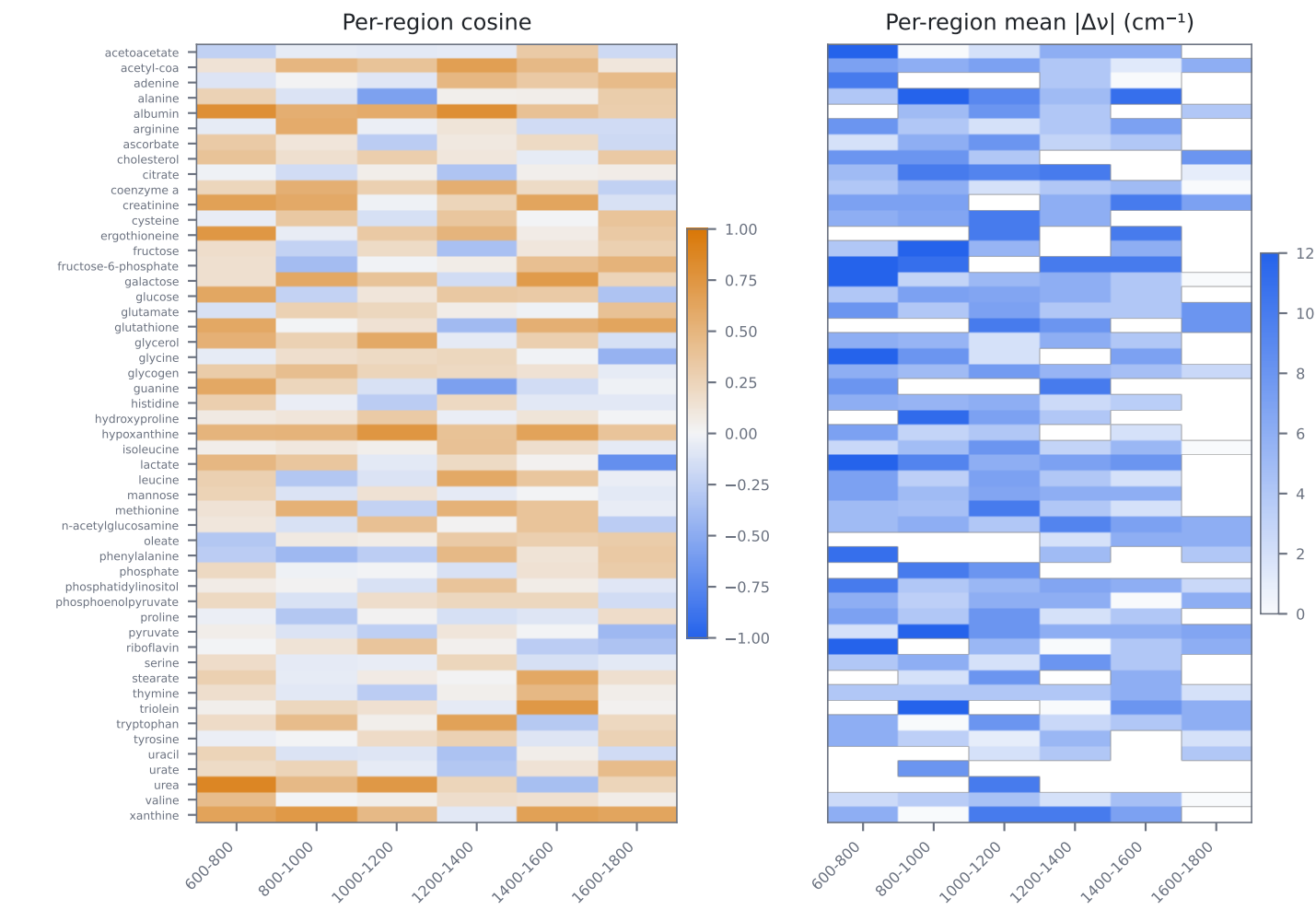
### Largest mean $|\Delta\nu|$

urea (10.00), ergothioneine (10.00), fructose-6-phosphate (9.71), glutathione (9.00), guanine (9.00), hydroxyproline (8.67), urate (8.00), phosphate (8.00)

### Strongest intensity redistribution

cholesterol (0.45), glutamate (0.37), glycine (0.36), thymine (0.33), riboflavin (0.33), hydroxyproline (0.32), citrate (0.31), fructose-6-phosphate (0.30)

Band-level comparison and chemical families (Parts 7, 10)



Chemical-family summary (sorted by peak-correspondence score)

| family           | n  | cosine | peak_f1 | mean_abs_shifl | pcs   | redistribution | rank_corr | matched_pct |
|------------------|----|--------|---------|----------------|-------|----------------|-----------|-------------|
| polysaccharide   | 1  | 0.232  | 0.556   | 5.5            | 0.301 | 0.186          | 0.233     | 74.074      |
| amino_acid       | 16 | 0.086  | 0.356   | 5.404          | 0.204 | 0.254          | 0.056     | 84.346      |
| protein          | 1  | 0.51   | 0.357   | 5.2            | 0.202 | 0.167          | 0.3       | 29.412      |
| polyol           | 1  | 0.348  | 0.314   | 5.0            | 0.183 | 0.202          | -0.31     | 100.0       |
| saccharide       | 6  | 0.075  | 0.383   | 6.727          | 0.177 | 0.229          | -0.201    | 86.015      |
| pyrimidine       | 2  | 0.013  | 0.274   | 5.0            | 0.164 | 0.272          | -0.429    | 72.222      |
| organic_acid     | 7  | -0.005 | 0.339   | 6.45           | 0.159 | 0.232          | 0.198     | 83.503      |
| cofactor         | 5  | 0.246  | 0.306   | 6.709          | 0.148 | 0.215          | 0.098     | 60.381      |
| purine           | 5  | 0.29   | 0.293   | 6.533          | 0.144 | 0.183          | 0.471     | 30.402      |
| lipid            | 5  | 0.179  | 0.266   | 5.986          | 0.131 | 0.216          | 0.059     | 61.099      |
| small_nitrogenou | 2  | 0.508  | 0.28    | 8.889          | 0.08  | 0.179          | 0.3       | 75.0        |

## **Per-analyte pages**

51 matched analytes — Parts 1, 2, 8 and 11



acetoacetate

family: organic\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.001    Pearson +0.001    Spearman -0.029    spectral angle 90.0°    NRMSE 0.265  
DTW 0.262    derivative r -0.061    peak F1 0.33    Jaccard 0.20    precision 0.24    recall 0.50  
peaks: Raman 16, Ag-SERS 33, matched 8    mean |Δν| 5.8 cm<sup>-1</sup>    correspondence score 0.17  
optimal rigid shift -10 cm<sup>-1</sup> (cosine gain +0.084)    replicate ceiling: Raman 0.97 / Ag-SERS 0.72

| Peak correspondence — strongest matches (8 Raman-only, 25 SERS-only) |         |     |         |        |                       |
|----------------------------------------------------------------------|---------|-----|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | Δν  | I ratio | conf.  | note                  |
| 1400                                                                 | 1392    | -8  | 0.34    | Medium | Intensity reduced     |
| 640                                                                  | 652     | +12 | 0.30    | Low    | Intensity reduced     |
| 930                                                                  | 930     | +0  | 0.56    | High   | Intensity reduced     |
| 1006                                                                 | 1004    | -2  | 0.66    | High   | Intensity reduced     |
| 1190                                                                 | 1188    | -2  | 1.23    | High   | Strong correspondence |
| 1444                                                                 | 1438    | -6  | 0.78    | Medium | Shifted band          |
| 1326                                                                 | 1320    | -6  | 2.63    | Medium | Intensity enhanced    |
| 552                                                                  | 562     | +10 | 1.69    | Low    | Intensity enhanced    |

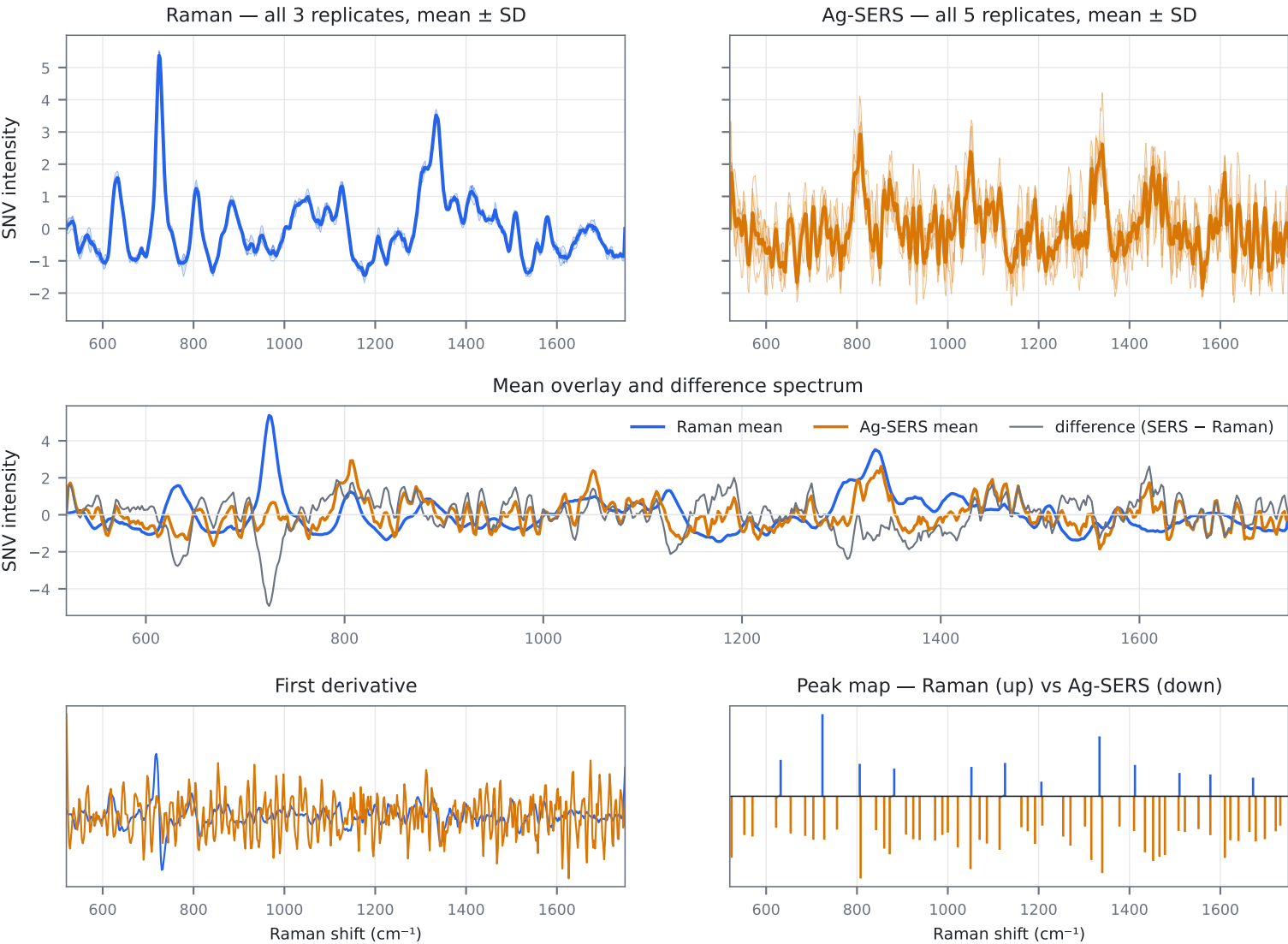
**Spectroscopic interpretation**  
Of 16 prominent Raman bands, 8 (50%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.33), while the full-profile cosine is only +0.00 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 1400→1392 cm<sup>-1</sup> (-8); 640→652 cm<sup>-1</sup> (+12); 930→930 cm<sup>-1</sup> (+0). Modest systematic displacement (mean |Δν| 5.8 cm<sup>-1</sup>, median -2.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank p -0.19, redistribution index 0.26): the same modes are

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

acetyl-coa

family: cofactor (family assignment ambiguous)

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520-1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.378    Pearson +0.378    Spearman +0.331    spectral angle 67.8°    NRMSE 0.149  
DTW 0.288    derivative r +0.164    peak F1 0.38    Jaccard 0.24    precision 0.24    recall 1.00  
peaks: Raman 12, Ag-SERS 51, matched 12    mean |Δν| 4.8 cm<sup>-1</sup>    correspondence score 0.23  
optimal rigid shift +0 cm<sup>-1</sup> (cosine gain +0.000)    replicate ceiling: Raman 0.98 / Ag-SERS 0.59

| Peak correspondence — strongest matches (0 Raman-only, 39 SERS-only) |         |     |         |        |                       |
|----------------------------------------------------------------------|---------|-----|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | Δν  | I ratio | conf.  | note                  |
| 724                                                                  | 728     | +4  | 0.53    | High   | Intensity reduced     |
| 1334                                                                 | 1340    | +6  | 1.28    | Medium | Shifted band          |
| 1126                                                                 | 1114    | -12 | 1.61    | Low    | Intensity enhanced    |
| 632                                                                  | 622     | -10 | 0.86    | Low    | Shifted band          |
| 806                                                                  | 808     | +2  | 2.53    | High   | Intensity enhanced    |
| 882                                                                  | 872     | -10 | 2.10    | Low    | Intensity enhanced    |
| 1578                                                                 | 1578    | +0  | 1.78    | High   | Intensity enhanced    |
| 1510                                                                 | 1508    | -2  | 1.49    | High   | Intensity enhanced    |
| 1412                                                                 | 1410    | -2  | 1.11    | High   | Strong correspondence |

Spectroscopic interpretation

Of 12 prominent Raman bands, 12 (100%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.38), while the full-profile cosine is only +0.38 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 724→728 cm<sup>-1</sup> (+4); 1334→1340 cm<sup>-1</sup> (+6); 1126→1114 cm<sup>-1</sup> (-12). Modest systematic displacement (mean |Δν| 4.8 cm<sup>-1</sup>, median -1.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank ρ 0.28, redistribution index 0.20): the same modes are

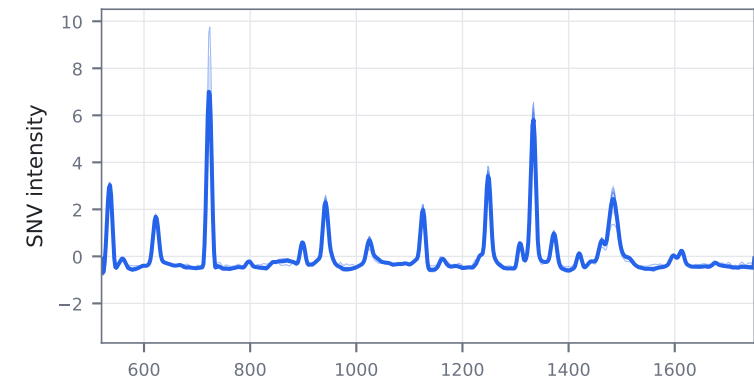
Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# adenine

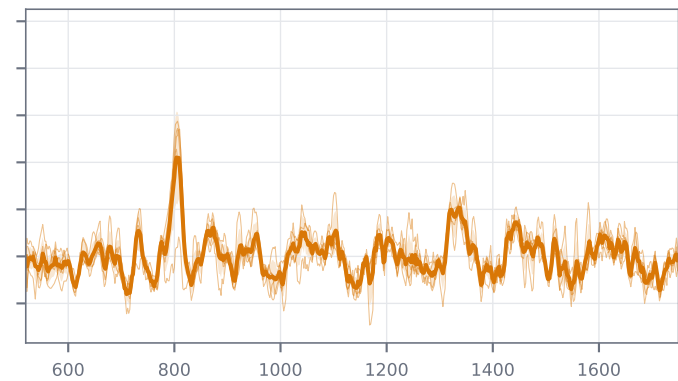
family: purine

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]

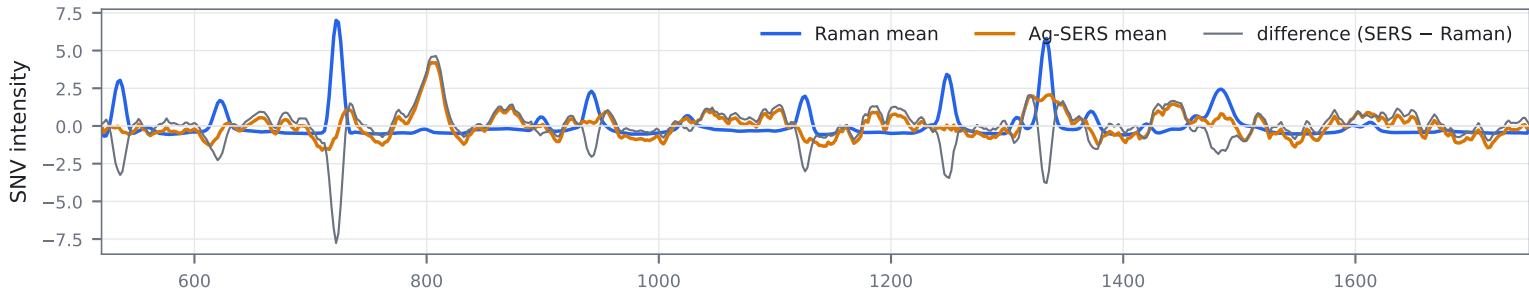
Raman — all 4 replicates, mean ± SD



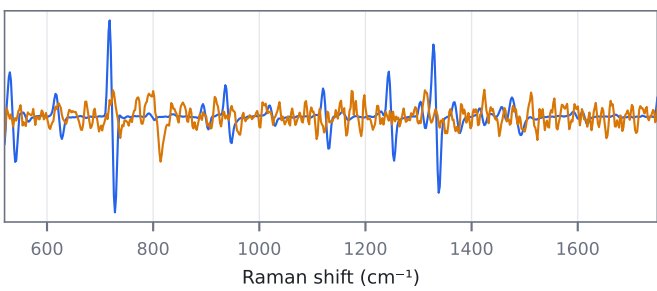
Ag-SERS — all 5 replicates, mean ± SD



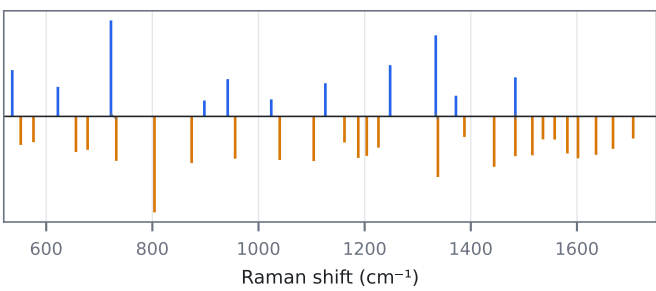
Mean overlay and difference spectrum



First derivative



Peak map — Raman (up) vs Ag-SERS (down)



cosine +0.141    Pearson +0.141    Spearman +0.372    spectral angle 81.9°    NRMSE 0.156  
DTW 0.324    derivative r +0.012    peak F1 0.16    Jaccard 0.09    precision 0.12    recall 0.27  
peaks: Raman 11, Ag-SERS 26, matched 3    mean |Δν| 4.7 cm<sup>-1</sup>    correspondence score 0.10  
optimal rigid shift -8 cm<sup>-1</sup> (cosine gain +0.070)    replicate ceiling: Raman 0.96 / Ag-SERS 0.63

Peak correspondence — strongest matches (8 Raman-only, 23 SERS-only)

| Raman | Ag-SERS | Δν  | I ratio | conf. | note                  |
|-------|---------|-----|---------|-------|-----------------------|
| 722   | 732     | +10 | 0.46    | Low   | Intensity reduced     |
| 1334  | 1338    | +4  | 0.75    | High  | Strong correspondence |
| 1484  | 1484    | +0  | 1.02    | High  | Strong correspondence |

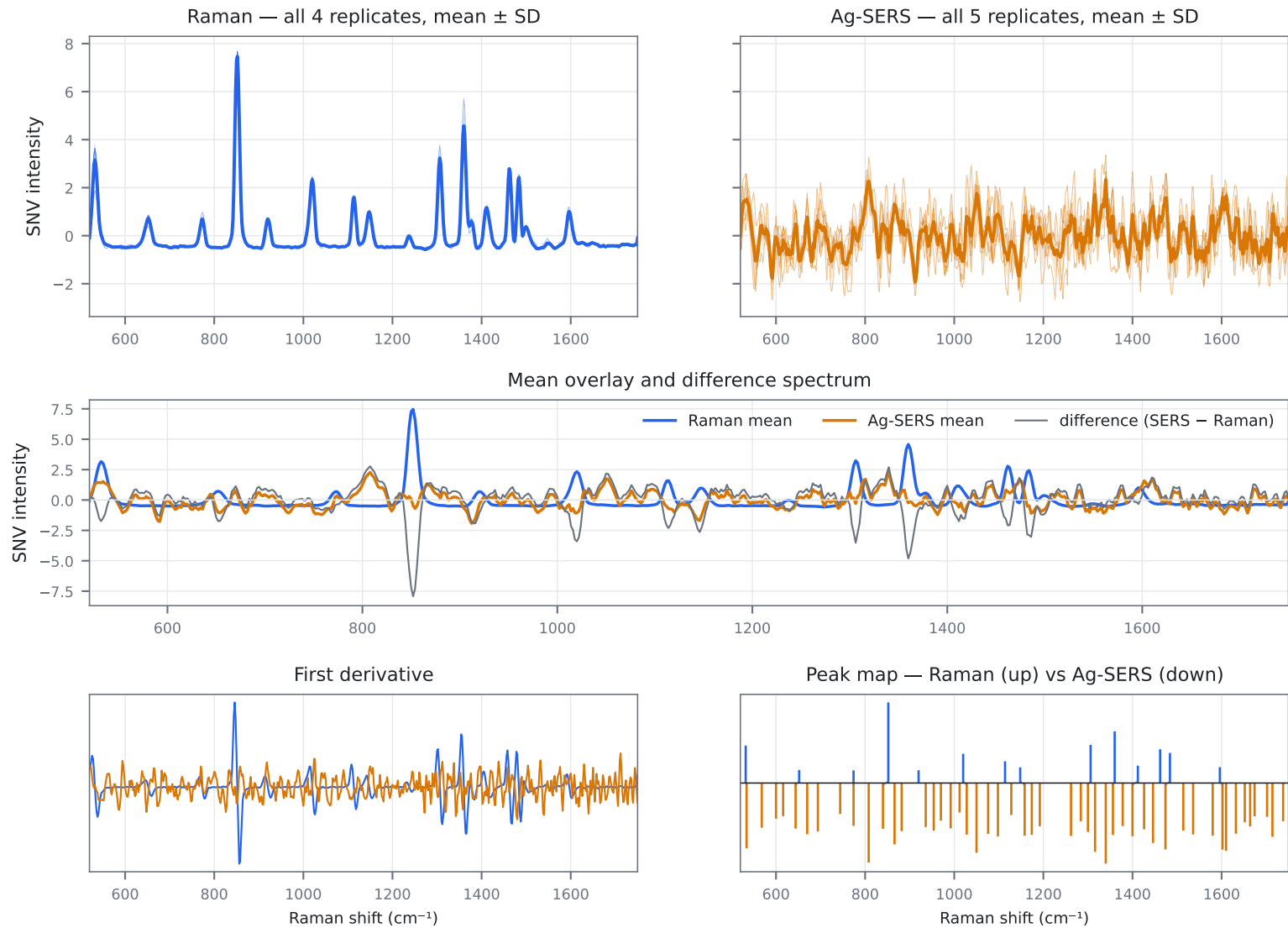
## Spectroscopic interpretation

Of 11 prominent Raman bands, 3 (27%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.16), while the full-profile cosine is only +0.14 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 725 cm<sup>-1</sup> (purine ring breathing) → 732 cm<sup>-1</sup> (+10); 1330 cm<sup>-1</sup> (C-N stretch) → 1338 cm<sup>-1</sup> (+4); 1480 cm<sup>-1</sup> (imidazole) → 1484 cm<sup>-1</sup> (+0). Modest systematic displacement (mean |Δν| 4.7 cm<sup>-1</sup>, median 4.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative band intensities are comparatively well

# alanine

family: amino\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.021    Pearson +0.021    Spearman +0.008    spectral angle 88.8°    NRMSE 0.150  
DTW 0.279    derivative r -0.134    peak F1 0.35    Jaccard 0.21    precision 0.22    recall 0.79  
peaks: Raman 14, Ag-SERS 49, matched 11    mean  $|\Delta\nu|$  7.1 cm<sup>-1</sup>    correspondence score 0.14  
optimal rigid shift -10 cm<sup>-1</sup> (cosine gain +0.148)    replicate ceiling: Raman 0.98 / Ag-SERS 0.38

Peak correspondence — strongest matches (3 Raman-only, 38 SERS-only)

| Raman | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note                  |
|-------|---------|-------------|---------|--------|-----------------------|
| 852   | 840     | -12         | 0.57    | Low    | Intensity reduced     |
| 1360  | 1356    | -4          | 1.00    | High   | Strong correspondence |
| 1306  | 1300    | -6          | 1.28    | Medium | Shifted band          |
| 532   | 534     | +2          | 1.74    | High   | Intensity enhanced    |
| 1020  | 1028    | +8          | 1.76    | Medium | Intensity enhanced    |
| 1484  | 1474    | -10         | 2.20    | Low    | Intensity enhanced    |
| 1412  | 1400    | -12         | 3.06    | Low    | Intensity enhanced    |
| 1596  | 1602    | +6          | 4.19    | Medium | Intensity enhanced    |
| 1148  | 1158    | +10         | 3.35    | Low    | Intensity enhanced    |

## Spectroscopic interpretation

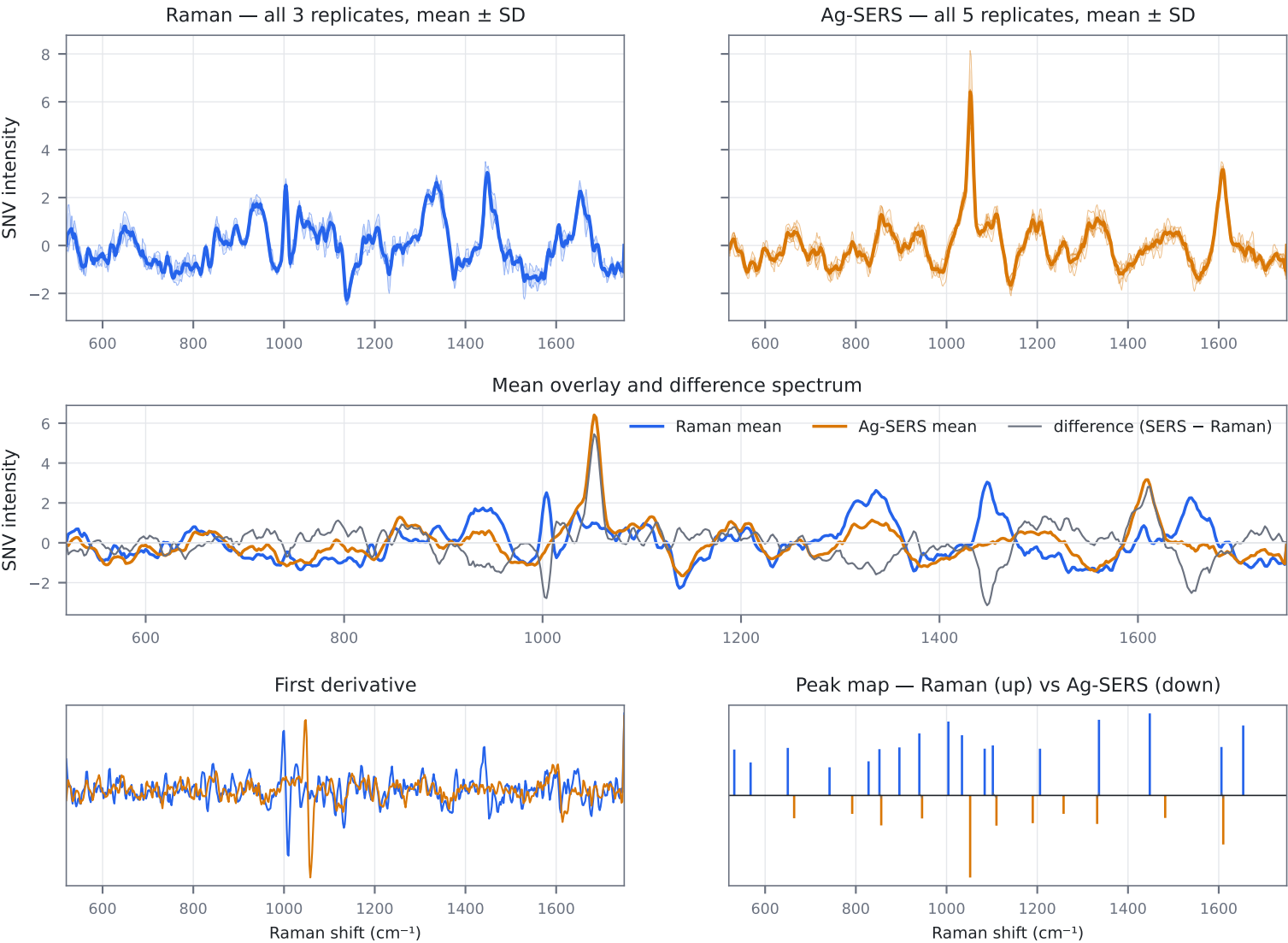
Of 14 prominent Raman bands, 11 (79%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.35), while the full-profile cosine is only +0.02 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 852→840 cm<sup>-1</sup> (-12); 1360→1356 cm<sup>-1</sup> (-4); 1306→1300 cm<sup>-1</sup> (-6). Modest systematic displacement (mean  $|\Delta\nu|$  7.1 cm<sup>-1</sup>, median -4.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank p < 0.05, redistribution index 0.25): the same modes are

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# albumin

family: protein

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.510    Pearson +0.510    Spearman +0.688    spectral angle 59.3°    NRMSE 0.179  
DTW 0.197    derivative r +0.286    peak F1 0.36    Jaccard 0.22    precision 0.45    recall 0.29  
peaks: Raman 17, Ag-SERS 11, matched 5    mean  $|\Delta\nu|$  5.2 cm<sup>-1</sup>    correspondence score 0.20  
optimal rigid shift -2 cm<sup>-1</sup> (cosine gain +0.002)    replicate ceiling: Raman 0.87 / Ag-SERS 0.92

Peak correspondence — strongest matches (12 Raman-only, 6 SERS-only)

| Raman | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note                  |
|-------|---------|-------------|---------|--------|-----------------------|
| 1336  | 1332    | -4          | 0.38    | High   | Intensity reduced     |
| 940   | 946     | +6          | 0.37    | Medium | Intensity reduced     |
| 1606  | 1610    | +4          | 1.01    | High   | Strong correspondence |
| 1102  | 1110    | +8          | 0.60    | Medium | Intensity reduced     |
| 852   | 856     | +4          | 0.65    | High   | Intensity reduced     |

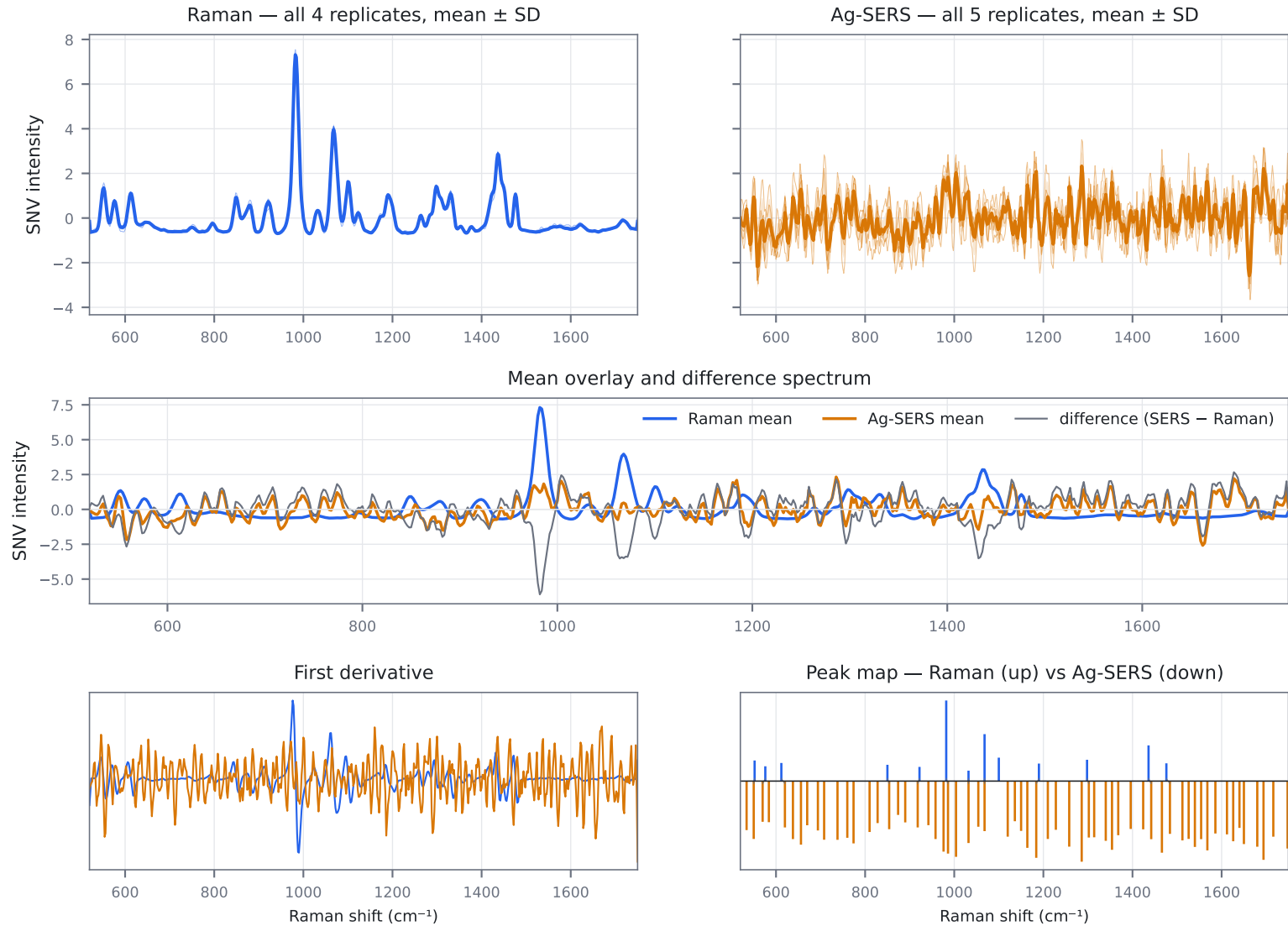
### Spectroscopic interpretation

Of 17 prominent Raman bands, 5 (29%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.36), while the full-profile cosine is only +0.51 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 1336→1332 cm<sup>-1</sup> (-4); 940→946 cm<sup>-1</sup> (+6); 1606→1610 cm<sup>-1</sup> (+4). Modest systematic displacement (mean  $|\Delta\nu|$  5.2 cm<sup>-1</sup>, median 4.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Intensity transfer is partial (peak-rank  $\rho$  0.30, redistribution 0.17), with enhancement favouring a subset of modes. Notes: 6/11 Ag-SERS

# arginine

family: amino\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.138    Pearson +0.138    Spearman +0.043    spectral angle 82.1°    NRMSE 0.145  
DTW 0.393    derivative r +0.033    peak F1 0.33    Jaccard 0.20    precision 0.20    recall 0.92  
peaks: Raman 13, Ag-SERS 60, matched 12    mean  $|\Delta\nu|$  4.3 cm<sup>-1</sup>    correspondence score 0.21  
optimal rigid shift +0 cm<sup>-1</sup> (cosine gain +0.000)    replicate ceiling: Raman 0.99 / Ag-SERS 0.46

| Peak correspondence — strongest matches (1 Raman-only, 48 SERS-only) |         |             |         |        |                       |
|----------------------------------------------------------------------|---------|-------------|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note                  |
| 982                                                                  | 986     | +4          | 0.90    | High   | Strong correspondence |
| 1068                                                                 | 1068    | +0          | 1.07    | High   | Strong correspondence |
| 1436                                                                 | 1442    | +6          | 1.63    | Medium | Intensity enhanced    |
| 1298                                                                 | 1302    | +4          | 2.65    | High   | Intensity enhanced    |
| 552                                                                  | 550     | -2          | 2.81    | High   | Intensity enhanced    |
| 1190                                                                 | 1184    | -6          | 4.38    | Medium | Intensity enhanced    |
| 850                                                                  | 854     | +4          | 2.95    | High   | Intensity enhanced    |
| 612                                                                  | 620     | +8          | 2.56    | Medium | Intensity enhanced    |
| 1476                                                                 | 1484    | +8          | 2.97    | Medium | Intensity enhanced    |

**Spectroscopic interpretation**

Of 13 prominent Raman bands, 12 (92%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.33), while the full-profile cosine is only +0.14 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 982→986 cm<sup>-1</sup> (+4); 1068→1068 cm<sup>-1</sup> (+0); 1436→1442 cm<sup>-1</sup> (+6). Modest systematic displacement (mean  $|\Delta\nu|$  4.3 cm<sup>-1</sup>, median 2.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  0.29, redistribution index 0.25): the same modes are observed but re-weighted

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.



ascorbate

family: organic\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine -0.037    Pearson -0.037    Spearman +0.057    spectral angle 92.1°    NRMSE 0.253  
DTW 0.350    derivative r +0.064    peak F1 0.36    Jaccard 0.22    precision 0.22    recall 0.92  
peaks: Raman 12, Ag-SERS 49, matched 11    mean |Δν| 6.0 cm<sup>-1</sup>    correspondence score 0.18  
optimal rigid shift +10 cm<sup>-1</sup> (cosine gain +0.091)    replicate ceiling: Raman 0.99 / Ag-SERS 0.48

| Peak correspondence — strongest matches (1 Raman-only, 38 SERS-only) |         |     |         |        |                       |
|----------------------------------------------------------------------|---------|-----|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | Δν  | I ratio | conf.  | note                  |
| 1130                                                                 | 1122    | -8  | 0.76    | Medium | Shifted band          |
| 628                                                                  | 628     | +0  | 1.26    | High   | Strong correspondence |
| 1258                                                                 | 1252    | -6  | 1.10    | Medium | Shifted band          |
| 822                                                                  | 812     | -10 | 1.80    | Low    | Intensity enhanced    |
| 1320                                                                 | 1320    | +0  | 1.19    | High   | Strong correspondence |
| 566                                                                  | 578     | +12 | 0.85    | Low    | Shifted band          |
| 710                                                                  | 706     | -4  | 2.62    | High   | Intensity enhanced    |
| 1498                                                                 | 1502    | +4  | 3.28    | High   | Intensity enhanced    |
| 1028                                                                 | 1020    | -8  | 3.91    | Medium | Intensity enhanced    |

Spectroscopic interpretation

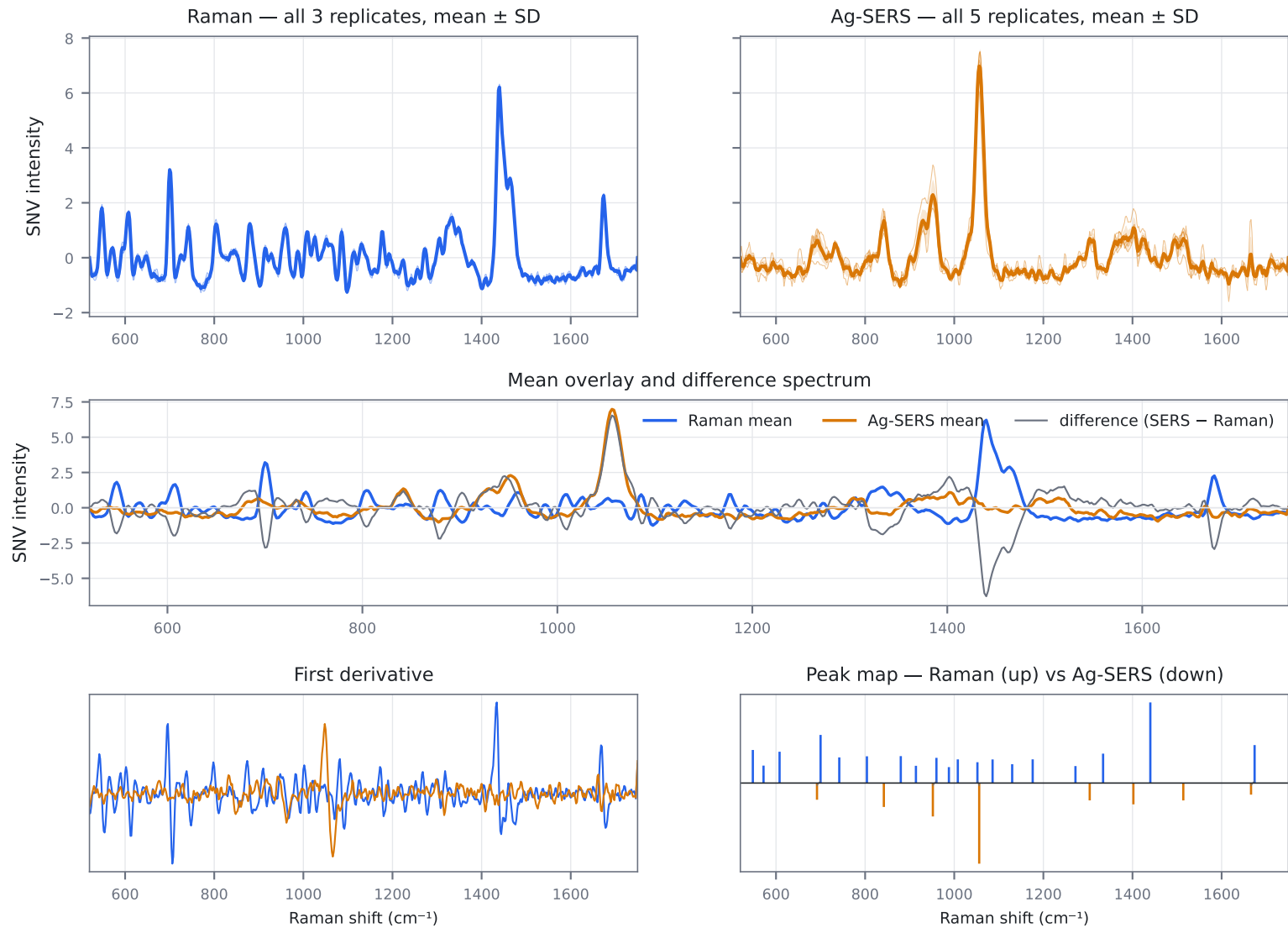
Of 12 prominent Raman bands, 11 (92%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.36), while the full-profile cosine is only -0.04 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 1130→1122 cm<sup>-1</sup> (-8); 628→628 cm<sup>-1</sup> (+0); 1258→1252 cm<sup>-1</sup> (-6). Modest systematic displacement (mean |Δν| 6.0 cm<sup>-1</sup>, median -2.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Intensity transfer is partial (peak-rank ρ 0.38, redistribution 0.25), with enhancement favouring a subset of modes.

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# cholesterol

family: lipid

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.109    Pearson +0.109    Spearman +0.205    spectral angle 83.7°    NRMSE 0.176  
DTW 0.363    derivative r +0.054    peak F1 0.30    Jaccard 0.17    precision 0.50    recall 0.21  
peaks: Raman 19, Ag-SERS 8, matched 4    mean  $|\Delta\nu|$  7.0 cm<sup>-1</sup>    correspondence score 0.12  
optimal rigid shift +6 cm<sup>-1</sup> (cosine gain +0.014)    replicate ceiling: Raman 0.99 / Ag-SERS 0.93

Peak correspondence — strongest matches (15 Raman-only, 4 SERS-only)

| Raman | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note               |
|-------|---------|-------------|---------|--------|--------------------|
| 700   | 692     | -8          | 0.35    | Medium | Intensity reduced  |
| 1674  | 1666    | -8          | 0.31    | Medium | Intensity reduced  |
| 960   | 952     | -8          | 1.33    | Medium | Shifted band       |
| 1052  | 1056    | +4          | 3.87    | High   | Intensity enhanced |

### Spectroscopic interpretation

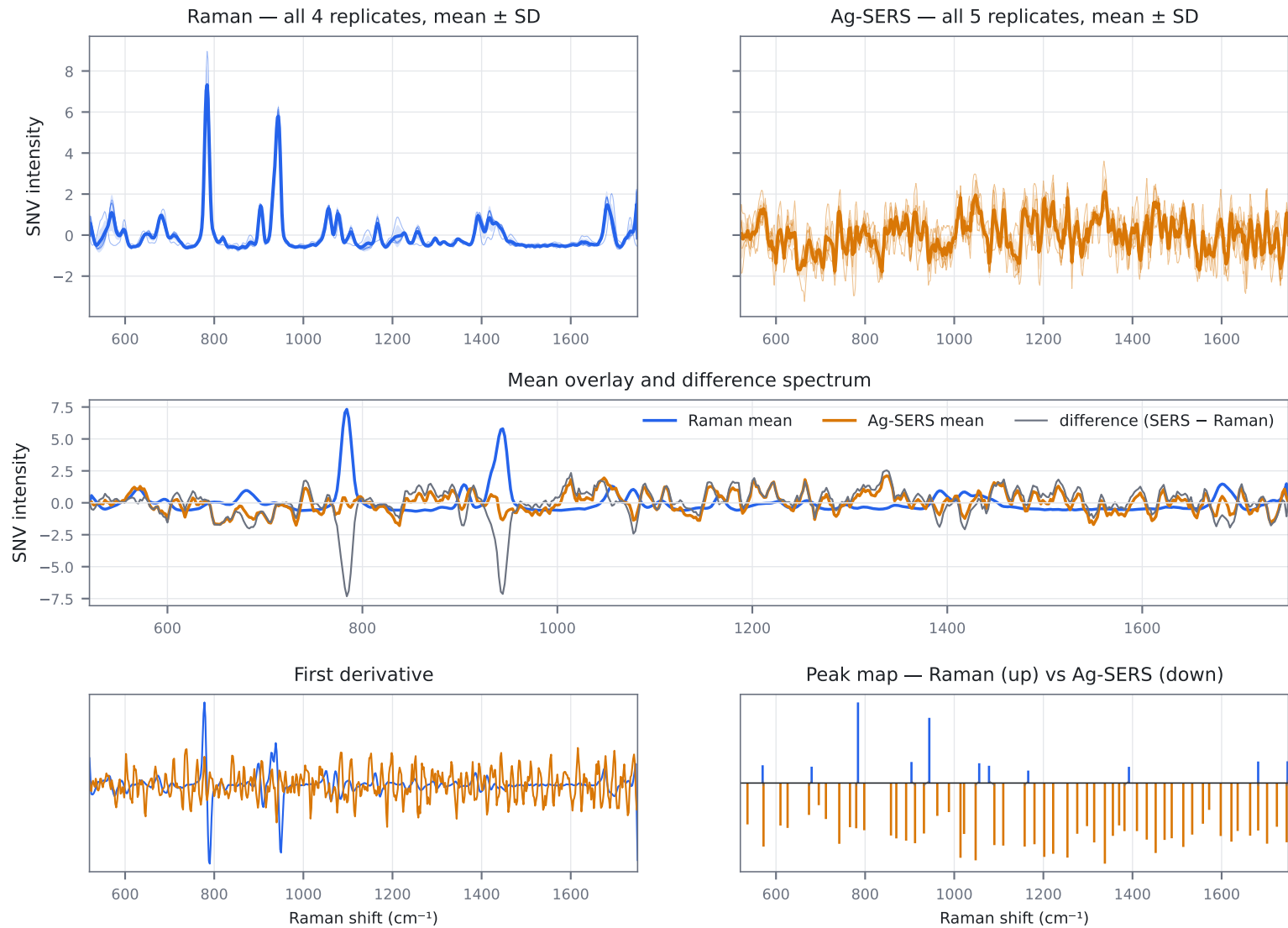
Of 19 prominent Raman bands, 4 (21%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.30), while the full-profile cosine is only +0.11 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 1670 cm<sup>-1</sup> (C=C stretch) → 1666 cm<sup>-1</sup> (-8). Modest systematic displacement (mean  $|\Delta\nu|$  7.0 cm<sup>-1</sup>, median -8.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Intensity transfer is partial (peak-rank  $\rho$  -0.80, redistribution 0.45), with enhancement favouring a subset of modes. Notes: a band in the 1380–1430



# citrate

family: organic\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine -0.032    Pearson -0.032    Spearman +0.059    spectral angle 91.8°    NRMSE 0.157  
DTW 0.397    derivative r -0.022    peak F1 0.35    Jaccard 0.21    precision 0.21    recall 1.00  
peaks: Raman 11, Ag-SERS 52, matched 11    mean | $\Delta\nu$ | 6.5 cm<sup>-1</sup>    correspondence score 0.16  
optimal rigid shift +10 cm<sup>-1</sup> (cosine gain +0.055)    replicate ceiling: Raman 0.91 / Ag-SERS 0.50

Peak correspondence — strongest matches (0 Raman-only, 41 SERS-only)

| Raman | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note               |
|-------|---------|-------------|---------|--------|--------------------|
| 784   | 780     | -4          | 0.56    | High   | Intensity reduced  |
| 944   | 932     | -12         | 0.77    | Low    | Shifted band       |
| 1682  | 1682    | +0          | 2.20    | High   | Intensity enhanced |
| 1056  | 1048    | -8          | 3.90    | Medium | Intensity enhanced |
| 904   | 912     | +8          | 2.86    | Medium | Intensity enhanced |
| 1748  | 1746    | -2          | 2.72    | High   | Intensity enhanced |
| 570   | 572     | +2          | 3.58    | High   | Intensity enhanced |
| 680   | 674     | -6          | 1.95    | Medium | Intensity enhanced |
| 1392  | 1382    | -10         | 2.96    | Low    | Intensity enhanced |

## Spectroscopic interpretation

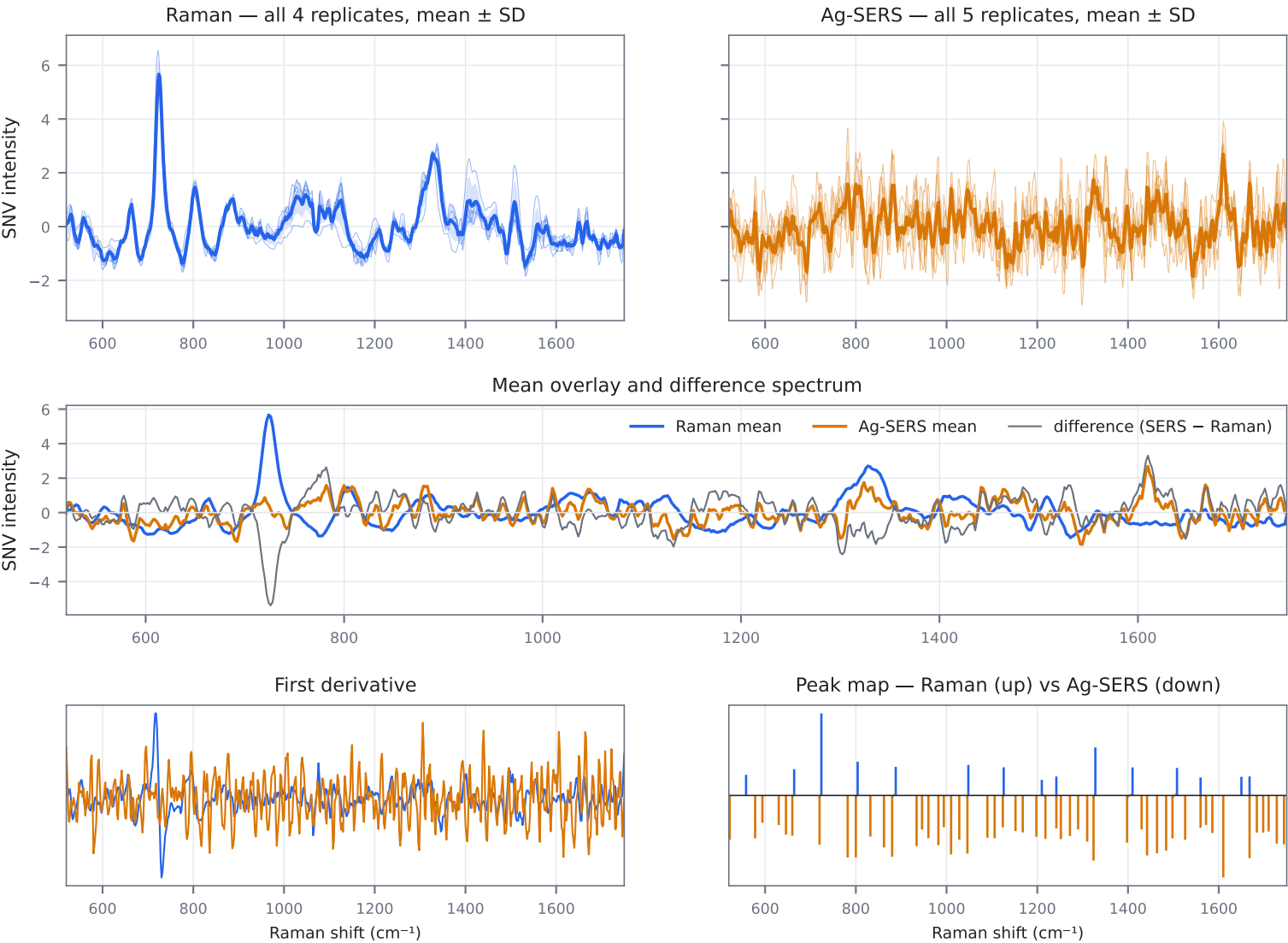
Of 11 prominent Raman bands, 11 (100%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.35), while the full-profile cosine is only -0.03 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 950 cm<sup>-1</sup> (C-C) → 932 cm<sup>-1</sup> (-12). Modest systematic displacement (mean | $\Delta\nu$ | 6.5 cm<sup>-1</sup>, median -4.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  -0.35, redistribution index 0.31): the same modes are observed but re-weighted by

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# coenzyme a

family: cofactor (family assignment ambiguous)

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.268    Pearson +0.268    Spearman +0.255    spectral angle 74.4°    NRMSE 0.139  
DTW 0.247    derivative r +0.093    peak F1 0.40    Jaccard 0.25    precision 0.26    recall 0.87  
peaks: Raman 15, Ag-SERS 50, matched 13    mean  $|\Delta\nu|$  4.5 cm<sup>-1</sup>    correspondence score 0.25  
optimal rigid shift +4 cm<sup>-1</sup> (cosine gain +0.006)    replicate ceiling: Raman 0.82 / Ag-SERS 0.33

| Peak correspondence — strongest matches (2 Raman-only, 37 SERS-only) |         |             |         |        |                       |
|----------------------------------------------------------------------|---------|-------------|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note                  |
| 724                                                                  | 720     | -4          | 0.60    | High   | Intensity reduced     |
| 1328                                                                 | 1324    | -4          | 1.36    | High   | Strong correspondence |
| 804                                                                  | 800     | -4          | 1.84    | High   | Intensity enhanced    |
| 1048                                                                 | 1046    | -2          | 1.91    | High   | Intensity enhanced    |
| 664                                                                  | 660     | -4          | 1.53    | High   | Intensity enhanced    |
| 1508                                                                 | 1498    | -10         | 1.56    | Low    | Intensity enhanced    |
| 888                                                                  | 880     | -8          | 2.14    | Medium | Intensity enhanced    |
| 1410                                                                 | 1398    | -12         | 1.69    | Low    | Intensity enhanced    |
| 1126                                                                 | 1124    | -2          | 1.14    | High   | Strong correspondence |

## Spectroscopic interpretation

Of 15 prominent Raman bands, 13 (87%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.40), while the full-profile cosine is only +0.27 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 724→720 cm<sup>-1</sup> (-4); 1328→1324 cm<sup>-1</sup> (-4); 804→800 cm<sup>-1</sup> (-4). Modest systematic displacement (mean  $|\Delta\nu|$  4.5 cm<sup>-1</sup>, median -4.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative band intensities are comparatively well preserved (peak-rank  $\rho$  0.55, redistribution 0.16), an unusually faithful

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# creatinine

family: small\_nitrogenous

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.475    Pearson +0.475    Spearman +0.455    spectral angle 61.7°    NRMSE 0.132  
DTW 0.279    derivative r +0.130    peak F1 0.36    Jaccard 0.22    precision 0.22    recall 1.00  
peaks: Raman 9, Ag-SERS 41, matched 9    mean | $\Delta\nu$ | 7.8 cm<sup>-1</sup>    correspondence score 0.13  
optimal rigid shift -4 cm<sup>-1</sup> (cosine gain +0.053)    replicate ceiling: Raman 1.00 / Ag-SERS 0.49

| Peak correspondence — strongest matches (0 Raman-only, 32 SERS-only) |         |             |         |        |                    |
|----------------------------------------------------------------------|---------|-------------|---------|--------|--------------------|
| Raman                                                                | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note               |
| 606                                                                  | 612     | +6          | 0.43    | Medium | Intensity reduced  |
| 674                                                                  | 682     | +8          | 1.07    | Medium | Shifted band       |
| 904                                                                  | 914     | +10         | 1.08    | Low    | Shifted band       |
| 838                                                                  | 842     | +4          | 1.57    | High   | Intensity enhanced |
| 1658                                                                 | 1652    | -6          | 1.23    | Medium | Shifted band       |
| 578                                                                  | 590     | +12         | 1.44    | Low    | Intensity enhanced |
| 1416                                                                 | 1426    | +10         | 2.88    | Low    | Intensity enhanced |
| 1246                                                                 | 1252    | +6          | 1.83    | Medium | Intensity enhanced |
| 1712                                                                 | 1704    | -8          | 1.59    | Medium | Intensity enhanced |

**Spectroscopic interpretation**  
Of 9 prominent Raman bands, 9 (100%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.36), while the full-profile cosine is only +0.47 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 680 cm<sup>-1</sup> (ring) → 682 cm<sup>-1</sup> (+8). Modest systematic displacement (mean | $\Delta\nu$ | 7.8 cm<sup>-1</sup>, median 6.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Intensity transfer is partial (peak-rank  $\rho$  0.30, redistribution 0.18), with enhancement favouring a subset of modes. Notes: 32/41 Ag-SERS bands have

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# cysteine

family: amino\_acid

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.083    Pearson +0.083    Spearman +0.190    spectral angle 85.3°    NRMSE 0.140  
DTW 0.325    derivative r +0.035    peak F1 0.34    Jaccard 0.20    precision 0.22    recall 0.71  
peaks: Raman 14, Ag-SERS 45, matched 10    mean | $\Delta\nu$ | 6.6 cm<sup>-1</sup>    correspondence score 0.15  
optimal rigid shift +10 cm<sup>-1</sup> (cosine gain +0.048)    replicate ceiling: Raman 1.00 / Ag-SERS 0.42

Peak correspondence — strongest matches (4 Raman-only, 35 SERS-only)

| Raman | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note               |
|-------|---------|-------------|---------|--------|--------------------|
| 640   | 640     | +0          | 0.46    | High   | Intensity reduced  |
| 1346  | 1344    | -2          | 1.79    | High   | Intensity enhanced |
| 772   | 760     | -12         | 2.17    | Low    | Intensity enhanced |
| 1200  | 1208    | +8          | 2.46    | Medium | Intensity enhanced |
| 822   | 812     | -10         | 4.41    | Low    | Intensity enhanced |
| 868   | 860     | -8          | 4.96    | Medium | Intensity enhanced |
| 1294  | 1298    | +4          | 2.76    | High   | Intensity enhanced |
| 942   | 940     | -2          | 4.44    | High   | Intensity enhanced |
| 1398  | 1388    | -10         | 2.19    | Low    | Intensity enhanced |

## Spectroscopic interpretation

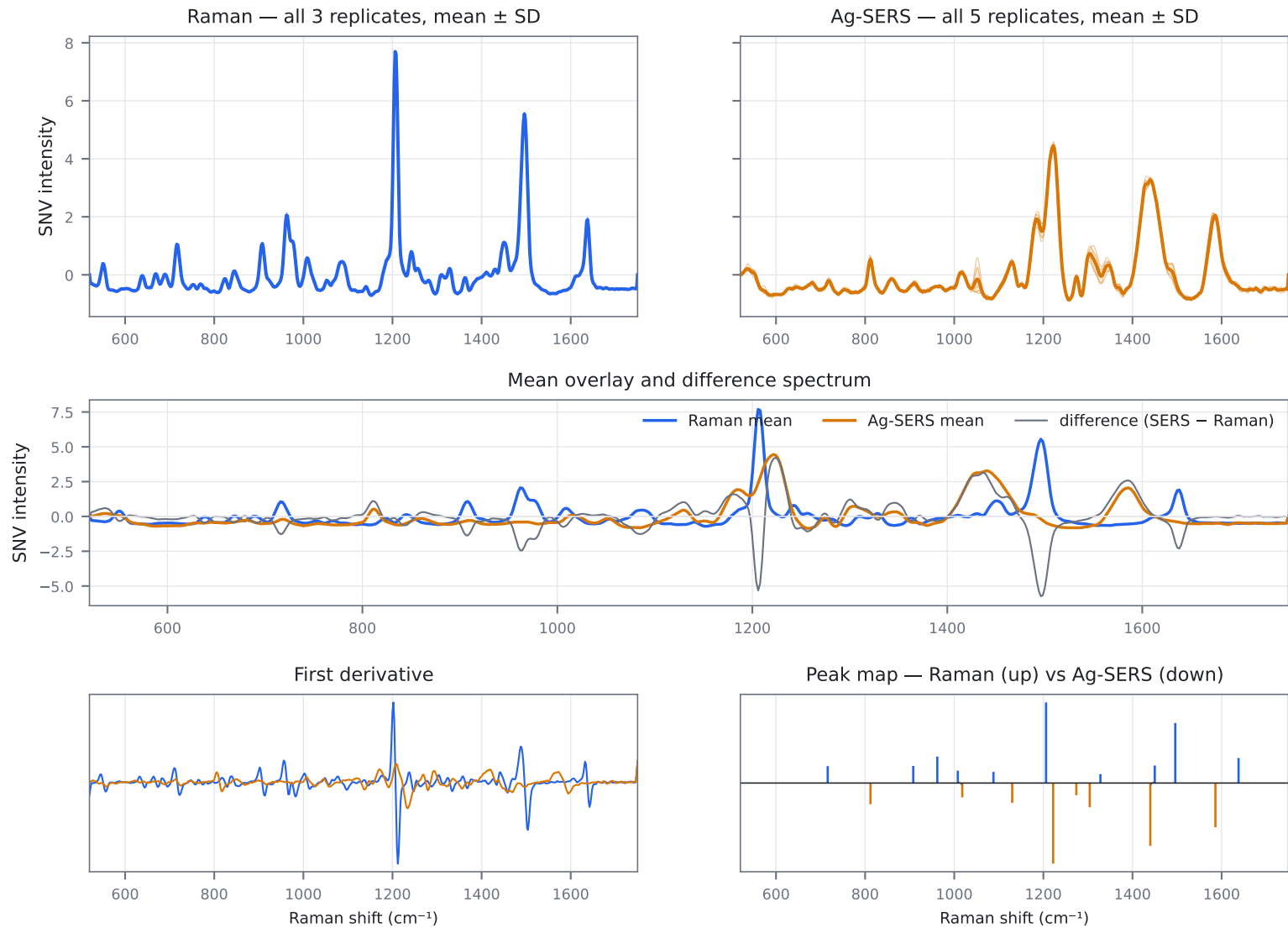
Of 14 prominent Raman bands, 10 (71%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.34), while the full-profile cosine is only +0.08 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 640→640 cm<sup>-1</sup> (+0); 1346→1344 cm<sup>-1</sup> (-2); 772→760 cm<sup>-1</sup> (-12). Modest systematic displacement (mean | $\Delta\nu$ | 6.6 cm<sup>-1</sup>, median -2.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  -0.33, redistribution index 0.30): the same modes are

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# ergothioneine

family: cofactor (family assignment ambiguous)

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.300    Pearson +0.300    Spearman +0.245    spectral angle 72.6°    NRMSE 0.140  
DTW 0.185    derivative r -0.036    peak F1 0.22    Jaccard 0.12    precision 0.25    recall 0.20  
peaks: Raman 10, Ag-SERS 8, matched 2    mean  $|\Delta v|$  10.0 cm<sup>-1</sup>    correspondence score 0.04  
optimal rigid shift -10 cm<sup>-1</sup> (cosine gain +0.036)    replicate ceiling: Raman 1.00 / Ag-SERS 0.99

Peak correspondence — strongest matches (8 Raman-only, 6 SERS-only)

| Raman | Ag-SERS | $\Delta v$ | I ratio | conf. | note               |
|-------|---------|------------|---------|-------|--------------------|
| 1008  | 1018    | +10        | 1.16    | Low   | Shifted band       |
| 1450  | 1440    | -10        | 3.58    | Low   | Intensity enhanced |

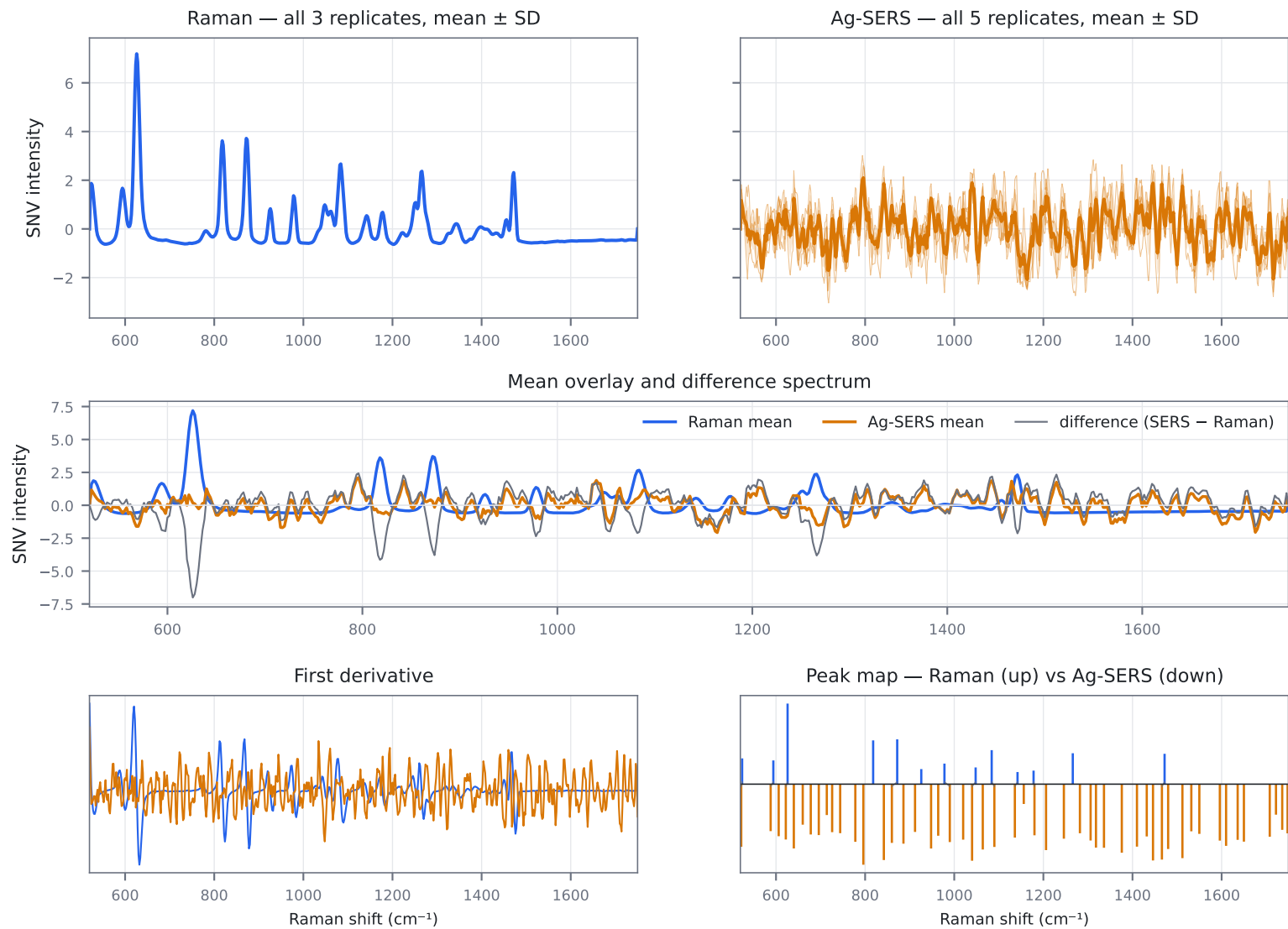
### Spectroscopic interpretation

Of 10 prominent Raman bands, 2 (20%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.22), while the full-profile cosine is only +0.30 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 1008→1018 cm<sup>-1</sup> (+10); 1450→1440 cm<sup>-1</sup> (-10). Larger displacements (mean  $|\Delta v|$  10.0 cm<sup>-1</sup>, max 10.0 cm<sup>-1</sup>) suggest strong surface interaction or ambiguous band assignment in a congested region. Notes: 6/8 Ag-SERS bands have no Raman counterpart, which may reflect surface-activated modes, adsorbate reorientation, or residual citrate/colloid features; 8/10 Raman

# fructose

family: saccharide

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.051    Pearson +0.051    Spearman +0.113    spectral angle 87.1°    NRMSE 0.156  
DTW 0.378    derivative r -0.034    peak F1 0.32    Jaccard 0.19    precision 0.20    recall 0.77  
peaks: Raman 13, Ag-SERS 50, matched 10    mean  $|\Delta\nu|$  6.4 cm<sup>-1</sup>    correspondence score 0.15  
optimal rigid shift +10 cm<sup>-1</sup> (cosine gain +0.078)    replicate ceiling: Raman 1.00 / Ag-SERS 0.46

Peak correspondence — strongest matches (3 Raman-only, 40 SERS-only)

| Raman | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note               |
|-------|---------|-------------|---------|--------|--------------------|
| 626   | 622     | -4          | 0.69    | High   | Intensity reduced  |
| 872   | 860     | -12         | 1.31    | Low    | Shifted band       |
| 1084  | 1090    | +6          | 1.85    | Medium | Intensity enhanced |
| 1472  | 1466    | -6          | 2.49    | Medium | Intensity enhanced |
| 978   | 990     | +12         | 2.82    | Low    | Intensity enhanced |
| 524   | 522     | -2          | 2.43    | High   | Intensity enhanced |
| 594   | 588     | -6          | 1.98    | Medium | Intensity enhanced |
| 1178  | 1180    | +2          | 3.78    | High   | Intensity enhanced |
| 1142  | 1136    | -6          | 4.41    | Medium | Intensity enhanced |

## Spectroscopic interpretation

Of 13 prominent Raman bands, 10 (77%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.32), while the full-profile cosine is only +0.05 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 626→622 cm<sup>-1</sup> (-4); 872→860 cm<sup>-1</sup> (-12); 1084→1090 cm<sup>-1</sup> (+6). Modest systematic displacement (mean  $|\Delta\nu|$  6.4 cm<sup>-1</sup>, median -5.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  -0.38, redistribution index 0.23): the same modes are

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.



# fructose-6-phosphate

family: saccharide

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine -0.052    Pearson -0.052    Spearman +0.060    spectral angle 93.0°    NRMSE 0.158  
DTW 0.289    derivative r -0.037    peak F1 0.22    Jaccard 0.12    precision 0.13    recall 0.88  
peaks: Raman 8, Ag-SERS 55, matched 7    mean  $|\Delta\nu|$  9.7 cm<sup>-1</sup>    correspondence score 0.04  
optimal rigid shift +10 cm<sup>-1</sup> (cosine gain +0.069)    replicate ceiling: Raman 0.98 / Ag-SERS 0.28

Peak correspondence — strongest matches (1 Raman-only, 48 SERS-only)

| Raman | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note               |
|-------|---------|-------------|---------|--------|--------------------|
| 972   | 962     | -10         | 0.57    | Low    | Intensity reduced  |
| 832   | 820     | -12         | 1.61    | Low    | Intensity enhanced |
| 1460  | 1470    | +10         | 1.93    | Low    | Intensity enhanced |
| 622   | 610     | -12         | 2.74    | Low    | Intensity enhanced |
| 534   | 530     | -4          | 2.85    | High   | Intensity enhanced |
| 1266  | 1254    | -12         | 4.29    | Low    | Intensity enhanced |
| 1346  | 1338    | -8          | 5.56    | Medium | Intensity enhanced |

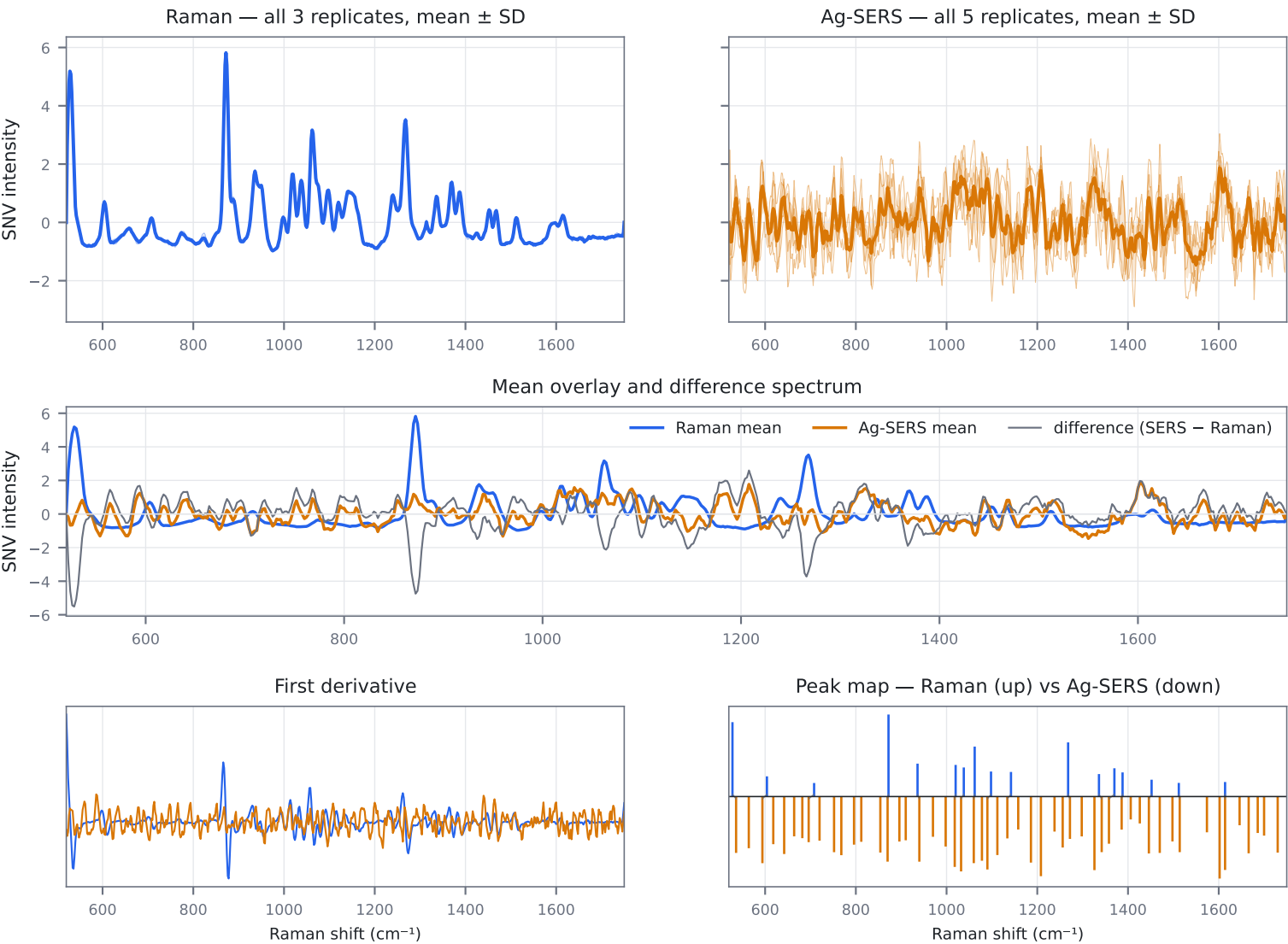
## Spectroscopic interpretation

Of 8 prominent Raman bands, 7 (88%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.22), while the full-profile cosine is only -0.05 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 972→962 cm<sup>-1</sup> (-10); 832→820 cm<sup>-1</sup> (-12); 1460→1470 cm<sup>-1</sup> (+10). Larger displacements (mean  $|\Delta\nu|$  9.7 cm<sup>-1</sup>, max 12.0 cm<sup>-1</sup>) suggest strong surface interaction or ambiguous band assignment in a congested region. Relative intensities are strongly reordered (peak-rank  $\rho$  -0.61, redistribution index 0.30): the same modes are observed but re-weighted by SERS surface selection

# galactose

family: saccharide

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.258    Pearson +0.258    Spearman +0.303    spectral angle 75.1°    NRMSE 0.155  
DTW 0.283    derivative r +0.068    peak F1 0.49    Jaccard 0.33    precision 0.33    recall 1.00  
peaks: Raman 17, Ag-SERS 52, matched 17    mean | $\Delta v$ | 5.5 cm<sup>-1</sup>    correspondence score 0.27  
optimal rigid shift +4 cm<sup>-1</sup> (cosine gain +0.010)    replicate ceiling: Raman 1.00 / Ag-SERS 0.33

| Peak correspondence — strongest matches (0 Raman-only, 35 SERS-only) |         |            |         |        |                       |
|----------------------------------------------------------------------|---------|------------|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | $\Delta v$ | I ratio | conf.  | note                  |
| 872                                                                  | 870     | -2         | 0.79    | High   | Strong correspondence |
| 528                                                                  | 536     | +8         | 0.76    | Medium | Shifted band          |
| 1268                                                                 | 1272    | +4         | 0.78    | High   | Strong correspondence |
| 1062                                                                 | 1060    | -2         | 1.33    | High   | Strong correspondence |
| 936                                                                  | 940     | +4         | 1.98    | High   | Intensity enhanced    |
| 1370                                                                 | 1358    | -12        | 1.67    | Low    | Intensity enhanced    |
| 1020                                                                 | 1018    | -2         | 2.22    | High   | Intensity enhanced    |
| 604                                                                  | 594     | -10        | 3.30    | Low    | Intensity enhanced    |
| 1142                                                                 | 1134    | -8         | 1.70    | Medium | Intensity enhanced    |

## Spectroscopic interpretation

Of 17 prominent Raman bands, 17 (100%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.49), while the full-profile cosine is only +0.26 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 872→870 cm<sup>-1</sup> (-2); 528→536 cm<sup>-1</sup> (+8); 1268→1272 cm<sup>-1</sup> (+4). Modest systematic displacement (mean  $|\Delta v|$  5.5 cm<sup>-1</sup>, median -2.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  0.12, redistribution index 0.25): the same modes are

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.



# glucose

family: saccharide

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.095    Pearson +0.095    Spearman +0.094    spectral angle 84.5°    NRMSE 0.200  
DTW 0.300    derivative r -0.065    peak F1 0.38    Jaccard 0.23    precision 0.24    recall 0.92  
peaks: Raman 12, Ag-SERS 46, matched 11    mean | $\Delta\nu$ | 5.8 cm<sup>-1</sup>    correspondence score 0.20  
optimal rigid shift +10 cm<sup>-1</sup> (cosine gain +0.187)    replicate ceiling: Raman 1.00 / Ag-SERS 0.39

| Peak correspondence — strongest matches (1 Raman-only, 35 SERS-only) |         |             |         |        |                       |
|----------------------------------------------------------------------|---------|-------------|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note                  |
| 542                                                                  | 548     | +6          | 0.52    | Medium | Intensity reduced     |
| 842                                                                  | 840     | -2          | 0.59    | High   | Intensity reduced     |
| 1120                                                                 | 1118    | -2          | 0.80    | High   | Strong correspondence |
| 1346                                                                 | 1350    | +4          | 0.97    | High   | Strong correspondence |
| 916                                                                  | 904     | -12         | 1.29    | Low    | Shifted band          |
| 1460                                                                 | 1464    | +4          | 2.13    | High   | Intensity enhanced    |
| 1272                                                                 | 1264    | -8          | 1.61    | Medium | Intensity enhanced    |
| 1150                                                                 | 1156    | +6          | 1.81    | Medium | Intensity enhanced    |
| 1022                                                                 | 1010    | -12         | 2.27    | Low    | Intensity enhanced    |

### Spectroscopic interpretation

Of 12 prominent Raman bands, 11 (92%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.38), while the full-profile cosine is only +0.10 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 1120 cm<sup>-1</sup> (C-O-C) → 1118 cm<sup>-1</sup> (-2). Modest systematic displacement (mean | $\Delta\nu$ | 5.8 cm<sup>-1</sup>, median 2.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  -0.32, redistribution index 0.25): the same modes are observed but re-weighted by

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

glutamate

family: amino\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520-1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.137    Pearson +0.137    Spearman +0.205    spectral angle 82.1°    NRMSE 0.149  
DTW 0.319    derivative r +0.086    peak F1 0.35    Jaccard 0.21    precision 0.23    recall 0.80  
peaks: Raman 15, Ag-SERS 53, matched 12    mean |Δν| 5.2 cm<sup>-1</sup>    correspondence score 0.20  
optimal rigid shift +4 cm<sup>-1</sup> (cosine gain +0.014)    replicate ceiling: Raman 0.68 / Ag-SERS 0.52

| Peak correspondence — strongest matches (3 Raman-only, 41 SERS-only) |         |     |         |        |                       |
|----------------------------------------------------------------------|---------|-----|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | Δν  | I ratio | conf.  | note                  |
| 866                                                                  | 866     | +0  | 0.73    | High   | Strong correspondence |
| 1408                                                                 | 1404    | -4  | 0.47    | High   | Intensity reduced     |
| 920                                                                  | 916     | -4  | 0.99    | High   | Strong correspondence |
| 1352                                                                 | 1358    | +6  | 0.87    | Medium | Shifted band          |
| 942                                                                  | 938     | -4  | 0.53    | High   | Intensity reduced     |
| 1312                                                                 | 1314    | +2  | 1.75    | High   | Intensity enhanced    |
| 804                                                                  | 800     | -4  | 3.95    | High   | Intensity enhanced    |
| 708                                                                  | 716     | +8  | 3.10    | Medium | Intensity enhanced    |
| 1090                                                                 | 1102    | +12 | 4.15    | Low    | Intensity enhanced    |

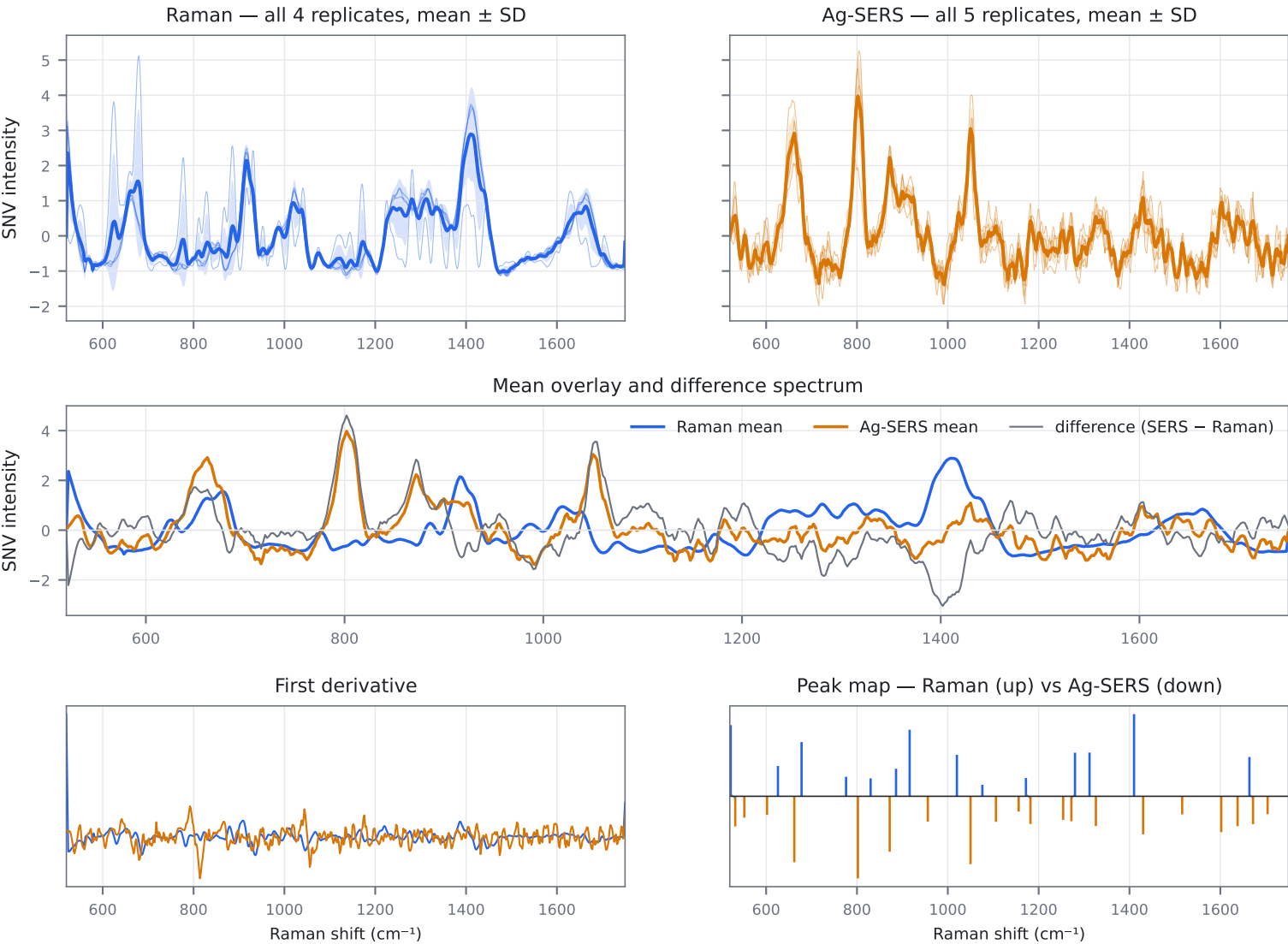
**Spectroscopic interpretation**  
Of 15 prominent Raman bands, 12 (80%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.35), while the full-profile cosine is only +0.14 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 866→866 cm<sup>-1</sup> (+0); 1408→1404 cm<sup>-1</sup> (-4); 920→916 cm<sup>-1</sup> (-4). Modest systematic displacement (mean |Δν| 5.2 cm<sup>-1</sup>, median -1.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank ρ -0.34, redistribution index 0.37): the same modes are

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# glutathione

family: cofactor

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.262    Pearson +0.262    Spearman +0.401    spectral angle 74.8°    NRMSE 0.274  
DTW 0.276    derivative r +0.090    peak F1 0.24    Jaccard 0.13    precision 0.20    recall 0.29  
peaks: Raman 14, Ag-SERS 20, matched 4    mean | $\Delta\nu$ | 9.0 cm<sup>-1</sup>    correspondence score 0.06  
optimal rigid shift -6 cm<sup>-1</sup> (cosine gain +0.008)    replicate ceiling: Raman 0.60 / Ag-SERS 0.84

| Peak correspondence — strongest matches (10 Raman-only, 16 SERS-only) |         |             |         |        |                    |
|-----------------------------------------------------------------------|---------|-------------|---------|--------|--------------------|
| Raman                                                                 | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note               |
| 522                                                                   | 532     | +10         | 0.42    | Low    | Intensity reduced  |
| 1664                                                                  | 1672    | +8          | 0.71    | Medium | Shifted band       |
| 1172                                                                  | 1182    | +10         | 1.52    | Low    | Intensity enhanced |
| 1280                                                                  | 1272    | -8          | 0.57    | Medium | Intensity reduced  |

### Spectroscopic interpretation

Of 14 prominent Raman bands, 4 (29%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.24), while the full-profile cosine is only +0.26 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 522→532 cm<sup>-1</sup> (+10); 1664→1672 cm<sup>-1</sup> (+8); 1172→1182 cm<sup>-1</sup> (+10). Larger displacements (mean | $\Delta\nu$ | 9.0 cm<sup>-1</sup>, max 10.0 cm<sup>-1</sup>) suggest strong surface interaction or ambiguous band assignment in a congested region. Intensity transfer is partial (peak-rank  $\rho$  -0.80, redistribution 0.17), with enhancement favouring a subset of modes. Notes: a band in the 1380–1430

# glycerol

family: polyol

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.348    Pearson +0.348    Spearman +0.302    spectral angle 69.7°    NRMSE 0.222  
DTW 0.314    derivative r +0.110    peak F1 0.31    Jaccard 0.19    precision 0.19    recall 1.00  
peaks: Raman 8, Ag-SERS 43, matched 8    mean | $\Delta\nu$ | 5.0 cm<sup>-1</sup>    correspondence score 0.18  
optimal rigid shift +0 cm<sup>-1</sup> (cosine gain +0.000)    replicate ceiling: Raman 0.99 / Ag-SERS 0.58

| Peak correspondence — strongest matches (0 Raman-only, 35 SERS-only) |         |             |         |        |                       |
|----------------------------------------------------------------------|---------|-------------|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note                  |
| 850                                                                  | 846     | -4          | 0.62    | High   | Intensity reduced     |
| 1058                                                                 | 1060    | +2          | 1.08    | High   | Strong correspondence |
| 1468                                                                 | 1464    | -4          | 0.86    | High   | Strong correspondence |
| 926                                                                  | 932     | +6          | 1.69    | Medium | Intensity enhanced    |
| 1250                                                                 | 1256    | +6          | 1.69    | Medium | Intensity enhanced    |
| 674                                                                  | 668     | -6          | 2.83    | Medium | Intensity enhanced    |
| 822                                                                  | 828     | +6          | 0.85    | Medium | Shifted band          |
| 976                                                                  | 982     | +6          | 2.57    | Medium | Intensity enhanced    |

## Spectroscopic interpretation

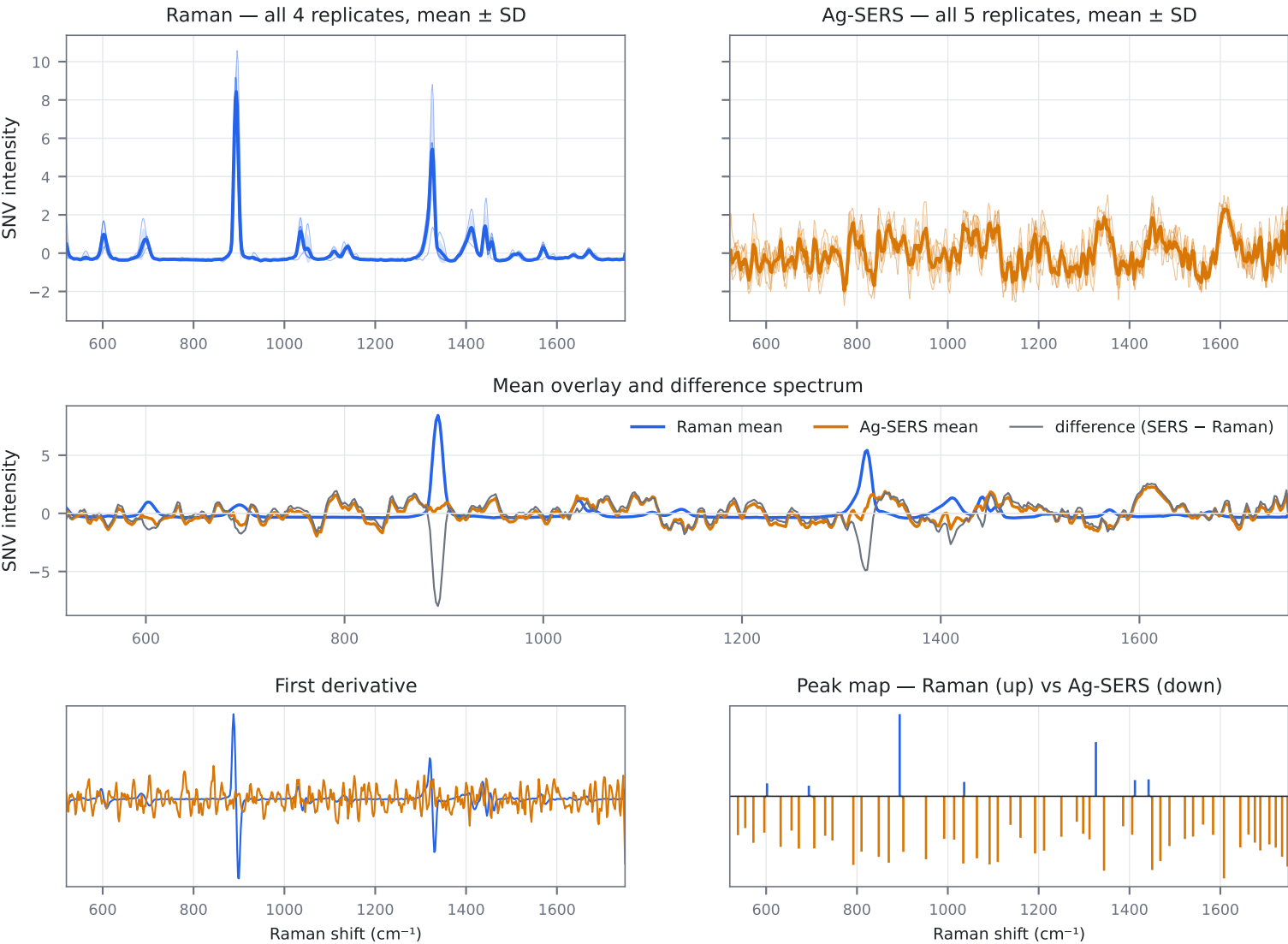
Of 8 prominent Raman bands, 8 (100%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.31), while the full-profile cosine is only +0.35 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 850→846 cm<sup>-1</sup> (-4); 1058→1060 cm<sup>-1</sup> (+2); 1468→1464 cm<sup>-1</sup> (-4). Modest systematic displacement (mean | $\Delta\nu$ | 5.0 cm<sup>-1</sup>, median 4.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  -0.31, redistribution index 0.20): the same modes are observed but re-weighted

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# glycine

family: amino\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.096    Pearson +0.096    Spearman +0.110    spectral angle 84.5°    NRMSE 0.132  
DTW 0.382    derivative r -0.038    peak F1 0.21    Jaccard 0.12    precision 0.12    recall 0.86  
peaks: Raman 7, Ag-SERS 49, matched 6    mean  $|\Delta\nu|$  7.0 cm<sup>-1</sup>    correspondence score 0.09  
optimal rigid shift -10 cm<sup>-1</sup> (cosine gain +0.101)    replicate ceiling: Raman 0.81 / Ag-SERS 0.56

Peak correspondence — strongest matches (1 Raman-only, 43 SERS-only)

| Raman | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note               |
|-------|---------|-------------|---------|--------|--------------------|
| 894   | 902     | +8          | 0.68    | Medium | Intensity reduced  |
| 1442  | 1450    | +8          | 4.35    | Medium | Intensity enhanced |
| 1036  | 1034    | -2          | 4.68    | High   | Intensity enhanced |
| 1412  | 1406    | -6          | 2.38    | Medium | Intensity enhanced |
| 602   | 596     | -6          | 2.82    | Medium | Intensity enhanced |
| 694   | 706     | +12         | 4.95    | Low    | Intensity enhanced |

### Spectroscopic interpretation

Of 7 prominent Raman bands, 6 (86%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.21), while the full-profile cosine is only +0.10 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 894→902 cm<sup>-1</sup> (+8); 1442→1450 cm<sup>-1</sup> (+8); 1036→1034 cm<sup>-1</sup> (-2). Modest systematic displacement (mean  $|\Delta\nu|$  7.0 cm<sup>-1</sup>, median 3.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  -0.20, redistribution index 0.36): the same modes are observed but re-weighted



# glycogen

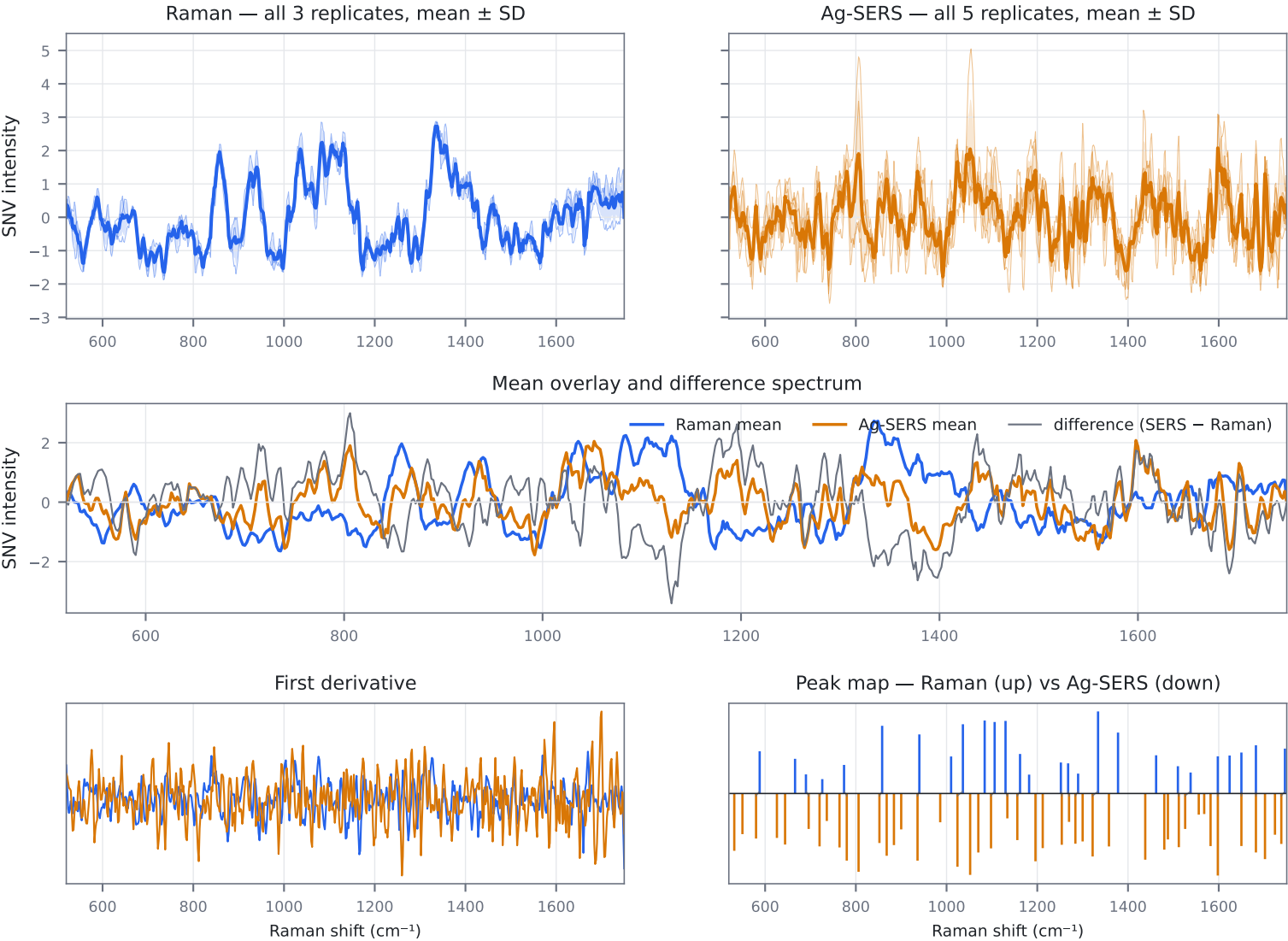
family: polysaccharide

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites

excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid

preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>

replicates: Raman n=3, Ag-SERS n=5



cosine +0.232    Pearson +0.232    Spearman +0.243    spectral angle 76.6°    NRMSE 0.244  
DTW 0.277    derivative r +0.011    peak F1 0.56    Jaccard 0.38    precision 0.44    recall 0.74  
peaks: Raman 27, Ag-SERS 45, matched 20    mean | $\Delta v$ | 5.5 cm<sup>-1</sup>    correspondence score 0.30  
optimal rigid shift -6 cm<sup>-1</sup> (cosine gain +0.011)    replicate ceiling: Raman 0.87 / Ag-SERS 0.46

| Peak correspondence — strongest matches (7 Raman-only, 25 SERS-only) |         |            |         |        |                       |
|----------------------------------------------------------------------|---------|------------|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | $\Delta v$ | I ratio | conf.  | note                  |
| 1334                                                                 | 1322    | -12        | 0.77    | Low    | Shifted band          |
| 858                                                                  | 852     | -6         | 0.73    | Medium | Shifted band          |
| 940                                                                  | 936     | -4         | 1.14    | High   | Strong correspondence |
| 588                                                                  | 580     | -8         | 1.07    | Medium | Shifted band          |
| 1036                                                                 | 1024    | -12        | 1.06    | Low    | Shifted band          |
| 1650                                                                 | 1650    | +0         | 1.28    | High   | Strong correspondence |
| 1252                                                                 | 1254    | +2         | 1.66    | High   | Intensity enhanced    |
| 774                                                                  | 780     | +6         | 2.35    | Medium | Intensity enhanced    |
| 1130                                                                 | 1134    | +4         | 0.34    | High   | Intensity reduced     |

## Spectroscopic interpretation

Of 27 prominent Raman bands, 20 (74%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.56), while the full-profile cosine is only +0.23 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 1334→1322 cm<sup>-1</sup> (-12); 858→852 cm<sup>-1</sup> (-6); 940→936 cm<sup>-1</sup> (-4). Modest systematic displacement (mean | $\Delta v$ | 5.5 cm<sup>-1</sup>, median -5.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  0.23, redistribution index 0.19): the same modes are

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# guanine

family: purine

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.297    Pearson +0.297    Spearman +0.097    spectral angle 72.7°    NRMSE 0.118  
DTW 0.260    derivative r +0.033    peak F1 0.21    Jaccard 0.12    precision 0.18    recall 0.25  
peaks: Raman 8, Ag-SERS 11, matched 2    mean  $|\Delta v|$  9.0 cm<sup>-1</sup>    correspondence score 0.05  
optimal rigid shift +8 cm<sup>-1</sup> (cosine gain +0.227)    replicate ceiling: Raman 0.98 / Ag-SERS 0.88

Peak correspondence — strongest matches (6 Raman-only, 9 SERS-only)

| Raman | Ag-SERS | $\Delta v$ | I ratio | conf.  | note              |
|-------|---------|------------|---------|--------|-------------------|
| 650   | 642     | -8         | 1.00    | Medium | Shifted band      |
| 1264  | 1254    | -10        | 0.50    | Low    | Intensity reduced |

### Spectroscopic interpretation

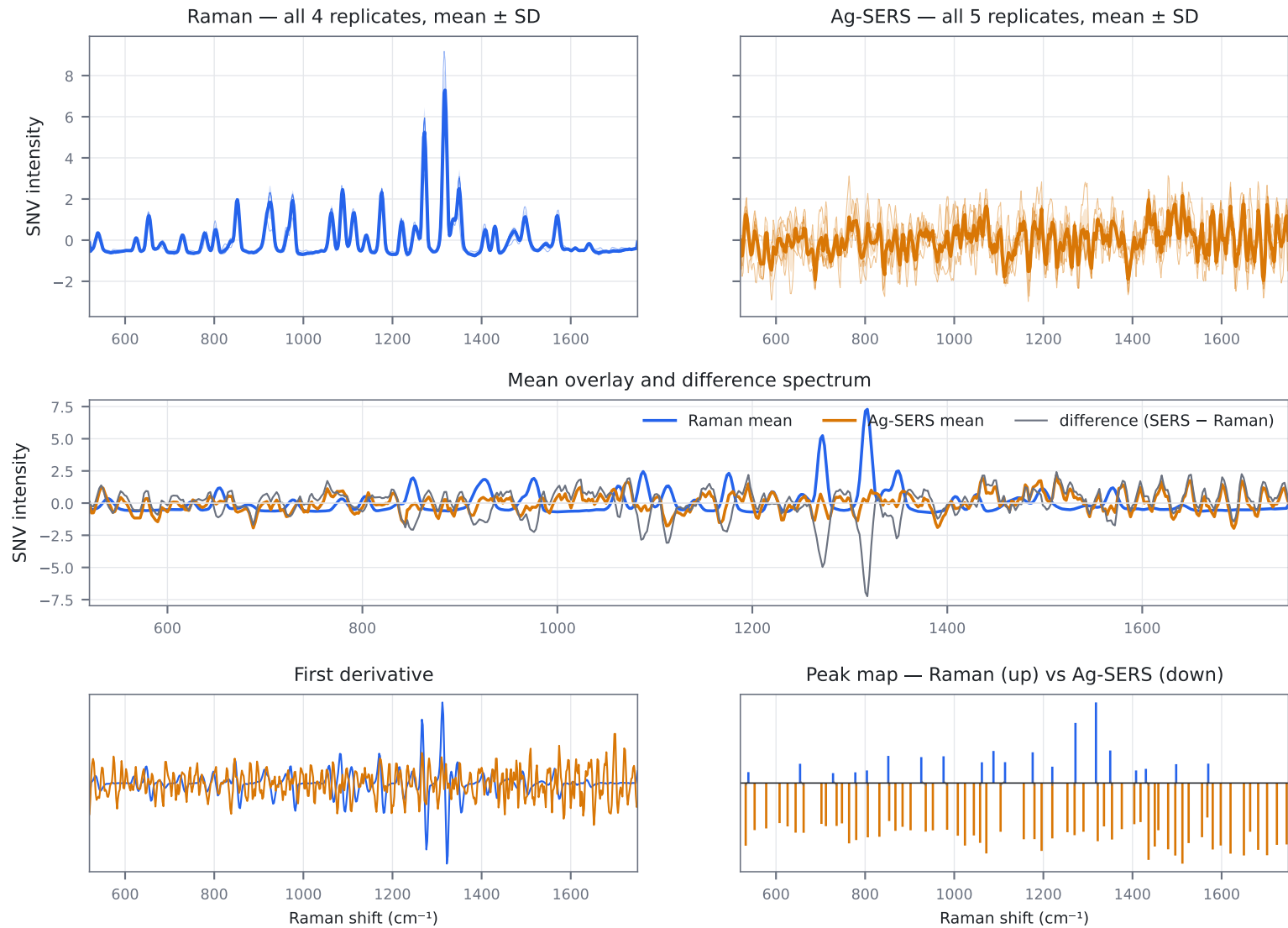
Of 8 prominent Raman bands, 2 (25%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.21), while the full-profile cosine is only +0.30 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 650 cm<sup>-1</sup> (purine ring)  $\rightarrow$  642 cm<sup>-1</sup> (-8). Larger displacements (mean  $|\Delta v|$  9.0 cm<sup>-1</sup>, max 10.0 cm<sup>-1</sup>) suggest strong surface interaction or ambiguous band assignment in a congested region. Notes: a band in the 1380–1430 cm<sup>-1</sup> carboxylate symmetric-stretch window is present in Ag-SERS, the expected signature of COO<sup>-</sup> anchoring to the colloid; 9/11 Ag-SERS bands have no Raman



# histidine

family: amino\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.045    Pearson +0.045    Spearman +0.071    spectral angle 87.4°    NRMSE 0.147  
DTW 0.335    derivative r +0.072    peak F1 0.48    Jaccard 0.32    precision 0.32    recall 0.95  
peaks: Raman 20, Ag-SERS 59, matched 19    mean | $\Delta\nu$ | 4.6 cm<sup>-1</sup>    correspondence score 0.30  
optimal rigid shift -10 cm<sup>-1</sup> (cosine gain +0.105)    replicate ceiling: Raman 0.93 / Ag-SERS 0.39

Peak correspondence — strongest matches (1 Raman-only, 40 SERS-only)

| Raman | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note                  |
|-------|---------|-------------|---------|--------|-----------------------|
| 1318  | 1322    | +4          | 0.72    | High   | Strong correspondence |
| 1272  | 1270    | -2          | 0.87    | High   | Strong correspondence |
| 1176  | 1180    | +4          | 1.82    | High   | Intensity enhanced    |
| 852   | 854     | +2          | 1.40    | High   | Strong correspondence |
| 976   | 984     | +8          | 1.75    | Medium | Intensity enhanced    |
| 926   | 936     | +10         | 1.88    | Low    | Intensity enhanced    |
| 1350  | 1354    | +4          | 1.73    | High   | Intensity enhanced    |
| 1570  | 1568    | -2          | 1.78    | High   | Intensity enhanced    |
| 654   | 662     | +8          | 2.58    | Medium | Intensity enhanced    |

### Spectroscopic interpretation

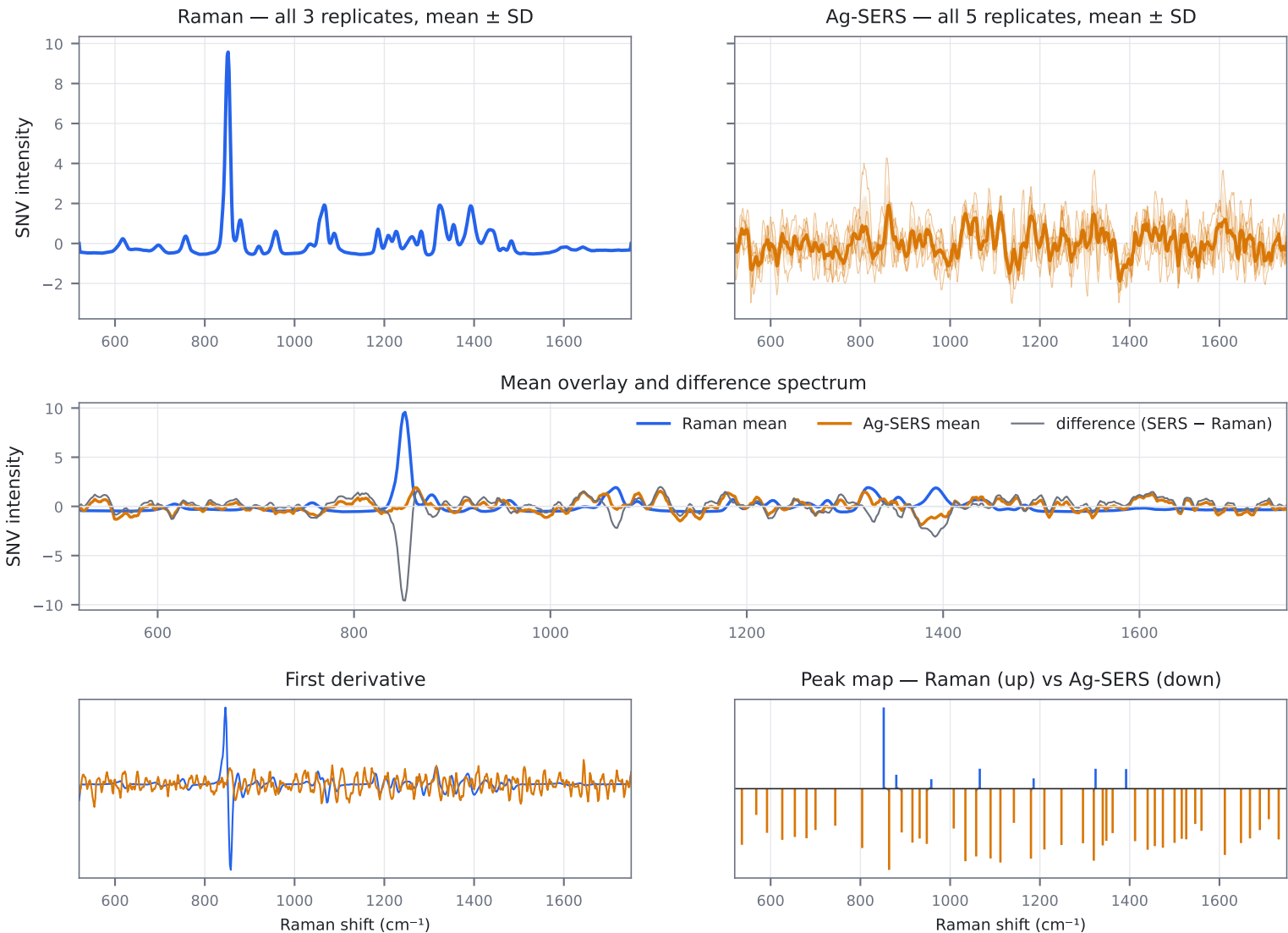
Of 20 prominent Raman bands, 19 (95%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.48), while the full-profile cosine is only +0.05 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 1318→1322 cm<sup>-1</sup> (+4); 1272→1270 cm<sup>-1</sup> (-2); 1176→1180 cm<sup>-1</sup> (+4). Modest systematic displacement (mean | $\Delta\nu$ | 4.6 cm<sup>-1</sup>, median 2.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  0.12, redistribution index 0.26): the same modes are observed but re-weighted

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# hydroxyproline

family: amino\_acid

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.101    Pearson +0.101    Spearman +0.218    spectral angle 84.2°    NRMSE 0.111  
DTW 0.262    derivative r -0.065    peak F1 0.24    Jaccard 0.14    precision 0.14    recall 0.86  
peaks: Raman 7, Ag-SERS 43, matched 6    mean | $\Delta\nu$ | 8.7 cm<sup>-1</sup>    correspondence score 0.07  
optimal rigid shift -10 cm<sup>-1</sup> (cosine gain +0.138)    replicate ceiling: Raman 1.00 / Ag-SERS 0.25

| Peak correspondence — strongest matches (1 Raman-only, 37 SERS-only) |         |             |         |        |                    |
|----------------------------------------------------------------------|---------|-------------|---------|--------|--------------------|
| Raman                                                                | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note               |
| 852                                                                  | 864     | +12         | 1.00    | Low    | Shifted band       |
| 1066                                                                 | 1058    | -8          | 3.39    | Medium | Intensity enhanced |
| 1324                                                                 | 1320    | -4          | 3.61    | High   | Intensity enhanced |
| 1186                                                                 | 1180    | -6          | 6.79    | Medium | Intensity enhanced |
| 958                                                                  | 948     | -10         | 5.82    | Low    | Intensity enhanced |
| 880                                                                  | 892     | +12         | 3.14    | Low    | Intensity enhanced |

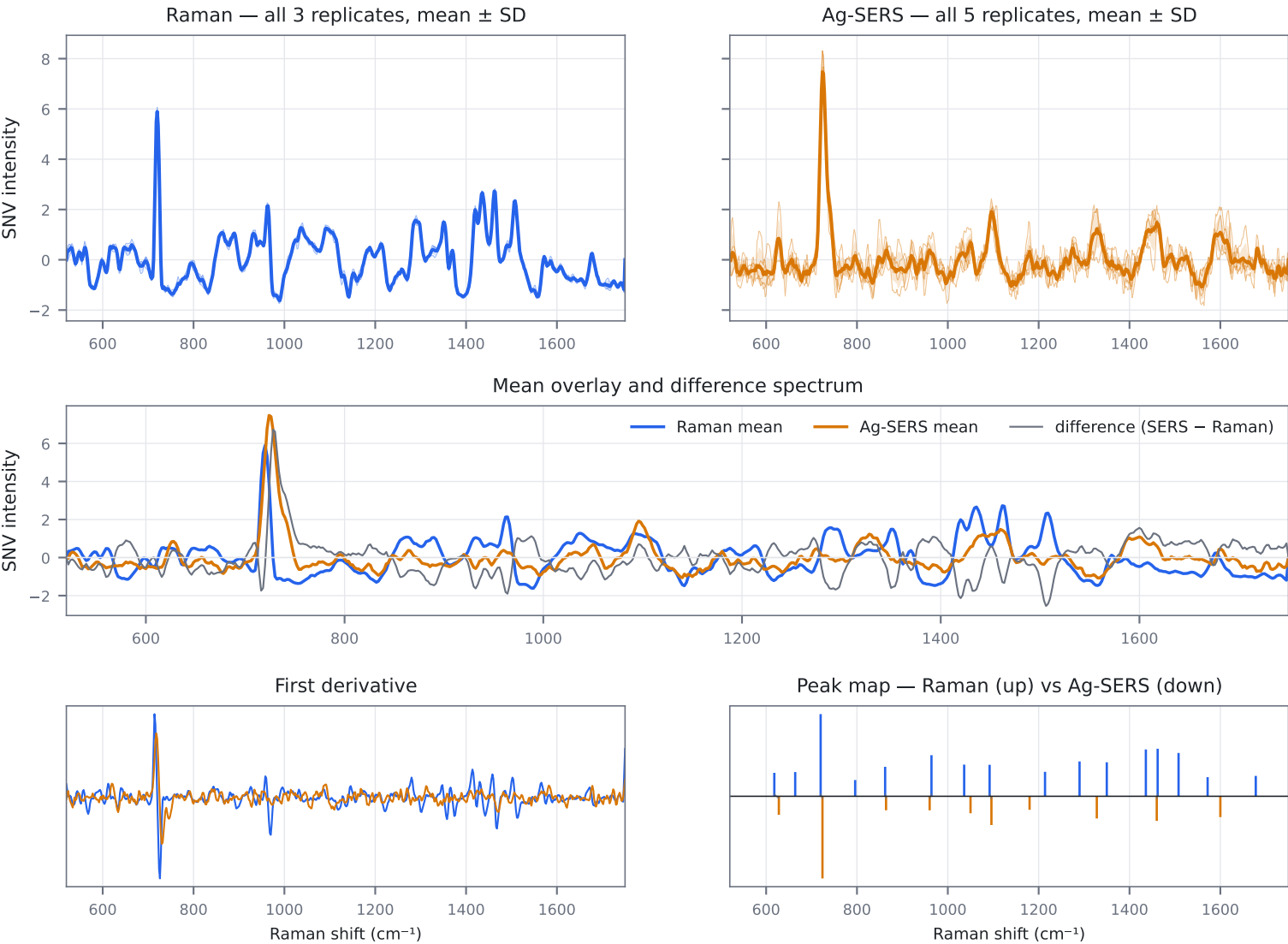
**Spectroscopic interpretation**

Of 7 prominent Raman bands, 6 (86%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.24), while the full-profile cosine is only +0.10 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 852→864 cm<sup>-1</sup> (+12); 1066→1058 cm<sup>-1</sup> (-8); 1324→1320 cm<sup>-1</sup> (-4). Larger displacements (mean | $\Delta\nu$ | 8.7 cm<sup>-1</sup>, max 12.0 cm<sup>-1</sup>) suggest strong surface interaction or ambiguous band assignment in a congested region. Relative band intensities are comparatively well preserved (peak-rank  $\rho$  0.66, redistribution 0.32), an unusually faithful transfer for a colloidal SERS

# hypoxanthine

family: purine

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.486    Pearson +0.486    Spearman +0.539    spectral angle 60.9°    NRMSE 0.129  
DTW 0.236    derivative r +0.371    peak F1 0.46    Jaccard 0.30    precision 0.60    recall 0.38  
peaks: Raman 16, Ag-SERS 10, matched 6    mean  $|\Delta\nu|$  4.3 cm<sup>-1</sup>    correspondence score 0.29  
optimal rigid shift -4 cm<sup>-1</sup> (cosine gain +0.095)    replicate ceiling: Raman 0.98 / Ag-SERS 0.79

| Peak correspondence — strongest matches (10 Raman-only, 4 SERS-only) |         |             |         |       |                       |
|----------------------------------------------------------------------|---------|-------------|---------|-------|-----------------------|
| Raman                                                                | Ag-SERS | $\Delta\nu$ | I ratio | conf. | note                  |
| 720                                                                  | 724     | +4          | 1.00    | High  | Strong correspondence |
| 1462                                                                 | 1460    | -2          | 0.52    | High  | Intensity reduced     |
| 964                                                                  | 960     | -4          | 0.34    | High  | Intensity reduced     |
| 862                                                                  | 864     | +2          | 0.47    | High  | Intensity reduced     |
| 1092                                                                 | 1096    | +4          | 0.91    | High  | Strong correspondence |
| 618                                                                  | 628     | +10         | 0.79    | Low   | Shifted band          |

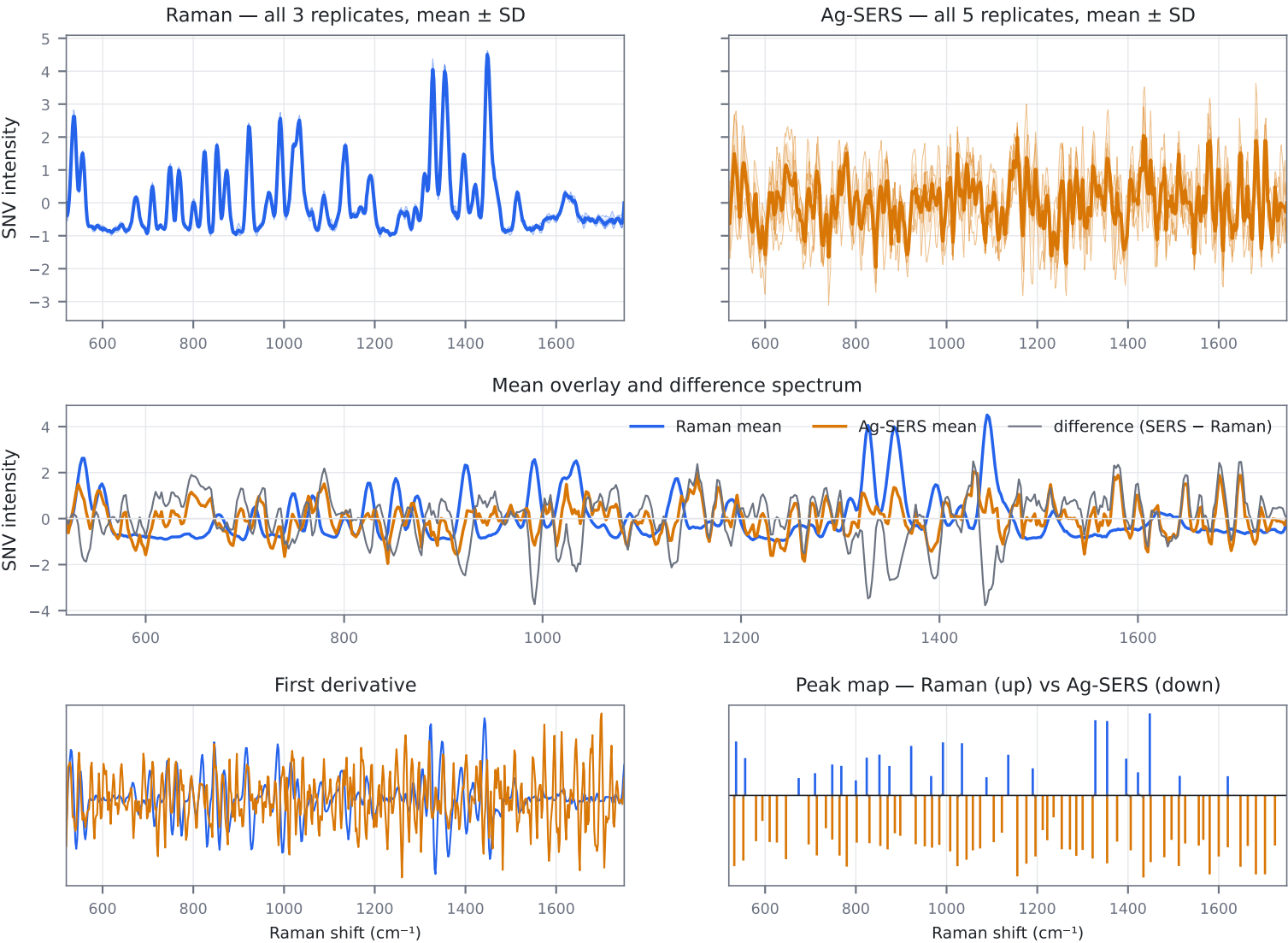
## Spectroscopic interpretation

Of 16 prominent Raman bands, 6 (38%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.46), while the full-profile cosine is only +0.49 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 725 cm<sup>-1</sup> (purine ring breathing) → 724 cm<sup>-1</sup> (+4). Modest systematic displacement (mean  $|\Delta\nu|$  4.3 cm<sup>-1</sup>, median 3.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Intensity transfer is partial (peak-rank  $\rho$  0.37, redistribution 0.17), with enhancement favouring a subset of modes. Notes: 10/16 Raman

# isoleucine

family: amino\_acid

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.210    Pearson +0.210    Spearman +0.234    spectral angle 77.9°    NRMSE 0.200  
DTW 0.328    derivative r +0.078    peak F1 0.50    Jaccard 0.33    precision 0.35    recall 0.88  
peaks: Raman 24, Ag-SERS 60, matched 21    mean | $\Delta\nu$ | 4.7 cm<sup>-1</sup>    correspondence score 0.31  
optimal rigid shift -4 cm<sup>-1</sup> (cosine gain +0.020)    replicate ceiling: Raman 0.99 / Ag-SERS 0.40

| Peak correspondence — strongest matches (3 Raman-only, 39 SERS-only) |         |             |         |        |                       |
|----------------------------------------------------------------------|---------|-------------|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note                  |
| 1448                                                                 | 1450    | +2          | 0.81    | High   | Strong correspondence |
| 1328                                                                 | 1322    | -6          | 0.84    | Medium | Shifted band          |
| 1354                                                                 | 1354    | +0          | 0.91    | High   | Strong correspondence |
| 992                                                                  | 984     | -8          | 0.93    | Medium | Shifted band          |
| 922                                                                  | 932     | +10         | 0.99    | Low    | Shifted band          |
| 536                                                                  | 532     | -4          | 1.31    | High   | Strong correspondence |
| 1034                                                                 | 1024    | -10         | 1.35    | Low    | Shifted band          |
| 852                                                                  | 854     | +2          | 1.24    | High   | Strong correspondence |
| 824                                                                  | 830     | +6          | 1.26    | Medium | Shifted band          |

**Spectroscopic interpretation**

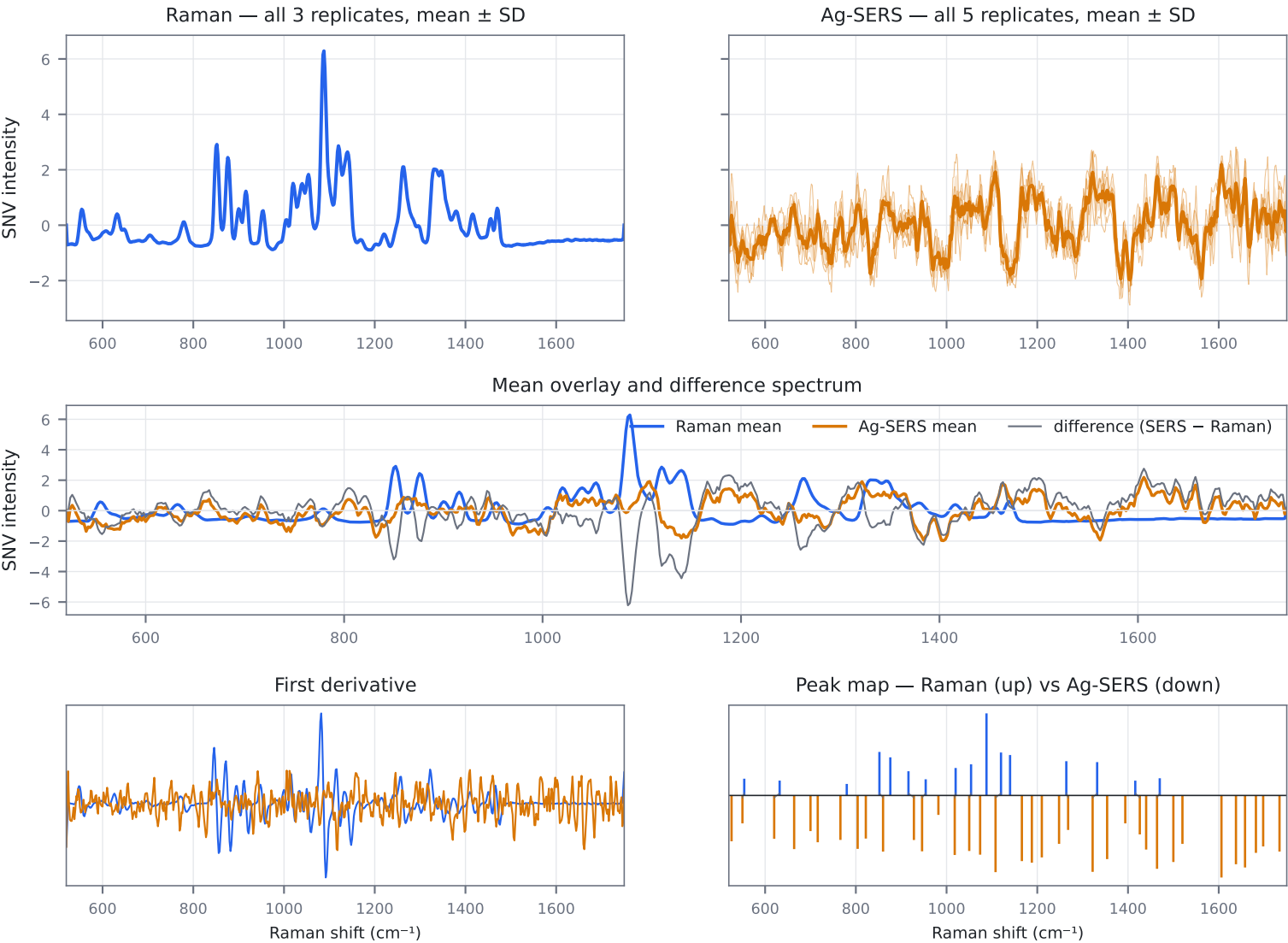
Of 24 prominent Raman bands, 21 (88%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.50), while the full-profile cosine is only +0.21 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 1448→1450 cm<sup>-1</sup> (+2); 1328→1322 cm<sup>-1</sup> (-6); 1354→1354 cm<sup>-1</sup> (+0). Modest systematic displacement (mean | $\Delta\nu$ | 4.7 cm<sup>-1</sup>, median 0.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  -0.27, redistribution index 0.20): the same modes are observed but re-weighted

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# lactate

family: organic\_acid

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.054    Pearson +0.054    Spearman +0.041    spectral angle 86.9°    NRMSE 0.176  
DTW 0.391    derivative r -0.098    peak F1 0.46    Jaccard 0.30    precision 0.33    recall 0.75  
peaks: Raman 16, Ag-SERS 36, matched 12    mean | $\Delta\nu$ | 7.7 cm<sup>-1</sup>    correspondence score 0.17  
optimal rigid shift +10 cm<sup>-1</sup> (cosine gain +0.069)    replicate ceiling: Raman 1.00 / Ag-SERS 0.61

| Peak correspondence — strongest matches (4 Raman-only, 24 SERS-only) |         |             |         |        |                       |
|----------------------------------------------------------------------|---------|-------------|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note                  |
| 852                                                                  | 860     | +8          | 1.29    | Medium | Shifted band          |
| 1264                                                                 | 1268    | +4          | 1.01    | High   | Strong correspondence |
| 1332                                                                 | 1322    | -10         | 2.29    | Low    | Intensity enhanced    |
| 1120                                                                 | 1108    | -12         | 1.79    | Low    | Intensity enhanced    |
| 1054                                                                 | 1050    | -4          | 1.79    | High   | Intensity enhanced    |
| 916                                                                  | 928     | +12         | 1.84    | Low    | Intensity enhanced    |
| 554                                                                  | 550     | -4          | 1.67    | High   | Intensity enhanced    |
| 954                                                                  | 946     | -8          | 3.46    | Medium | Intensity enhanced    |
| 1470                                                                 | 1464    | -6          | 4.27    | Medium | Intensity enhanced    |

## Spectroscopic interpretation

Of 16 prominent Raman bands, 12 (75%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.46), while the full-profile cosine is only +0.05 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 852→860 cm<sup>-1</sup> (+8); 1264→1268 cm<sup>-1</sup> (+4); 1332→1322 cm<sup>-1</sup> (-10). Modest systematic displacement (mean | $\Delta\nu$ | 7.7 cm<sup>-1</sup>, median -4.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  0.11, redistribution index 0.15): the same modes are

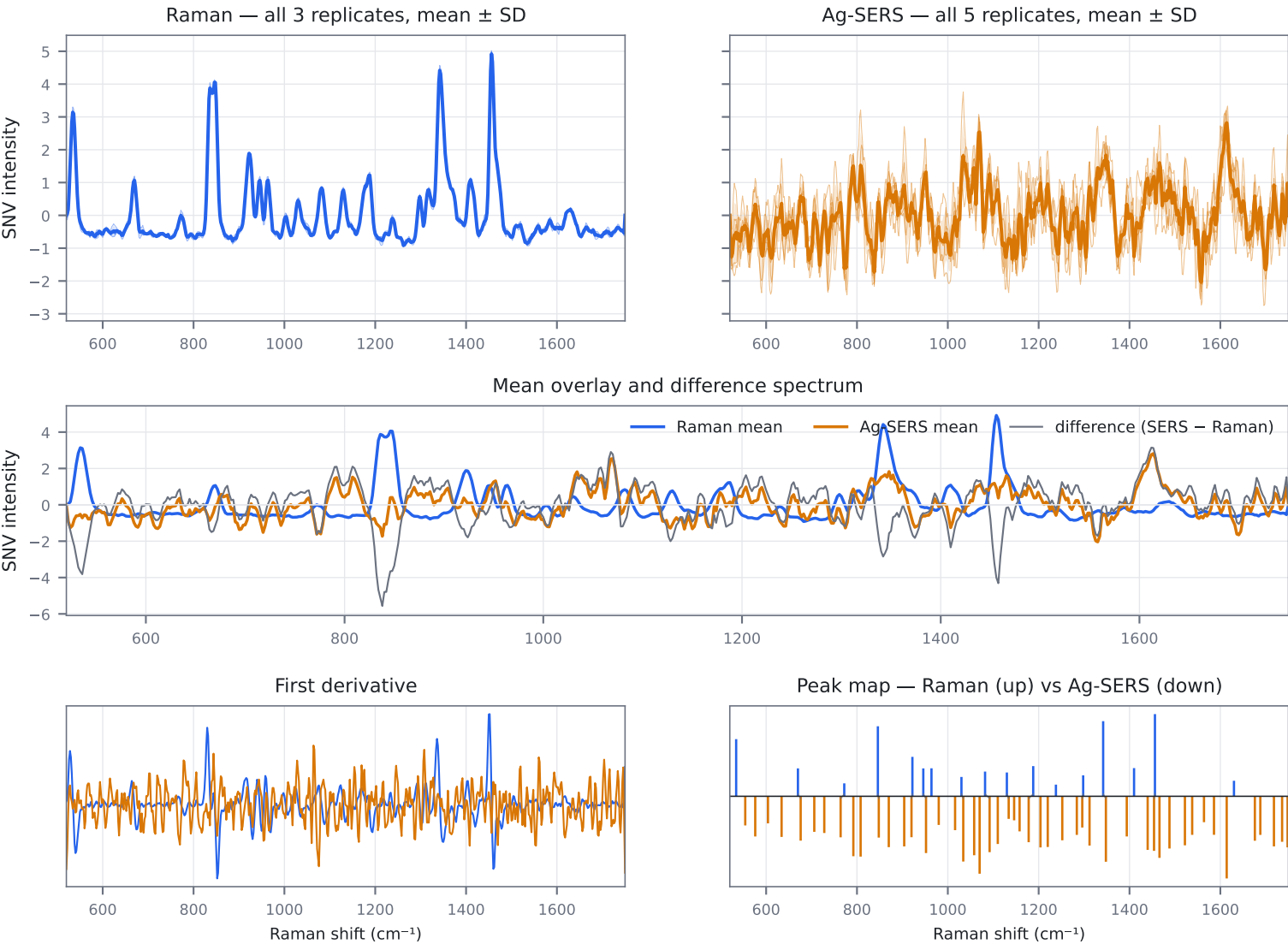
Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.



# leucine

family: amino\_acid

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.119    Pearson +0.119    Spearman +0.034    spectral angle 83.2°    NRMSE 0.206  
DTW 0.323    derivative r -0.011    peak F1 0.37    Jaccard 0.23    precision 0.25    recall 0.71  
peaks: Raman 17, Ag-SERS 48, matched 12    mean | $\Delta v$ | 5.2 cm<sup>-1</sup>    correspondence score 0.21  
optimal rigid shift -10 cm<sup>-1</sup> (cosine gain +0.046)    replicate ceiling: Raman 0.99 / Ag-SERS 0.56

| Peak correspondence — strongest matches (5 Raman-only, 36 SERS-only) |         |            |         |        |                       |
|----------------------------------------------------------------------|---------|------------|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | $\Delta v$ | I ratio | conf.  | note                  |
| 1456                                                                 | 1454    | -2         | 0.66    | High   | Intensity reduced     |
| 1342                                                                 | 1348    | +6         | 0.87    | Medium | Shifted band          |
| 846                                                                  | 848     | +2         | 0.59    | High   | Intensity reduced     |
| 922                                                                  | 920     | -2         | 1.04    | High   | Strong correspondence |
| 1188                                                                 | 1178    | -10        | 1.52    | Low    | Intensity enhanced    |
| 670                                                                  | 676     | +6         | 1.59    | Medium | Intensity enhanced    |
| 1082                                                                 | 1092    | +10        | 2.26    | Low    | Intensity enhanced    |
| 1130                                                                 | 1134    | +4         | 0.95    | High   | Strong correspondence |
| 946                                                                  | 952     | +6         | 2.04    | Medium | Intensity enhanced    |

## Spectroscopic interpretation

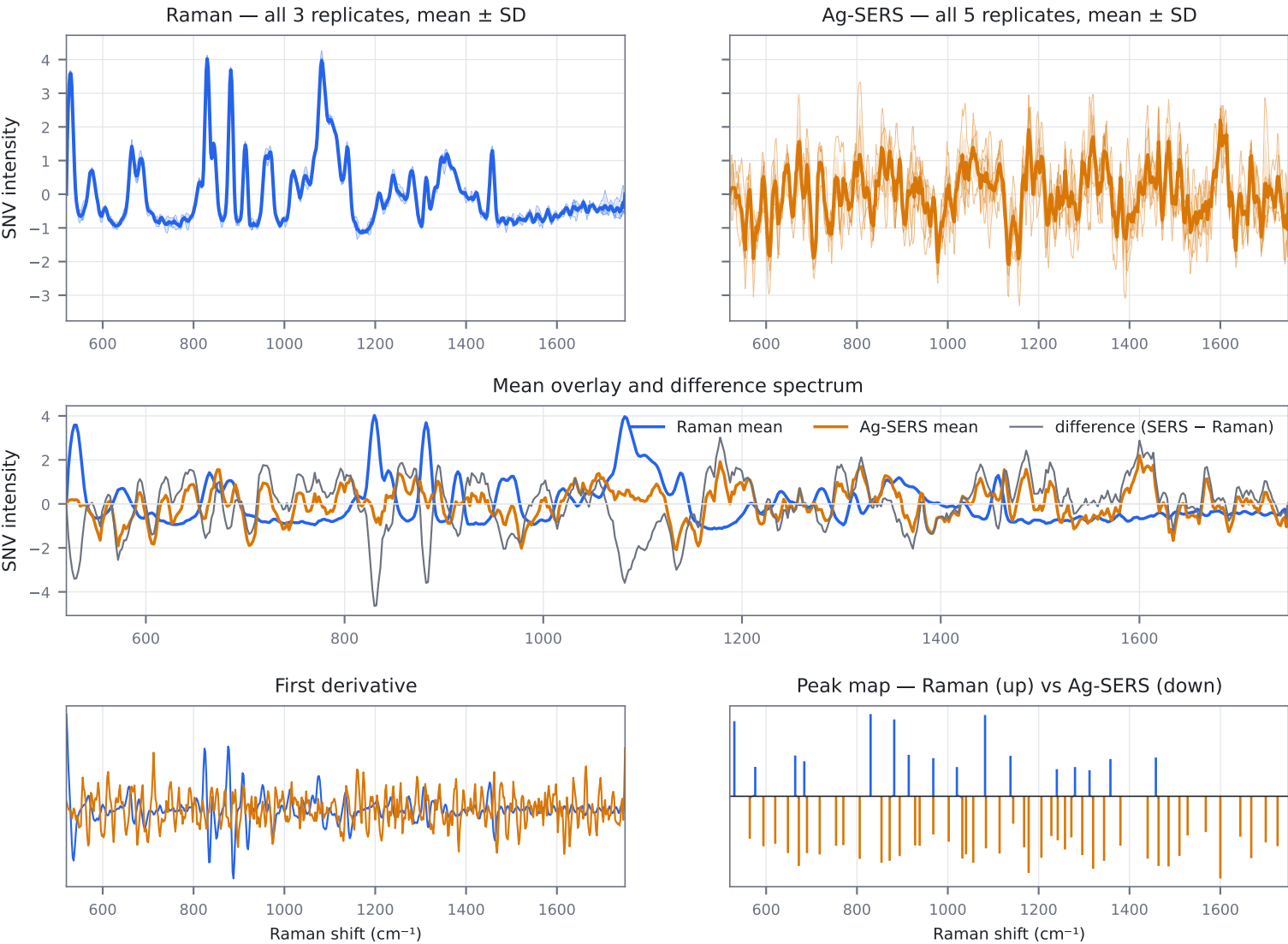
Of 17 prominent Raman bands, 12 (71%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.37), while the full-profile cosine is only +0.12 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 1456→1454 cm<sup>-1</sup> (-2); 1342→1348 cm<sup>-1</sup> (+6); 846→848 cm<sup>-1</sup> (+2). Modest systematic displacement (mean | $\Delta v$ | 5.2 cm<sup>-1</sup>, median 3.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  -0.20, redistribution index 0.24): the same modes are observed but re-weighted

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

mannose

family: saccharide

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520-1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.043    Pearson +0.043    Spearman +0.019    spectral angle 87.5°    NRMSE 0.237  
DTW 0.335    derivative r -0.042    peak F1 0.39    Jaccard 0.24    precision 0.27    recall 0.75  
peaks: Raman 16, Ag-SERS 45, matched 12    mean |Δv| 6.7 cm<sup>-1</sup>    correspondence score 0.17  
optimal rigid shift +10 cm<sup>-1</sup> (cosine gain +0.107)    replicate ceiling: Raman 0.98 / Ag-SERS 0.48

| Peak correspondence — strongest matches (4 Raman-only, 33 SERS-only) |         |     |         |        |                       |
|----------------------------------------------------------------------|---------|-----|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | Δv  | I ratio | conf.  | note                  |
| 1082                                                                 | 1084    | +2  | 0.64    | High   | Intensity reduced     |
| 882                                                                  | 872     | -10 | 0.84    | Low    | Shifted band          |
| 664                                                                  | 672     | +8  | 1.71    | Medium | Intensity enhanced    |
| 1458                                                                 | 1464    | +6  | 1.80    | Medium | Intensity enhanced    |
| 968                                                                  | 968     | +0  | 1.00    | High   | Strong correspondence |
| 1280                                                                 | 1272    | -8  | 1.41    | Medium | Intensity enhanced    |
| 576                                                                  | 564     | -12 | 1.44    | Low    | Intensity enhanced    |
| 1138                                                                 | 1144    | +6  | 0.68    | Medium | Intensity reduced     |
| 1312                                                                 | 1320    | +8  | 2.79    | Medium | Intensity enhanced    |

**Spectroscopic interpretation**  
Of 16 prominent Raman bands, 12 (75%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.39), while the full-profile cosine is only +0.04 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 1082→1084 cm<sup>-1</sup> (+2); 882→872 cm<sup>-1</sup> (-10); 664→672 cm<sup>-1</sup> (+8). Modest systematic displacement (mean |Δv| 6.7 cm<sup>-1</sup>, median 4.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank ρ 0.00, redistribution index 0.20): the same modes are observed but re-weighted

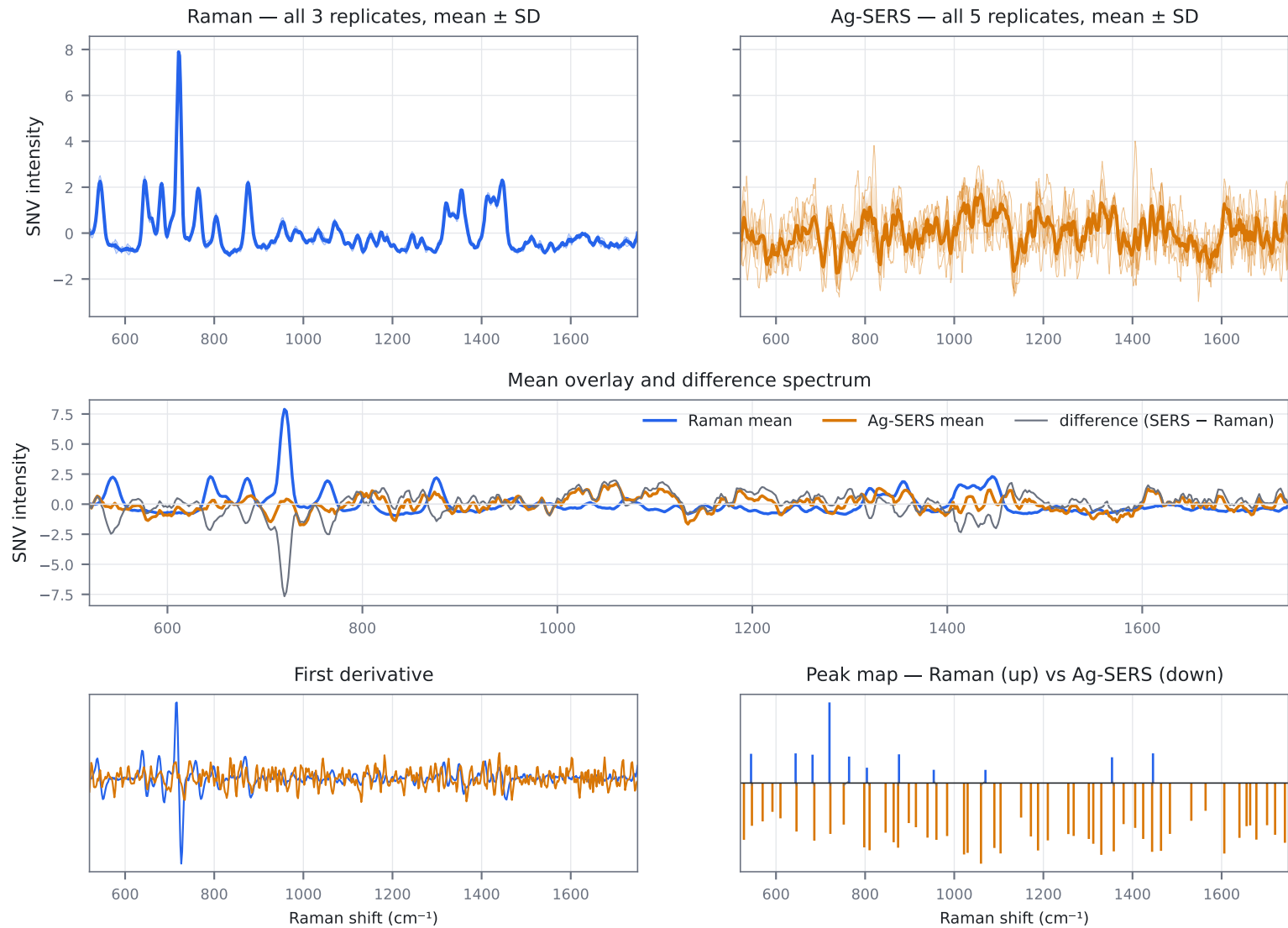
Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.



# methionine

family: amino\_acid

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.201    Pearson +0.201    Spearman +0.323    spectral angle 78.4°    NRMSE 0.121  
DTW 0.287    derivative r +0.170    peak F1 0.36    Jaccard 0.22    precision 0.22    recall 1.00  
peaks: Raman 11, Ag-SERS 50, matched 11    mean |Δν| 4.7 cm<sup>-1</sup>    correspondence score 0.22  
optimal rigid shift +0 cm<sup>-1</sup> (cosine gain +0.000)    replicate ceiling: Raman 0.99 / Ag-SERS 0.28

| Peak correspondence — strongest matches (0 Raman-only, 39 SERS-only) |         |     |         |        |                    |
|----------------------------------------------------------------------|---------|-----|---------|--------|--------------------|
| Raman                                                                | Ag-SERS | Δν  | I ratio | conf.  | note               |
| 720                                                                  | 722     | +2  | 0.63    | High   | Intensity reduced  |
| 1446                                                                 | 1444    | -2  | 2.32    | High   | Intensity enhanced |
| 876                                                                  | 874     | -2  | 2.25    | High   | Intensity enhanced |
| 1354                                                                 | 1358    | +4  | 2.65    | High   | Intensity enhanced |
| 644                                                                  | 646     | +2  | 1.64    | High   | Intensity enhanced |
| 544                                                                  | 546     | +2  | 1.46    | High   | Intensity enhanced |
| 764                                                                  | 752     | -12 | 1.57    | Low    | Intensity enhanced |
| 682                                                                  | 686     | +4  | 2.03    | High   | Intensity enhanced |
| 804                                                                  | 810     | +6  | 4.38    | Medium | Intensity enhanced |

## Spectroscopic interpretation

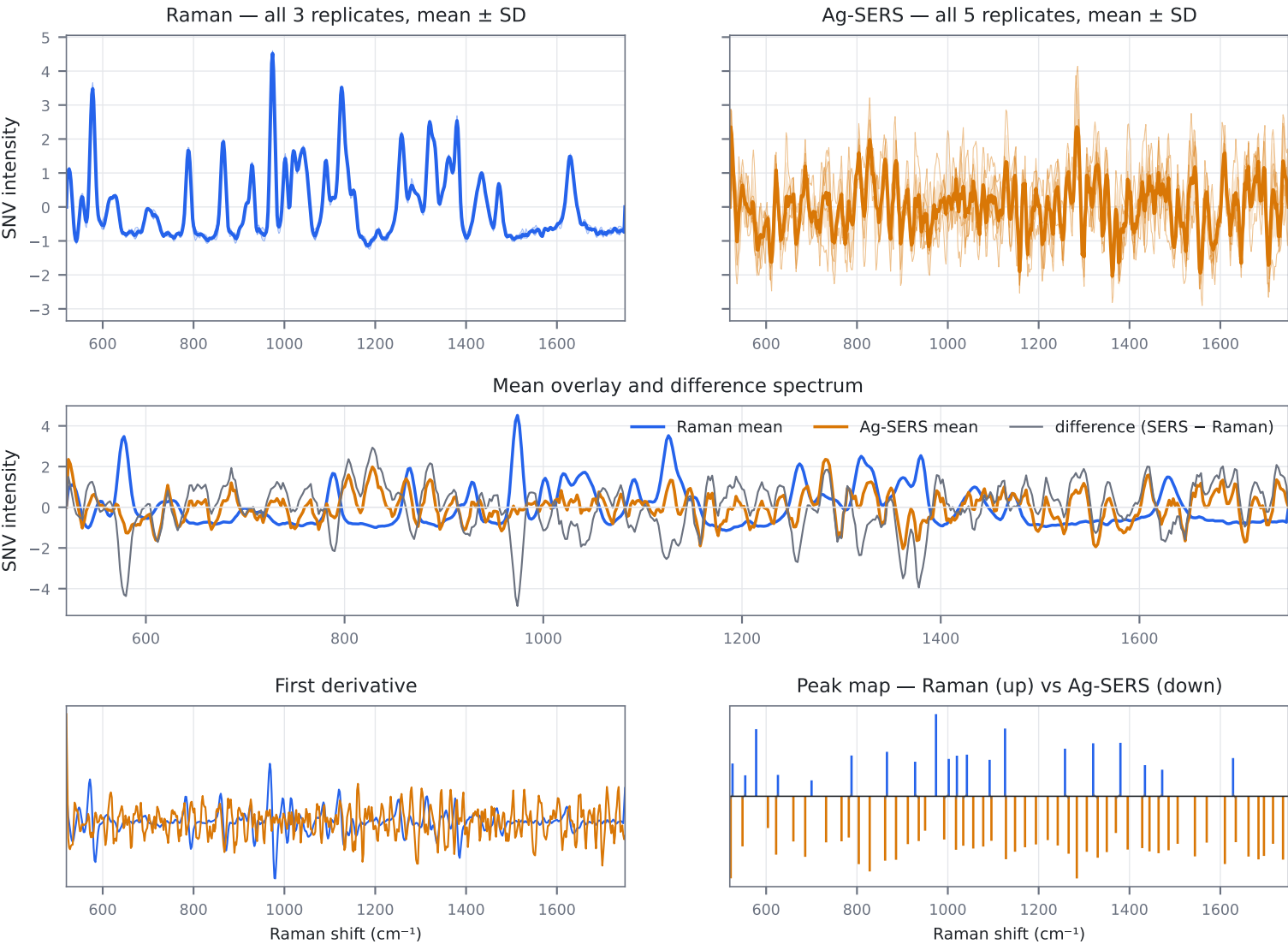
Of 11 prominent Raman bands, 11 (100%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.36), while the full-profile cosine is only +0.20 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 720→722 cm<sup>-1</sup> (+2); 1446→1444 cm<sup>-1</sup> (-2); 876→874 cm<sup>-1</sup> (-2). Modest systematic displacement (mean |Δν| 4.7 cm<sup>-1</sup>, median 2.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank ρ -0.25, redistribution index 0.23): the same modes are observed but re-weighted

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# n-acetylglucosamine

family: saccharide

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.056    Pearson +0.056    Spearman +0.067    spectral angle 86.8°    NRMSE 0.212  
DTW 0.324    derivative r +0.056    peak F1 0.49    Jaccard 0.33    precision 0.35    recall 0.85  
peaks: Raman 20, Ag-SERS 49, matched 17    mean  $|\Delta\nu|$  6.2 cm<sup>-1</sup>    correspondence score 0.24  
optimal rigid shift +4 cm<sup>-1</sup> (cosine gain +0.005)    replicate ceiling: Raman 0.99 / Ag-SERS 0.42

| Peak correspondence — strongest matches (3 Raman-only, 32 SERS-only) |         |             |         |        |                       |
|----------------------------------------------------------------------|---------|-------------|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note                  |
| 1126                                                                 | 1128    | +2          | 0.93    | High   | Strong correspondence |
| 1380                                                                 | 1370    | -10         | 0.69    | Low    | Intensity reduced     |
| 866                                                                  | 862     | -4          | 1.45    | High   | Intensity enhanced    |
| 788                                                                  | 782     | -6          | 1.01    | Medium | Shifted band          |
| 1320                                                                 | 1330    | +10         | 1.16    | Low    | Shifted band          |
| 1042                                                                 | 1034    | -8          | 1.19    | Medium | Shifted band          |
| 1628                                                                 | 1634    | +6          | 1.20    | Medium | Shifted band          |
| 1258                                                                 | 1266    | +8          | 1.21    | Medium | Shifted band          |
| 928                                                                  | 936     | +8          | 1.30    | Medium | Shifted band          |

## Spectroscopic interpretation

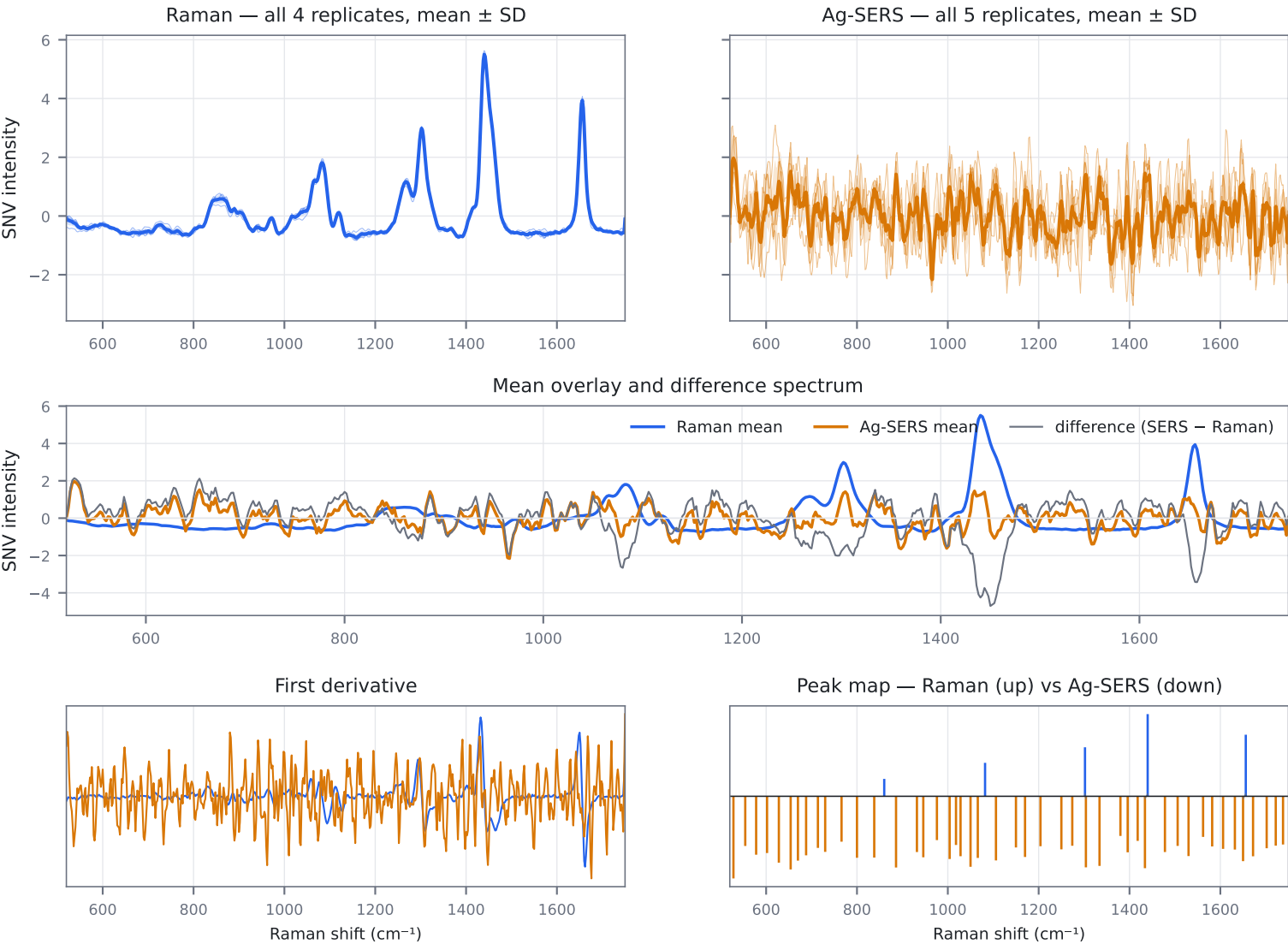
Of 20 prominent Raman bands, 17 (85%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.49), while the full-profile cosine is only +0.06 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 1126→1128 cm<sup>-1</sup> (+2); 1380→1370 cm<sup>-1</sup> (-10); 866→862 cm<sup>-1</sup> (-4). Modest systematic displacement (mean  $|\Delta\nu|$  6.2 cm<sup>-1</sup>, median -4.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank p < 0.01, redistribution index 0.14): the same modes are

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# oleate

family: lipid

Raman source: RamanBioLib[gobbato\_raman\_metabolites] | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.170    Pearson +0.170    Spearman +0.096    spectral angle 80.2°    NRMSE 0.175  
DTW 0.372    derivative r +0.140    peak F1 0.12    Jaccard 0.06    precision 0.07    recall 0.60  
peaks: Raman 5, Ag-SERS 46, matched 3    mean |Δν| 4.7 cm<sup>-1</sup>    correspondence score 0.07  
optimal rigid shift +4 cm<sup>-1</sup> (cosine gain +0.017)    replicate ceiling: Raman 0.99 / Ag-SERS 0.27

| Peak correspondence — strongest matches (2 Raman-only, 43 SERS-only) |         |    |         |        |                    |
|----------------------------------------------------------------------|---------|----|---------|--------|--------------------|
| Raman                                                                | Ag-SERS | Δν | I ratio | conf.  | note               |
| 1440                                                                 | 1434    | -6 | 0.88    | Medium | Shifted band       |
| 1656                                                                 | 1650    | -6 | 1.05    | Medium | Shifted band       |
| 1302                                                                 | 1304    | +2 | 1.45    | High   | Intensity enhanced |

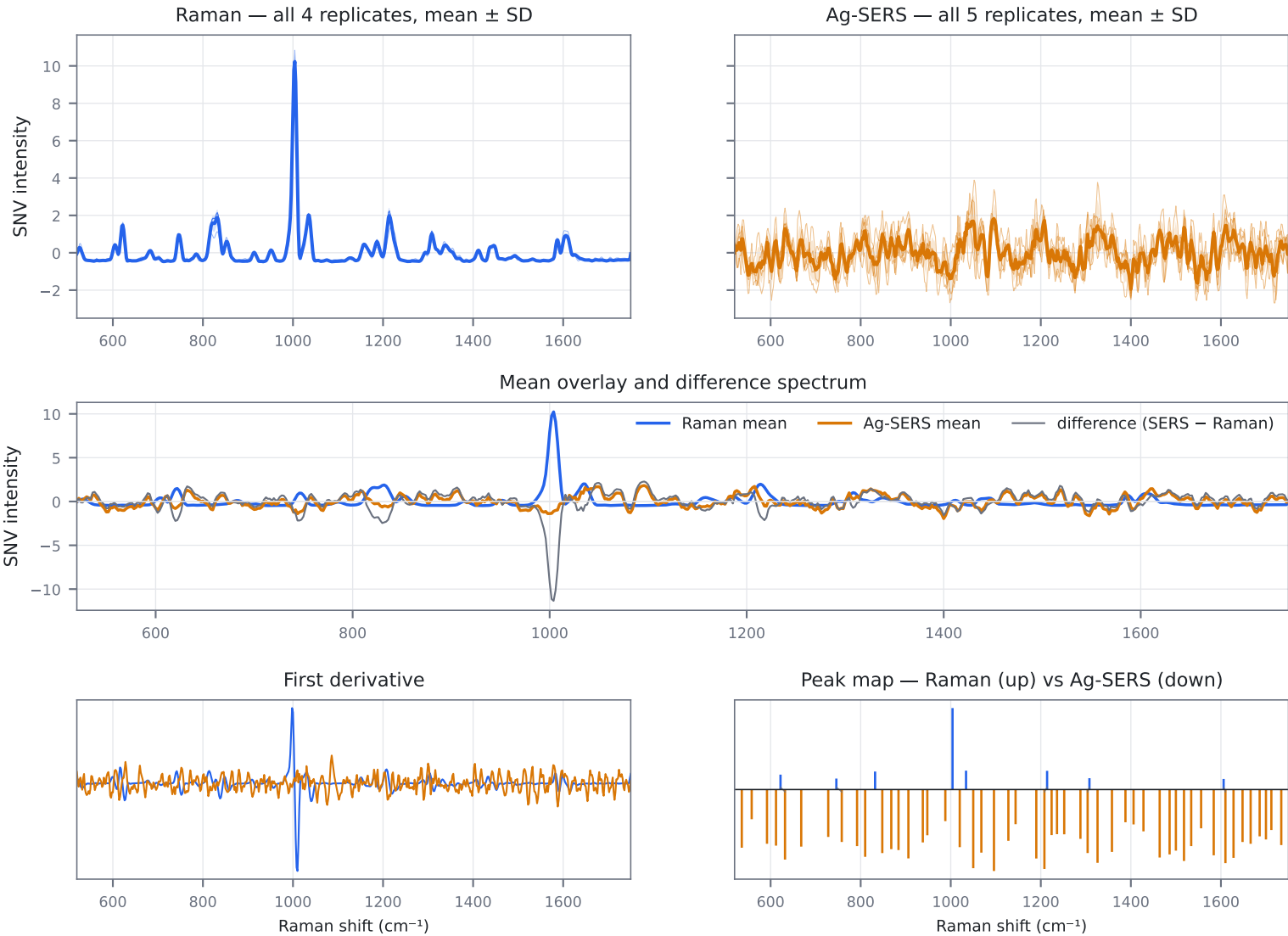
**Spectroscopic interpretation**  
Of 5 prominent Raman bands, 3 (60%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.12), while the full-profile cosine is only +0.17 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 1440 cm<sup>-1</sup> (CH2 scissoring) → 1434 cm<sup>-1</sup> (-6); 1655 cm<sup>-1</sup> (cis C=C stretch) → 1650 cm<sup>-1</sup> (-6). Modest systematic displacement (mean |Δν| 4.7 cm<sup>-1</sup>, median -6.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank ρ -0.50, redistribution index 0.09):

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# phenylalanine

family: amino\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine -0.092    Pearson -0.092    Spearman +0.137    spectral angle 95.3°    NRMSE 0.118  
DTW 0.293    derivative r -0.107    peak F1 0.17    Jaccard 0.09    precision 0.10    recall 0.62  
peaks: Raman 8, Ag-SERS 50, matched 5    mean  $|\Delta\nu|$  7.2 cm<sup>-1</sup>    correspondence score 0.07  
optimal rigid shift -10 cm<sup>-1</sup> (cosine gain +0.203)    replicate ceiling: Raman 0.99 / Ag-SERS 0.36

| Peak correspondence — strongest matches (3 Raman-only, 45 SERS-only) |         |             |         |        |                    |
|----------------------------------------------------------------------|---------|-------------|---------|--------|--------------------|
| Raman                                                                | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note               |
| 1214                                                                 | 1208    | -6          | 4.25    | Medium | Intensity enhanced |
| 622                                                                  | 632     | +10         | 4.70    | Low    | Intensity enhanced |
| 1308                                                                 | 1304    | -4          | 5.59    | High   | Intensity enhanced |
| 746                                                                  | 758     | +12         | 4.79    | Low    | Intensity enhanced |
| 1606                                                                 | 1610    | +4          | 6.96    | High   | Intensity enhanced |

## Spectroscopic interpretation

Of 8 prominent Raman bands, 5 (62%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.17), while the full-profile cosine is only -0.09 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 1206 cm<sup>-1</sup> (C-C ring) → 1208 cm<sup>-1</sup> (-6). Modest systematic displacement (mean  $|\Delta\nu|$  7.2 cm<sup>-1</sup>, median 4.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  -0.10, redistribution index 0.07): the same modes are observed but re-weighted by

# phosphate

family: organic\_acid (family assignment ambiguous)

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine -0.020    Pearson -0.020    Spearman -0.073    spectral angle 91.2°    NRMSE 0.177  
DTW 0.427    derivative r -0.069    peak F1 0.11    Jaccard 0.06    precision 0.06    recall 0.75  
peaks: Raman 4, Ag-SERS 51, matched 3    mean  $|\Delta v|$  8.0 cm<sup>-1</sup>    correspondence score 0.04  
optimal rigid shift +10 cm<sup>-1</sup> (cosine gain +0.025)    replicate ceiling: Raman 0.99 / Ag-SERS 0.49

| Peak correspondence — strongest matches (1 Raman-only, 48 SERS-only) |         |            |         |        |                    |
|----------------------------------------------------------------------|---------|------------|---------|--------|--------------------|
| Raman                                                                | Ag-SERS | $\Delta v$ | I ratio | conf.  | note               |
| 936                                                                  | 926     | -10        | 0.65    | Low    | Intensity reduced  |
| 560                                                                  | 566     | +6         | 0.77    | Medium | Shifted band       |
| 1066                                                                 | 1058    | -8         | 2.24    | Medium | Intensity enhanced |

## Spectroscopic interpretation

Of 4 prominent Raman bands, 3 (75%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.11), while the full-profile cosine is only -0.02 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 936→926 cm<sup>-1</sup> (-10); 560→566 cm<sup>-1</sup> (+6); 1066→1058 cm<sup>-1</sup> (-8). Modest systematic displacement (mean  $|\Delta v|$  8.0 cm<sup>-1</sup>, median -8.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative band intensities are comparatively well preserved (peak-rank p 0.50, redistribution 0.26), an unusually faithful

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.



# phosphatidylinositol

family: lipid

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.070    Pearson +0.070    Spearman +0.058    spectral angle 86.0°    NRMSE 0.265  
DTW 0.333    derivative r +0.065    peak F1 0.48    Jaccard 0.31    precision 0.31    recall 1.00  
peaks: Raman 17, Ag-SERS 54, matched 17    mean |Δν| 5.8 cm<sup>-1</sup>    correspondence score 0.25  
optimal rigid shift -10 cm<sup>-1</sup> (cosine gain +0.001)    replicate ceiling: Raman 0.87 / Ag-SERS 0.51

| Peak correspondence — strongest matches (0 Raman-only, 37 SERS-only) |         |     |         |        |                       |
|----------------------------------------------------------------------|---------|-----|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | Δν  | I ratio | conf.  | note                  |
| 862                                                                  | 864     | +2  | 0.75    | High   | Strong correspondence |
| 1090                                                                 | 1080    | -10 | 0.81    | Low    | Shifted band          |
| 650                                                                  | 660     | +10 | 0.74    | Low    | Shifted band          |
| 1182                                                                 | 1180    | -2  | 0.63    | High   | Intensity reduced     |
| 530                                                                  | 522     | -8  | 0.79    | Medium | Shifted band          |
| 1502                                                                 | 1508    | +6  | 0.91    | Medium | Shifted band          |
| 1346                                                                 | 1350    | +4  | 1.20    | High   | Strong correspondence |
| 928                                                                  | 934     | +6  | 0.83    | Medium | Shifted band          |
| 796                                                                  | 804     | +8  | 1.83    | Medium | Intensity enhanced    |

## Spectroscopic interpretation

Of 17 prominent Raman bands, 17 (100%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.48), while the full-profile cosine is only +0.07 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 862→864 cm<sup>-1</sup> (+2); 1090→1080 cm<sup>-1</sup> (-10); 650→660 cm<sup>-1</sup> (+10). Modest systematic displacement (mean |Δν| 5.8 cm<sup>-1</sup>, median 2.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Intensity transfer is partial (peak-rank ρ 0.39, redistribution 0.17), with enhancement favouring a subset of modes. Notes: a band in the

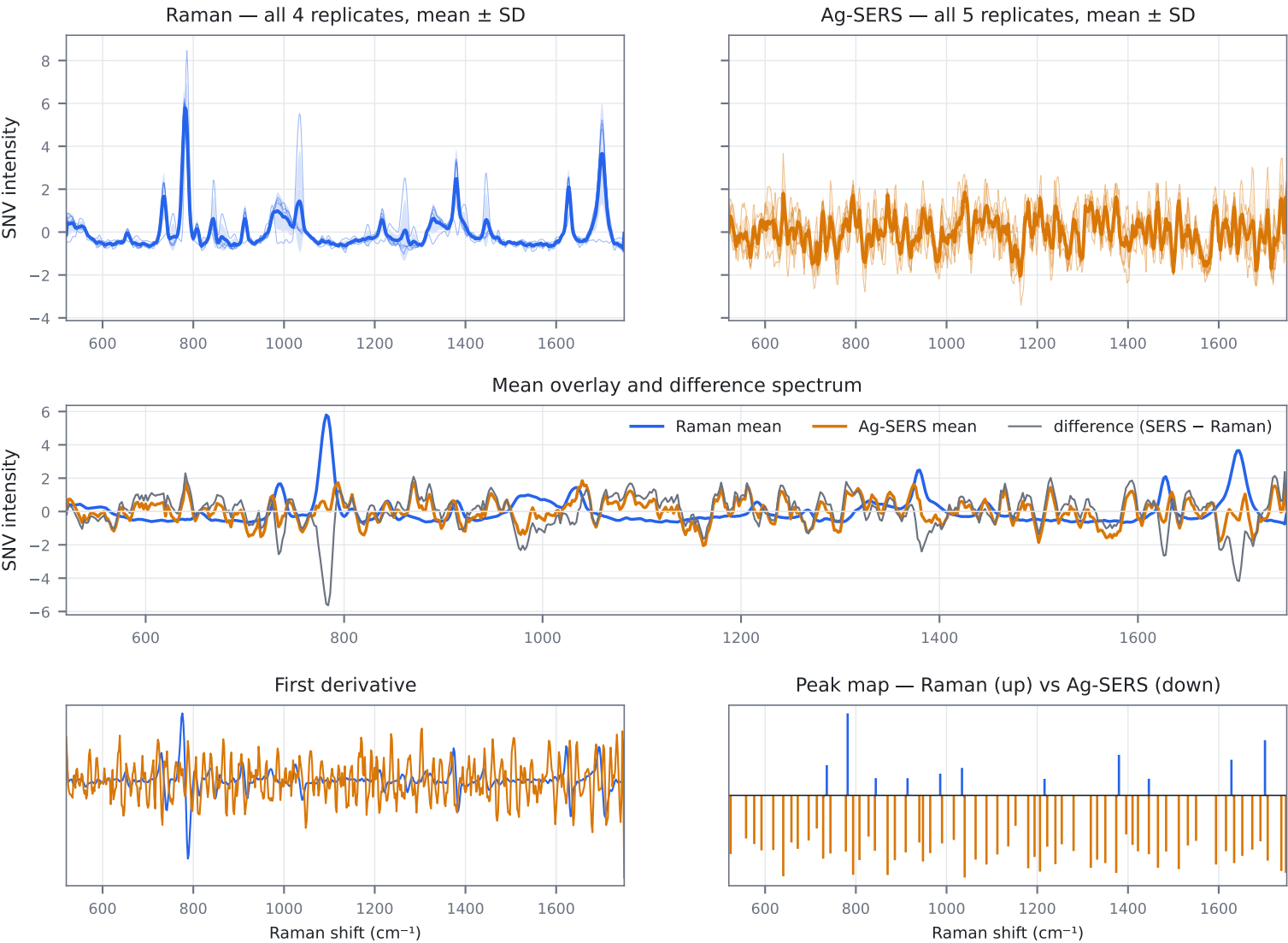
Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.



# phosphoenolpyruvate

family: organic\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.089    Pearson +0.089    Spearman +0.090    spectral angle 84.9°    NRMSE 0.163  
DTW 0.348    derivative r -0.052    peak F1 0.31    Jaccard 0.18    precision 0.18    recall 1.00  
peaks: Raman 11, Ag-SERS 60, matched 11    mean  $|\Delta\nu|$  4.7 cm<sup>-1</sup>    correspondence score 0.19  
optimal rigid shift -6 cm<sup>-1</sup> (cosine gain +0.015)    replicate ceiling: Raman 0.63 / Ag-SERS 0.41

| Peak correspondence — strongest matches (0 Raman-only, 49 SERS-only) |         |             |         |        |                    |
|----------------------------------------------------------------------|---------|-------------|---------|--------|--------------------|
| Raman                                                                | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note               |
| 782                                                                  | 778     | -4          | 0.68    | High   | Intensity reduced  |
| 1702                                                                 | 1708    | +6          | 1.18    | Medium | Shifted band       |
| 1380                                                                 | 1374    | -6          | 1.91    | Medium | Intensity enhanced |
| 1628                                                                 | 1634    | +6          | 1.90    | Medium | Intensity enhanced |
| 1034                                                                 | 1040    | +6          | 2.98    | Medium | Intensity enhanced |
| 736                                                                  | 728     | -8          | 2.09    | Medium | Intensity enhanced |
| 844                                                                  | 842     | -2          | 3.20    | High   | Intensity enhanced |
| 1216                                                                 | 1222    | +6          | 3.09    | Medium | Intensity enhanced |
| 914                                                                  | 910     | -4          | 3.30    | High   | Intensity enhanced |

## Spectroscopic interpretation

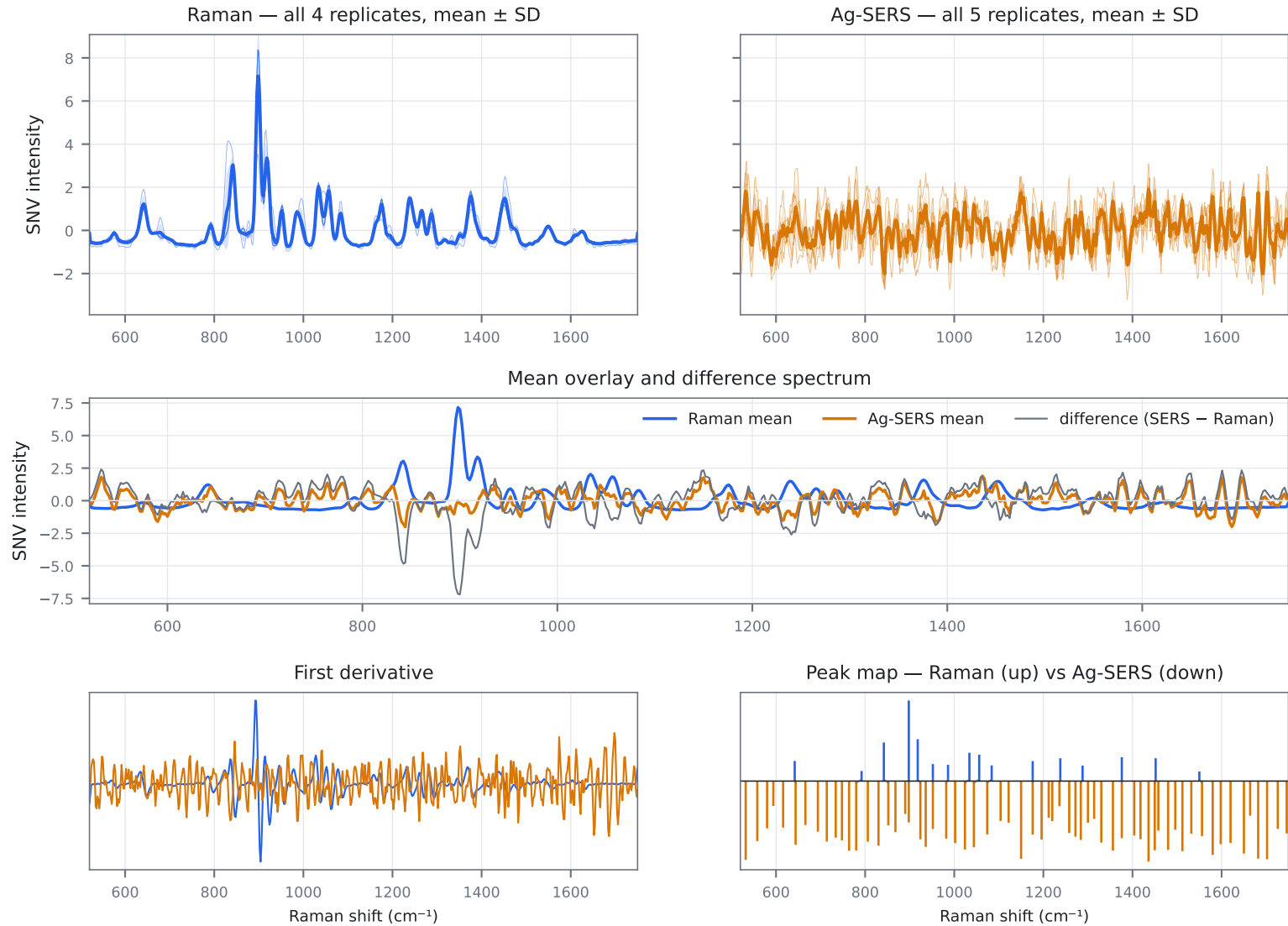
Of 11 prominent Raman bands, 11 (100%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.31), while the full-profile cosine is only +0.09 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 782→778 cm<sup>-1</sup> (-4); 1702→1708 cm<sup>-1</sup> (+6); 1380→1374 cm<sup>-1</sup> (-6). Modest systematic displacement (mean  $|\Delta\nu|$  4.7 cm<sup>-1</sup>, median 0.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Intensity transfer is partial (peak-rank  $\rho$  0.34, redistribution 0.20), with enhancement favouring a subset of modes. Notes: a band in the

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

proline

family: amino\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520-1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine -0.113    Pearson -0.113    Spearman -0.083    spectral angle 96.5°    NRMSE 0.157  
DTW 0.333    derivative r -0.044    peak F1 0.41    Jaccard 0.26    precision 0.26    recall 1.00  
peaks: Raman 16, Ag-SERS 62, matched 16    mean |Δv| 5.0 cm<sup>-1</sup>    correspondence score 0.24  
optimal rigid shift +10 cm<sup>-1</sup> (cosine gain +0.122)    replicate ceiling: Raman 0.86 / Ag-SERS 0.38

| Peak correspondence — strongest matches (0 Raman-only, 46 SERS-only) |         |     |         |        |                       |
|----------------------------------------------------------------------|---------|-----|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | Δv  | I ratio | conf.  | note                  |
| 898                                                                  | 898     | +0  | 0.51    | High   | Intensity reduced     |
| 842                                                                  | 852     | +10 | 1.15    | Low    | Shifted band          |
| 1034                                                                 | 1024    | -10 | 2.41    | Low    | Intensity enhanced    |
| 1376                                                                 | 1376    | +0  | 2.39    | High   | Intensity enhanced    |
| 1238                                                                 | 1236    | -2  | 1.10    | High   | Strong correspondence |
| 1452                                                                 | 1450    | -2  | 3.02    | High   | Intensity enhanced    |
| 1176                                                                 | 1176    | +0  | 2.67    | High   | Intensity enhanced    |
| 642                                                                  | 644     | +2  | 3.15    | High   | Intensity enhanced    |
| 918                                                                  | 924     | +6  | 1.39    | Medium | Shifted band          |

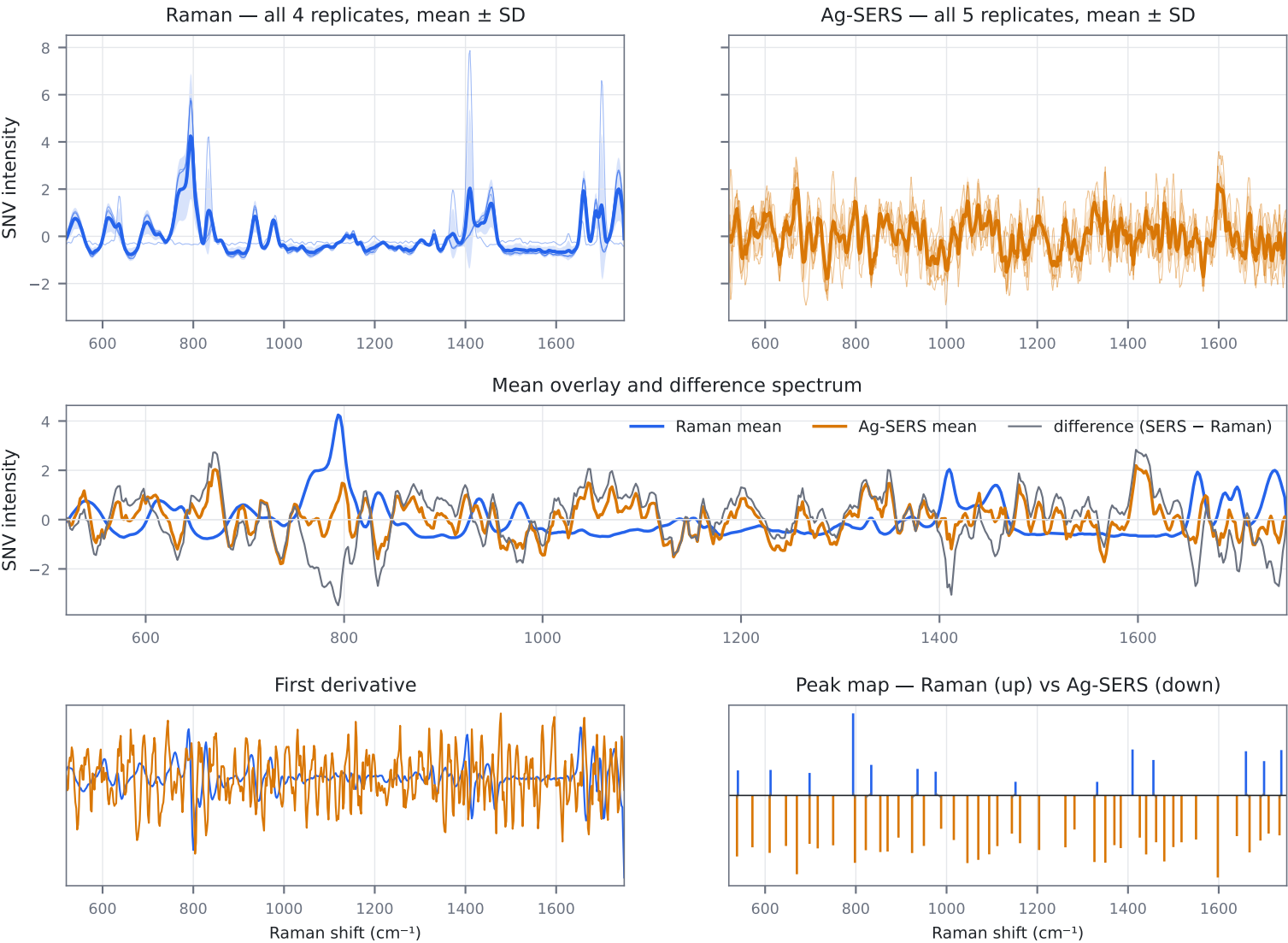
**Spectroscopic interpretation**  
Of 16 prominent Raman bands, 16 (100%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.41), while the full-profile cosine is only -0.11 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 898→898 cm<sup>-1</sup> (+0); 842→852 cm<sup>-1</sup> (+10); 1034→1024 cm<sup>-1</sup> (-10). Modest systematic displacement (mean |Δv| 5.0 cm<sup>-1</sup>, median -1.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank ρ -0.35, redistribution index 0.26): the same modes are

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# pyruvate

family: organic\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine -0.087    Pearson -0.087    Spearman -0.168    spectral angle 95.0°    NRMSE 0.219  
DTW 0.337    derivative r -0.074    peak F1 0.46    Jaccard 0.30    precision 0.30    recall 0.93  
peaks: Raman 14, Ag-SERS 43, matched 13    mean |Δν| 6.5 cm<sup>-1</sup>    correspondence score 0.21  
optimal rigid shift -8 cm<sup>-1</sup> (cosine gain +0.036)    replicate ceiling: Raman 0.49 / Ag-SERS 0.37

| Peak correspondence — strongest matches (1 Raman-only, 30 SERS-only) |         |     |         |        |                       |
|----------------------------------------------------------------------|---------|-----|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | Δν  | I ratio | conf.  | note                  |
| 794                                                                  | 798     | +4  | 0.82    | High   | Strong correspondence |
| 1660                                                                 | 1668    | +8  | 1.29    | Medium | Shifted band          |
| 1738                                                                 | 1734    | -4  | 0.88    | High   | Strong correspondence |
| 1700                                                                 | 1692    | -8  | 1.31    | Medium | Shifted band          |
| 936                                                                  | 924     | -12 | 2.17    | Low    | Intensity enhanced    |
| 612                                                                  | 610     | -2  | 2.25    | High   | Intensity enhanced    |
| 834                                                                  | 822     | -12 | 1.79    | Low    | Intensity enhanced    |
| 976                                                                  | 988     | +12 | 1.40    | Low    | Intensity enhanced    |
| 1456                                                                 | 1462    | +6  | 1.32    | Medium | Shifted band          |

## Spectroscopic interpretation

Of 14 prominent Raman bands, 13 (93%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.46), while the full-profile cosine is only -0.09 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 794→798 cm<sup>-1</sup> (+4); 1660→1668 cm<sup>-1</sup> (+8); 1738→1734 cm<sup>-1</sup> (-4). Modest systematic displacement (mean |Δν| 6.5 cm<sup>-1</sup>, median -2.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative band intensities are comparatively well preserved (peak-rank p 0.60, redistribution 0.20), an unusually faithful

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# riboflavin

family: cofactor

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.020    Pearson +0.020    Spearman +0.117    spectral angle 88.8°    NRMSE 0.146  
DTW 0.351    derivative r +0.030    peak F1 0.29    Jaccard 0.17    precision 0.19    recall 0.67  
peaks: Raman 12, Ag-SERS 43, matched 8    mean | $\Delta\nu$ | 5.2 cm<sup>-1</sup>    correspondence score 0.16  
optimal rigid shift +10 cm<sup>-1</sup> (cosine gain +0.019)    replicate ceiling: Raman 1.00 / Ag-SERS 0.57

| Peak correspondence — strongest matches (4 Raman-only, 35 SERS-only) |         |             |         |        |                       |
|----------------------------------------------------------------------|---------|-------------|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note                  |
| 1348                                                                 | 1348    | +0          | 0.48    | High   | Intensity reduced     |
| 1180                                                                 | 1178    | -2          | 1.68    | High   | Intensity enhanced    |
| 1536                                                                 | 1528    | -8          | 1.62    | Medium | Intensity enhanced    |
| 1464                                                                 | 1464    | +0          | 3.74    | High   | Intensity enhanced    |
| 1060                                                                 | 1050    | -10         | 4.28    | Low    | Intensity enhanced    |
| 1156                                                                 | 1152    | -4          | 1.29    | High   | Strong correspondence |
| 1730                                                                 | 1736    | +6          | 3.98    | Medium | Intensity enhanced    |
| 788                                                                  | 776     | -12         | 3.28    | Low    | Intensity enhanced    |

## Spectroscopic interpretation

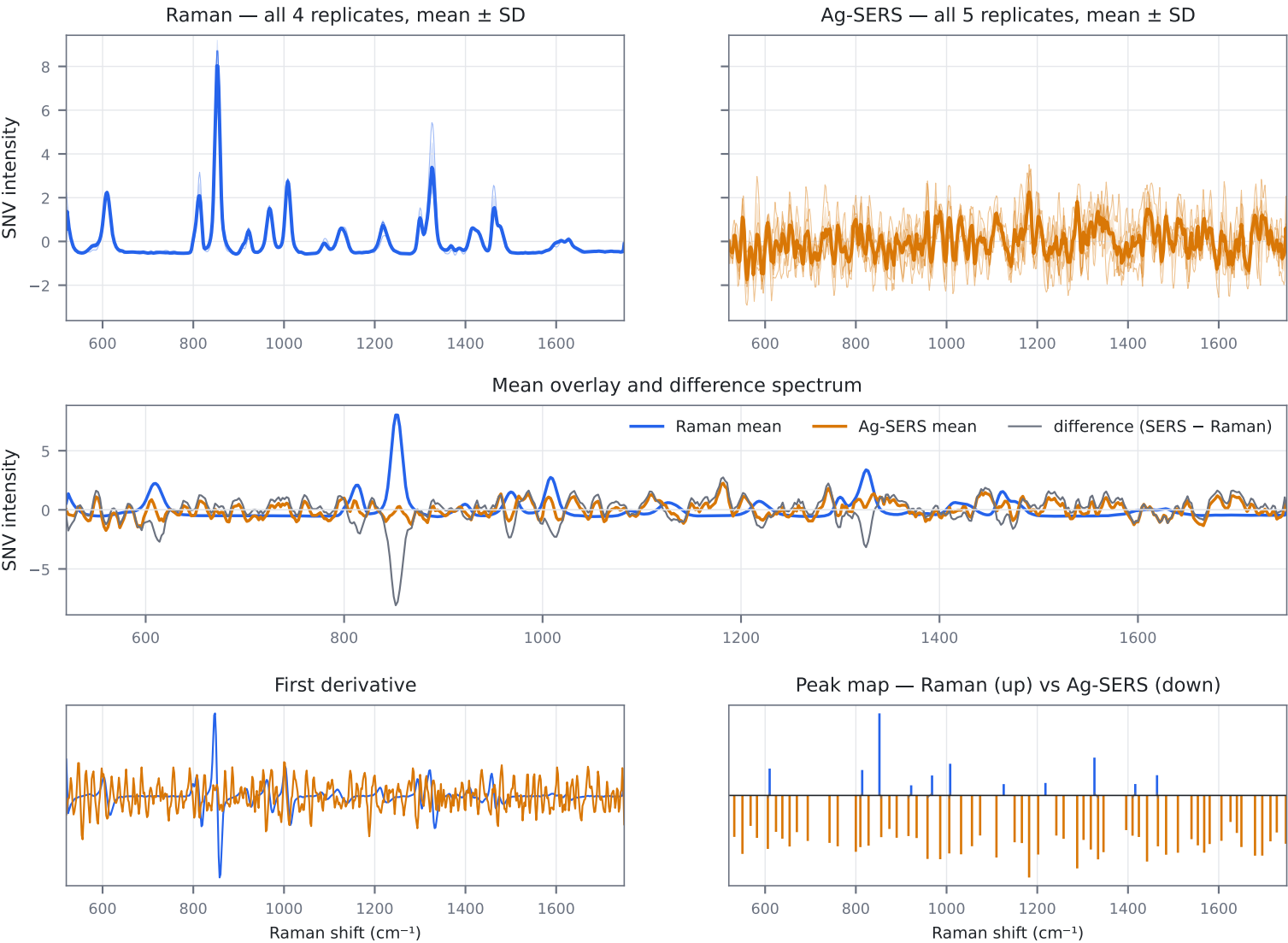
Of 12 prominent Raman bands, 8 (67%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.29), while the full-profile cosine is only +0.02 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 1350 cm<sup>-1</sup> (isoalloxazine) → 1348 cm<sup>-1</sup> (+0). Modest systematic displacement (mean | $\Delta\nu$ | 5.2 cm<sup>-1</sup>, median -3.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Intensity transfer is partial (peak-rank  $\rho$  0.36, redistribution 0.33), with enhancement favouring a subset of modes. Notes: 35/43 Ag-SERS bands have

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

serine

family: amino\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.019    Pearson +0.019    Spearman +0.018    spectral angle 88.9°    NRMSE 0.136  
DTW 0.343    derivative r +0.029    peak F1 0.26    Jaccard 0.15    precision 0.16    recall 0.82  
peaks: Raman 11, Ag-SERS 58, matched 9    mean |Δν| 5.1 cm<sup>-1</sup>    correspondence score 0.15  
optimal rigid shift +4 cm<sup>-1</sup> (cosine gain +0.005)    replicate ceiling: Raman 0.95 / Ag-SERS 0.28

| Peak correspondence — strongest matches (2 Raman-only, 49 SERS-only) |         |     |         |        |                    |
|----------------------------------------------------------------------|---------|-----|---------|--------|--------------------|
| Raman                                                                | Ag-SERS | Δν  | I ratio | conf.  | note               |
| 852                                                                  | 856     | +4  | 0.51    | High   | Intensity reduced  |
| 1326                                                                 | 1334    | +8  | 1.69    | Medium | Intensity enhanced |
| 1008                                                                 | 1006    | -2  | 1.83    | High   | Intensity enhanced |
| 610                                                                  | 606     | -4  | 1.99    | High   | Intensity enhanced |
| 814                                                                  | 810     | -4  | 2.03    | High   | Intensity enhanced |
| 1464                                                                 | 1466    | +2  | 2.51    | High   | Intensity enhanced |
| 968                                                                  | 958     | -10 | 3.18    | Low    | Intensity enhanced |
| 1416                                                                 | 1410    | -6  | 3.52    | Medium | Intensity enhanced |
| 922                                                                  | 916     | -6  | 4.01    | Medium | Intensity enhanced |

Spectroscopic interpretation

Of 11 prominent Raman bands, 9 (82%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.26), while the full-profile cosine is only +0.02 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 852→856 cm<sup>-1</sup> (+4); 1326→1334 cm<sup>-1</sup> (+8); 1008→1006 cm<sup>-1</sup> (-2). Modest systematic displacement (mean |Δν| 5.1 cm<sup>-1</sup>, median -4.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Intensity transfer is partial (peak-rank ρ 0.48, redistribution 0.22), with enhancement favouring a subset of modes.

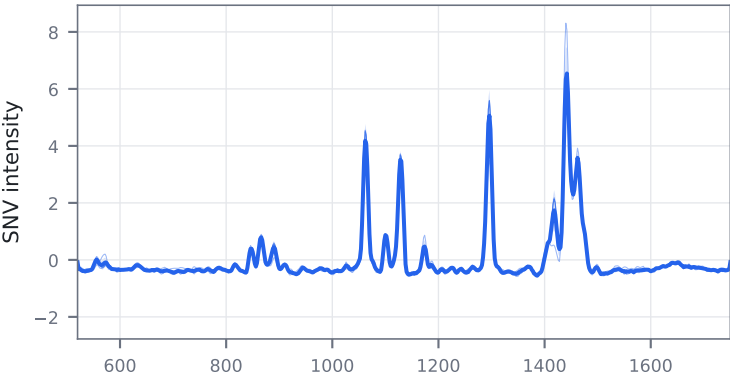
Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# stearate

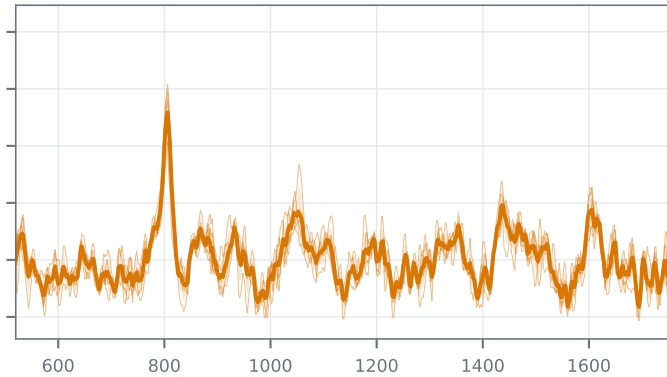
family: lipid

Raman source: RamanBioLib[gobbato\_raman\_metabolites] | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]

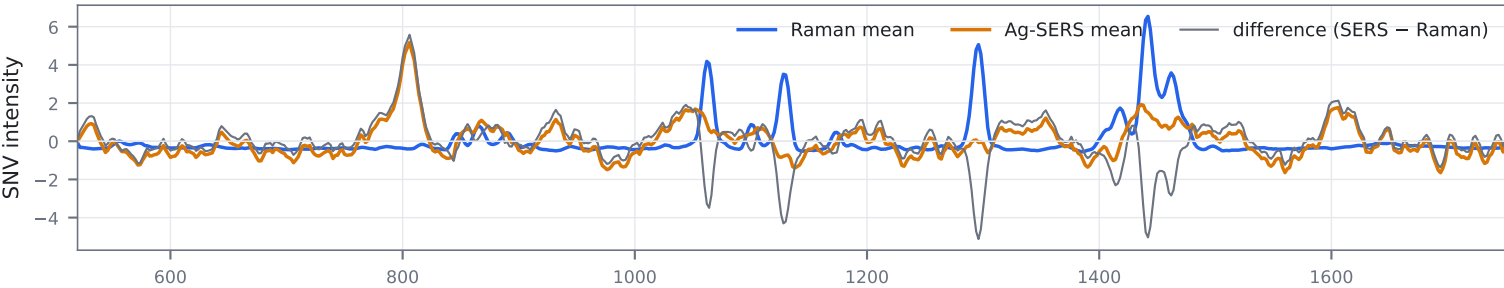
Raman — all 4 replicates, mean ± SD



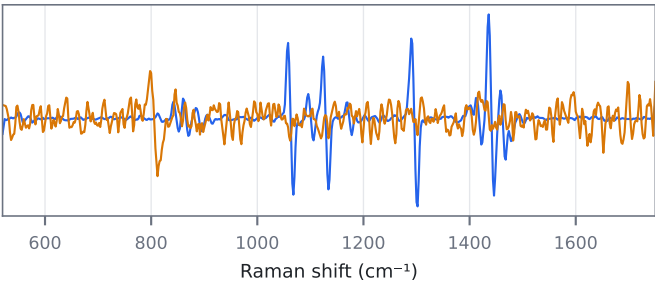
Ag-SERS — all 5 replicates, mean ± SD



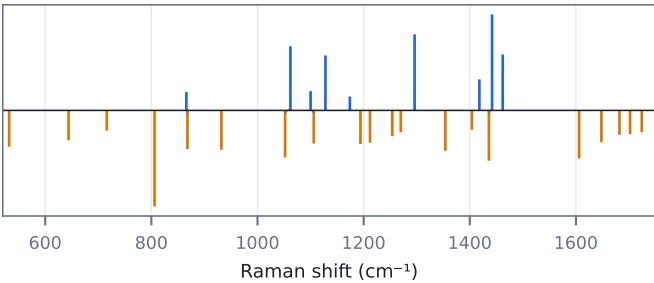
Mean overlay and difference spectrum



First derivative



Peak map — Raman (up) vs Ag-SERS (down)



cosine +0.202    Pearson +0.202    Spearman +0.198    spectral angle 78.4°    NRMSE 0.168  
DTW 0.356    derivative r +0.040    peak F1 0.28    Jaccard 0.16    precision 0.20    recall 0.44  
peaks: Raman 9, Ag-SERS 20, matched 4    mean |Δν| 6.0 cm<sup>-1</sup>    correspondence score 0.14  
optimal rigid shift +8 cm<sup>-1</sup> (cosine gain +0.029)    replicate ceiling: Raman 0.97 / Ag-SERS 0.77

Peak correspondence — strongest matches (5 Raman-only, 16 SERS-only)

| Raman | Ag-SERS | Δν  | I ratio | conf.  | note               |
|-------|---------|-----|---------|--------|--------------------|
| 1442  | 1436    | -6  | 0.52    | Medium | Intensity reduced  |
| 1062  | 1052    | -10 | 0.73    | Low    | Shifted band       |
| 866   | 868     | +2  | 2.12    | High   | Intensity enhanced |
| 1100  | 1106    | +6  | 1.72    | Medium | Intensity enhanced |

## Spectroscopic interpretation

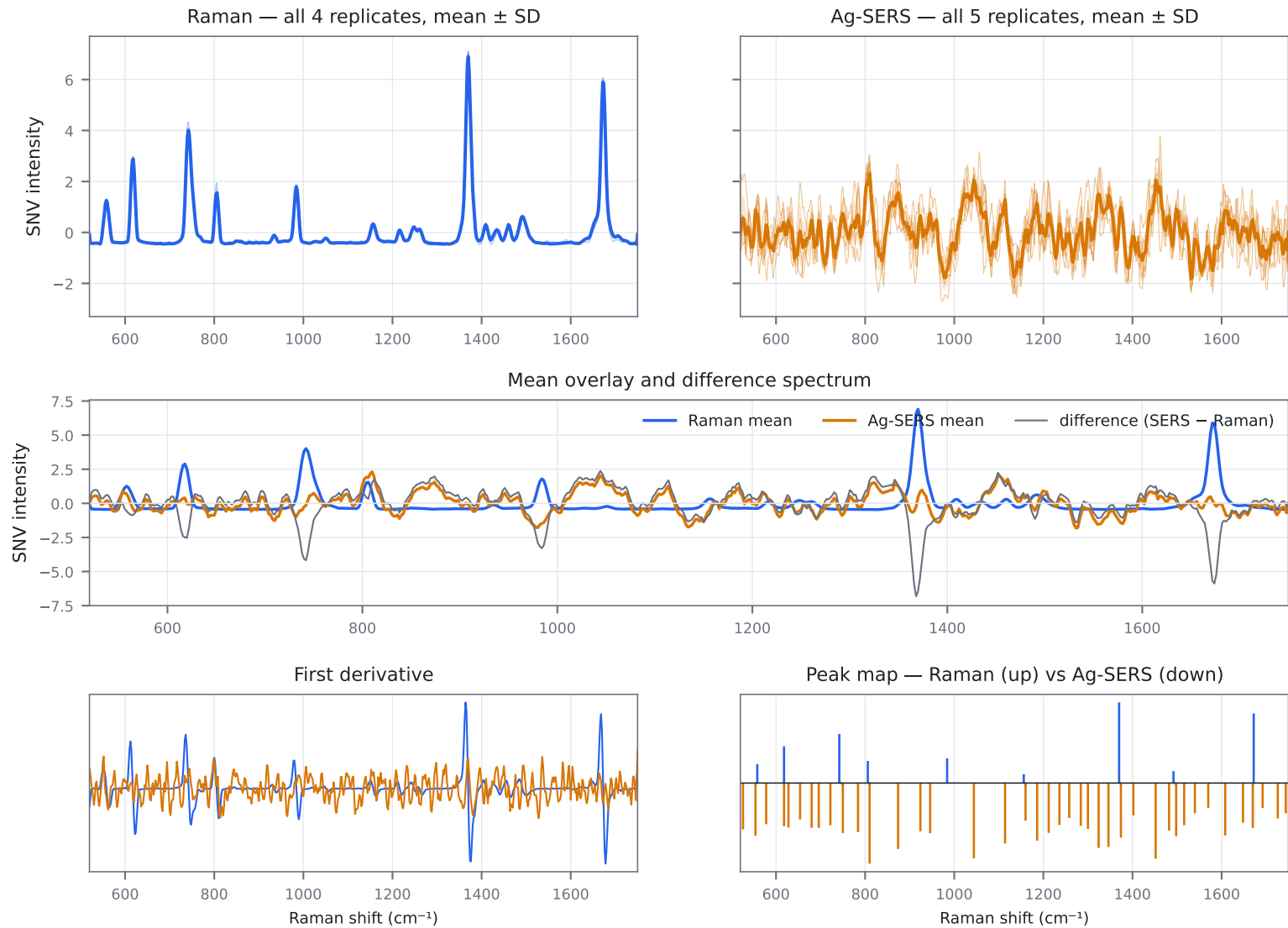
Of 9 prominent Raman bands, 4 (44%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.28), while the full-profile cosine is only +0.20 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 1440 cm<sup>-1</sup> (CH<sub>2</sub> scissoring) → 1436 cm<sup>-1</sup> (-6). Modest systematic displacement (mean |Δν| 6.0 cm<sup>-1</sup>, median -2.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative band intensities are comparatively well preserved (peak-rank ρ 1.00, redistribution 0.23), an unusually faithful transfer for a colloidal



# thymine

family: pyrimidine

Raman source: RamanBioLib[gobbato\_raman\_metabolites] | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.046    Pearson +0.046    Spearman +0.086    spectral angle 87.3°    NRMSE 0.167  
DTW 0.399    derivative r +0.121    peak F1 0.33    Jaccard 0.20    precision 0.20    recall 0.89  
peaks: Raman 9, Ag-SERS 40, matched 8    mean  $|\Delta\nu|$  4.0 cm<sup>-1</sup>    correspondence score 0.22  
optimal rigid shift -4 cm<sup>-1</sup> (cosine gain +0.013)    replicate ceiling: Raman 0.99 / Ag-SERS 0.49

Peak correspondence — strongest matches (1 Raman-only, 32 SERS-only)

| Raman | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note                  |
|-------|---------|-------------|---------|--------|-----------------------|
| 1370  | 1374    | +4          | 0.68    | High   | Intensity reduced     |
| 1672  | 1670    | -2          | 0.65    | High   | Intensity reduced     |
| 742   | 750     | +8          | 1.02    | Medium | Shifted band          |
| 618   | 618     | +0          | 1.18    | High   | Strong correspondence |
| 806   | 810     | +4          | 3.65    | High   | Intensity enhanced    |
| 558   | 554     | -4          | 2.80    | High   | Intensity enhanced    |
| 1492  | 1498    | +6          | 4.46    | Medium | Intensity enhanced    |
| 1156  | 1160    | +4          | 4.32    | High   | Intensity enhanced    |

## Spectroscopic interpretation

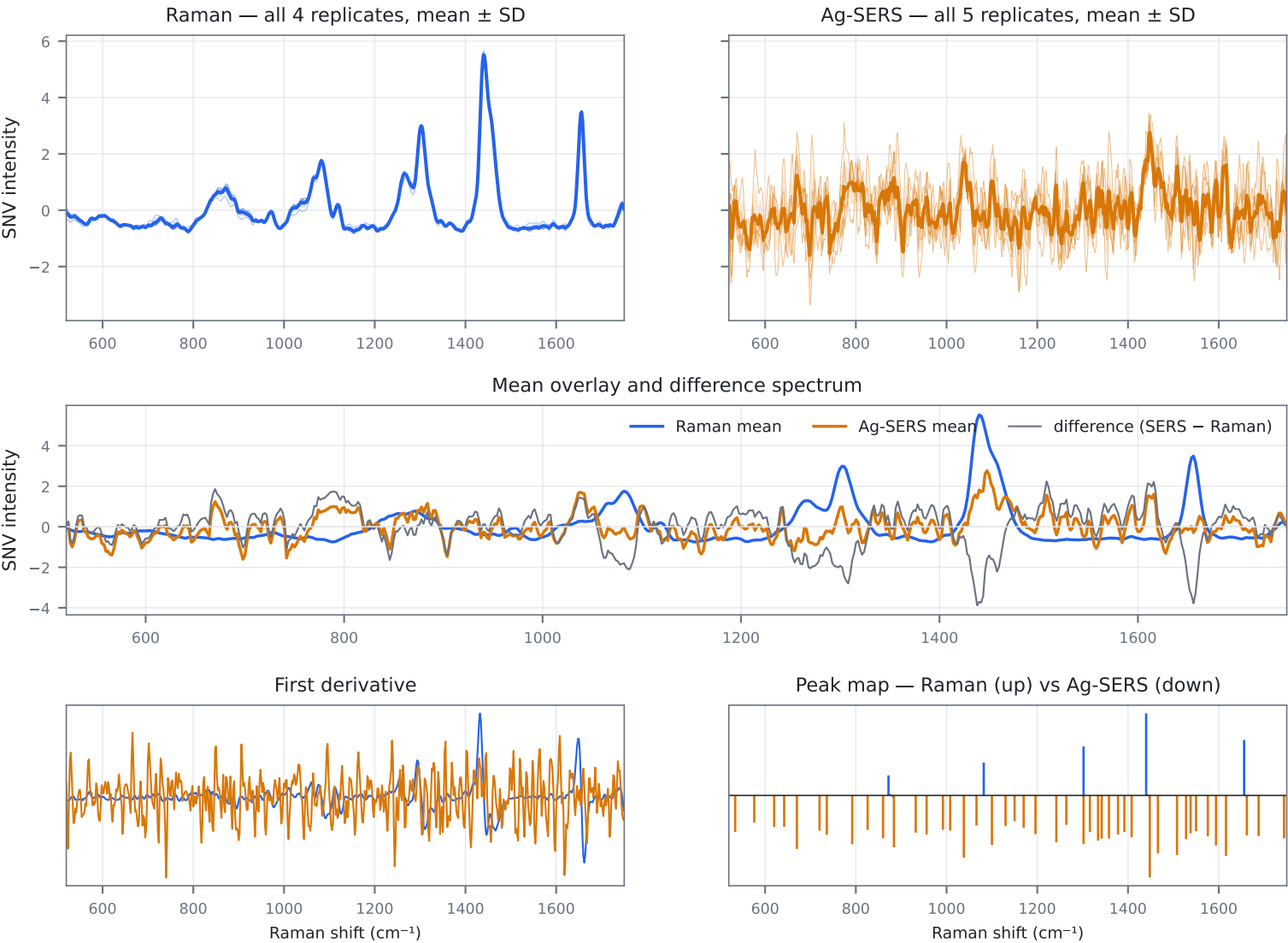
Of 9 prominent Raman bands, 8 (89%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.33), while the full-profile cosine is only +0.05 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 1370 cm<sup>-1</sup> (CH<sub>3</sub> deformation) → 1374 cm<sup>-1</sup> (+4); 745 cm<sup>-1</sup> (ring breathing) → 750 cm<sup>-1</sup> (+8). Band positions are essentially conserved (mean  $|\Delta\nu|$  4.0 cm<sup>-1</sup>, median 4.0 cm<sup>-1</sup>), i.e. within the 2 cm<sup>-1</sup> grid and typical of physisorption without strong bond perturbation. Relative intensities are strongly reordered (peak-rank p 0.14, redistribution index 0.33): the same modes

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# triolein

family: lipid

Raman source: RamanBioLib[gobbato\_raman\_metabolites] | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.345    Pearson +0.345    Spearman +0.069    spectral angle 69.8°    NRMSE 0.157  
DTW 0.280    derivative r +0.082    peak F1 0.16    Jaccard 0.09    precision 0.09    recall 0.80  
peaks: Raman 5, Ag-SERS 44, matched 4    mean  $|\Delta v|$  6.5 cm<sup>-1</sup>    correspondence score 0.07  
optimal rigid shift -6 cm<sup>-1</sup> (cosine gain +0.027)    replicate ceiling: Raman 0.99 / Ag-SERS 0.29

Peak correspondence — strongest matches (1 Raman-only, 40 SERS-only)

| Raman | Ag-SERS | $\Delta v$ | I ratio | conf.  | note                  |
|-------|---------|------------|---------|--------|-----------------------|
| 1440  | 1448    | +8         | 1.00    | Medium | Shifted band          |
| 1656  | 1662    | +6         | 0.72    | Medium | Shifted band          |
| 1302  | 1302    | +0         | 0.99    | High   | Strong correspondence |
| 872   | 884     | +12        | 2.59    | Low    | Intensity enhanced    |

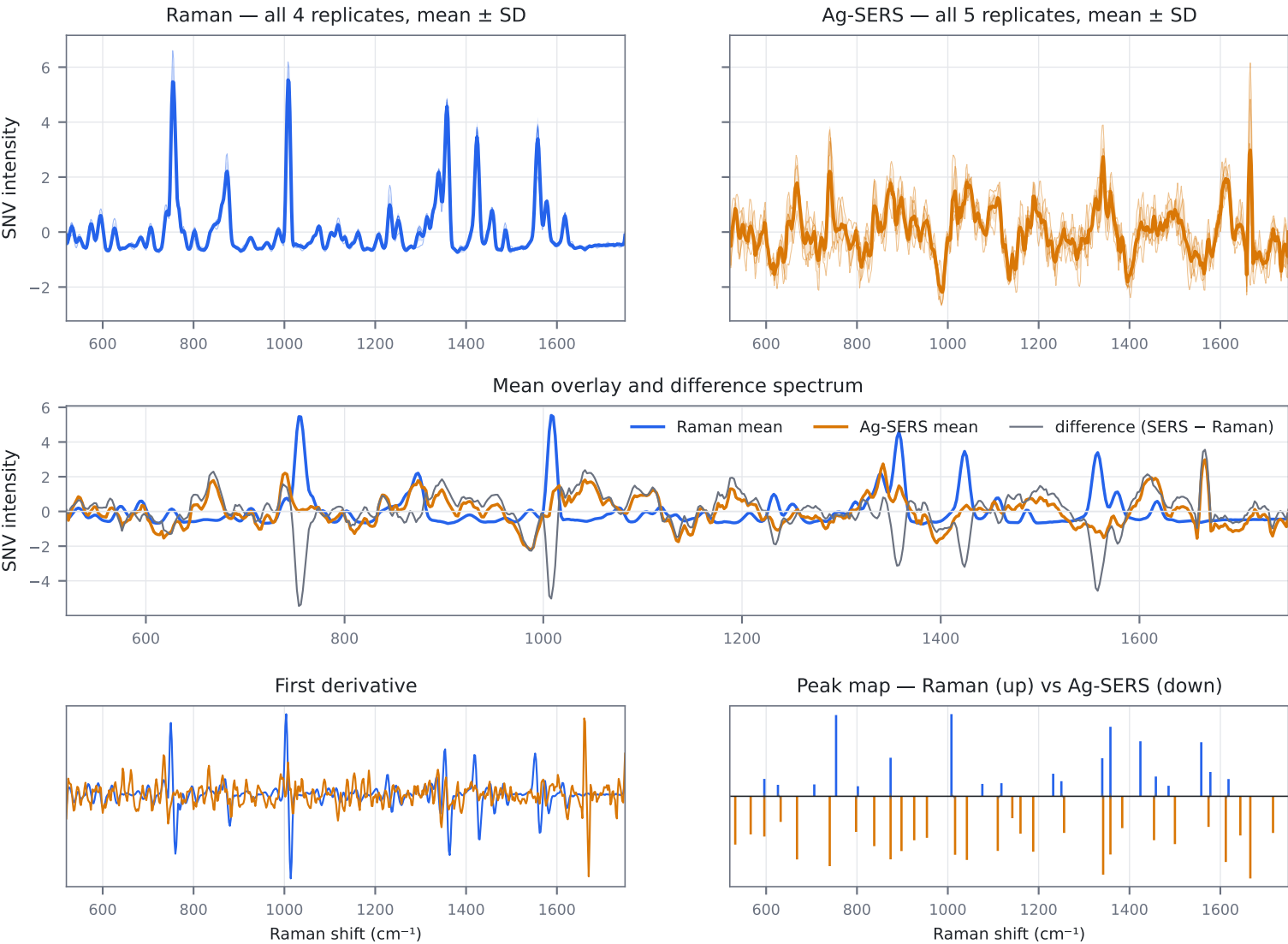
### Spectroscopic interpretation

Of 5 prominent Raman bands, 4 (80%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.16), while the full-profile cosine is only +0.35 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 1440→1448 cm<sup>-1</sup> (+8); 1656→1662 cm<sup>-1</sup> (+6); 1302→1302 cm<sup>-1</sup> (+0). Modest systematic displacement (mean  $|\Delta v|$  6.5 cm<sup>-1</sup>, median 7.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank  $\rho$  0.20, redistribution index 0.14): the same modes are observed but re-weighted

# tryptophan

family: amino\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.198    Pearson +0.198    Spearman +0.169    spectral angle 78.6°    NRMSE 0.186  
DTW 0.302    derivative r +0.129    peak F1 0.50    Jaccard 0.33    precision 0.41    recall 0.63  
peaks: Raman 19, Ag-SERS 29, matched 12    mean | $\Delta\nu$ | 4.0 cm<sup>-1</sup>    correspondence score 0.33  
optimal rigid shift -2 cm<sup>-1</sup> (cosine gain +0.007)    replicate ceiling: Raman 0.97 / Ag-SERS 0.65

Peak correspondence — strongest matches (7 Raman-only, 17 SERS-only)

| Raman | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note                  |
|-------|---------|-------------|---------|--------|-----------------------|
| 1008  | 1016    | +8          | 0.71    | Medium | Shifted band          |
| 1358  | 1358    | +0          | 0.84    | High   | Strong correspondence |
| 874   | 874     | +0          | 1.64    | High   | Intensity enhanced    |
| 1458  | 1454    | -4          | 2.22    | High   | Intensity enhanced    |
| 596   | 596     | +0          | 2.32    | High   | Intensity enhanced    |
| 1340  | 1342    | +2          | 2.06    | High   | Intensity enhanced    |
| 1618  | 1612    | -6          | 3.81    | Medium | Intensity enhanced    |
| 1578  | 1574    | -4          | 1.26    | High   | Strong correspondence |
| 1118  | 1110    | -8          | 4.18    | Medium | Intensity enhanced    |

### Spectroscopic interpretation

Of 19 prominent Raman bands, 12 (63%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.50), while the full-profile cosine is only +0.20 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 1010 cm<sup>-1</sup> (indole) → 1016 cm<sup>-1</sup> (+8). Band positions are essentially conserved (mean | $\Delta\nu$ | 4.0 cm<sup>-1</sup>, median 0.0 cm<sup>-1</sup>), i.e. within the 2 cm<sup>-1</sup> grid and typical of physisorption without strong bond perturbation. Intensity transfer is partial (peak-rank  $\rho$  0.30, redistribution 0.25), with enhancement favouring a subset of modes. Notes: a band in the 1380–1430 cm<sup>-1</sup> carboxylate

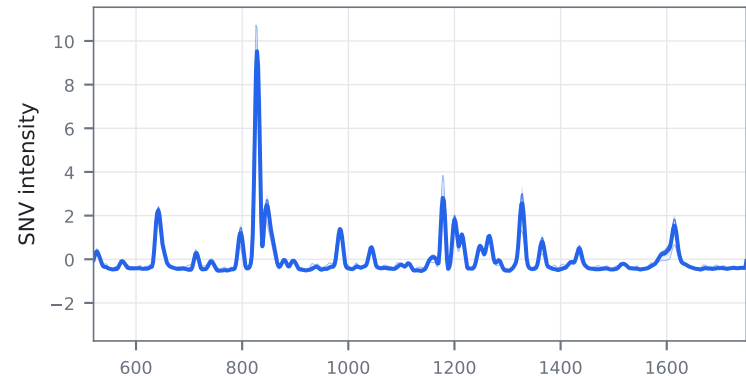
Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# tyrosine

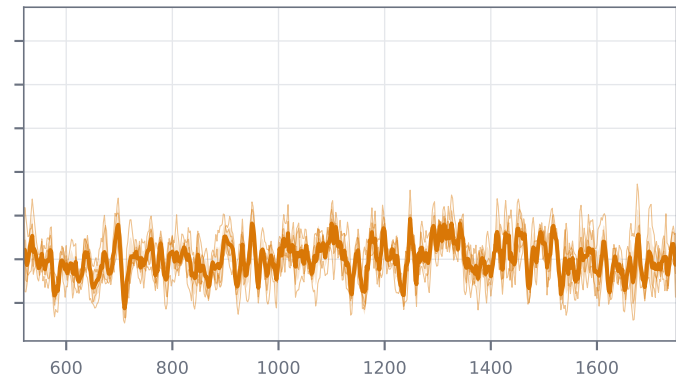
family: amino\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]

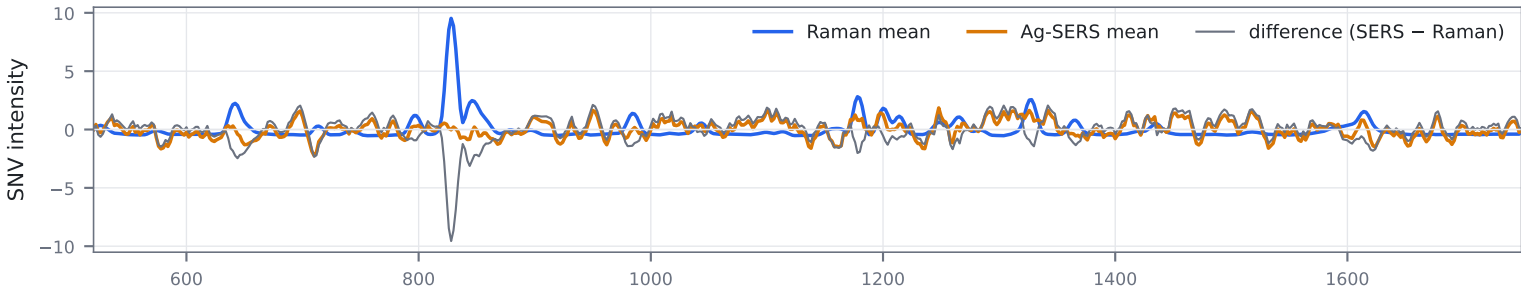
Raman — all 4 replicates, mean ± SD



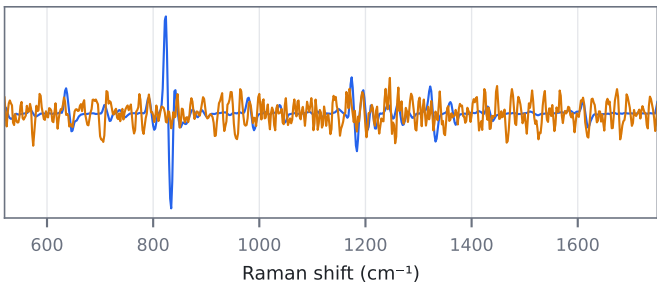
Ag-SERS — all 5 replicates, mean ± SD



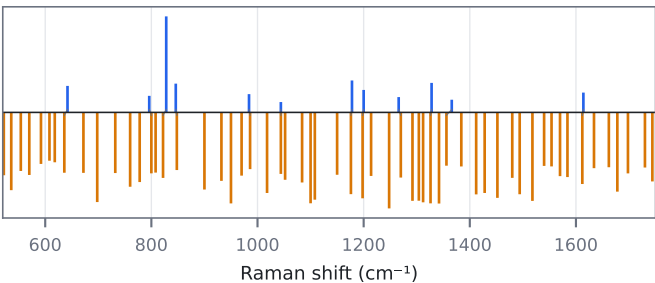
Mean overlay and difference spectrum



First derivative



Peak map — Raman (up) vs Ag-SERS (down)



cosine +0.072    Pearson +0.072    Spearman +0.039    spectral angle 85.9°    NRMSE 0.116  
DTW 0.356    derivative r +0.134    peak F1 0.34    Jaccard 0.21    precision 0.21    recall 1.00  
peaks: Raman 12, Ag-SERS 58, matched 12    mean |Δv| 3.5 cm<sup>-1</sup>    correspondence score 0.24  
optimal rigid shift +2 cm<sup>-1</sup> (cosine gain +0.015)    replicate ceiling: Raman 0.95 / Ag-SERS 0.39

Peak correspondence — strongest matches (0 Raman-only, 46 SERS-only)

| Raman | Ag-SERS | Δv | I ratio | conf.  | note               |
|-------|---------|----|---------|--------|--------------------|
| 828   | 822     | -6 | 0.68    | Medium | Intensity reduced  |
| 1178  | 1176    | -2 | 2.56    | High   | Intensity enhanced |
| 1328  | 1326    | -2 | 3.08    | High   | Intensity enhanced |
| 642   | 636     | -6 | 2.28    | Medium | Intensity enhanced |
| 1200  | 1198    | -2 | 3.82    | High   | Intensity enhanced |
| 1614  | 1612    | -2 | 3.62    | High   | Intensity enhanced |
| 984   | 986     | +2 | 3.11    | High   | Intensity enhanced |
| 846   | 848     | +2 | 2.00    | High   | Intensity enhanced |
| 796   | 800     | +4 | 3.68    | High   | Intensity enhanced |

## Spectroscopic interpretation

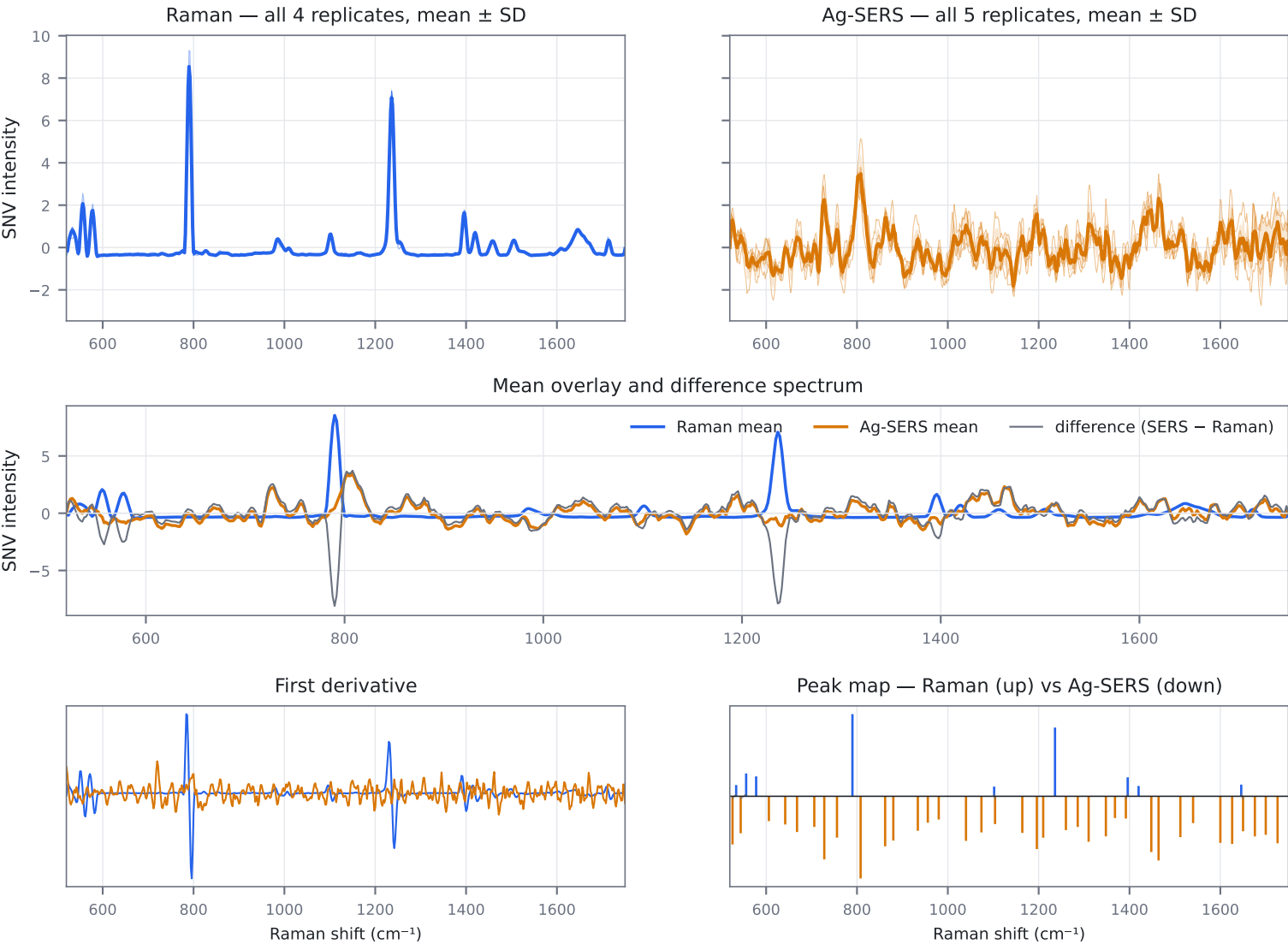
Of 12 prominent Raman bands, 12 (100%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.34), while the full-profile cosine is only +0.07 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 830 cm<sup>-1</sup> (Fermi doublet) → 822 cm<sup>-1</sup> (-6). Band positions are essentially conserved (mean |Δv| 3.5 cm<sup>-1</sup>, median -2.0 cm<sup>-1</sup>), i.e. within the 2 cm<sup>-1</sup> grid and typical of physisorption without strong bond perturbation. Relative band intensities are comparatively well preserved (peak-rank ρ 0.76, redistribution 0.23), an unusually faithful transfer for a colloidal

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# uracil

family: pyrimidine

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine -0.021    Pearson -0.021    Spearman +0.080    spectral angle 91.2°    NRMSE 0.145  
DTW 0.358    derivative r -0.110    peak F1 0.22    Jaccard 0.12    precision 0.14    recall 0.56  
peaks: Raman 9, Ag-SERS 36, matched 5    mean |Δν| 6.0 cm<sup>-1</sup>    correspondence score 0.11  
optimal rigid shift -10 cm<sup>-1</sup> (cosine gain +0.229)    replicate ceiling: Raman 0.99 / Ag-SERS 0.57

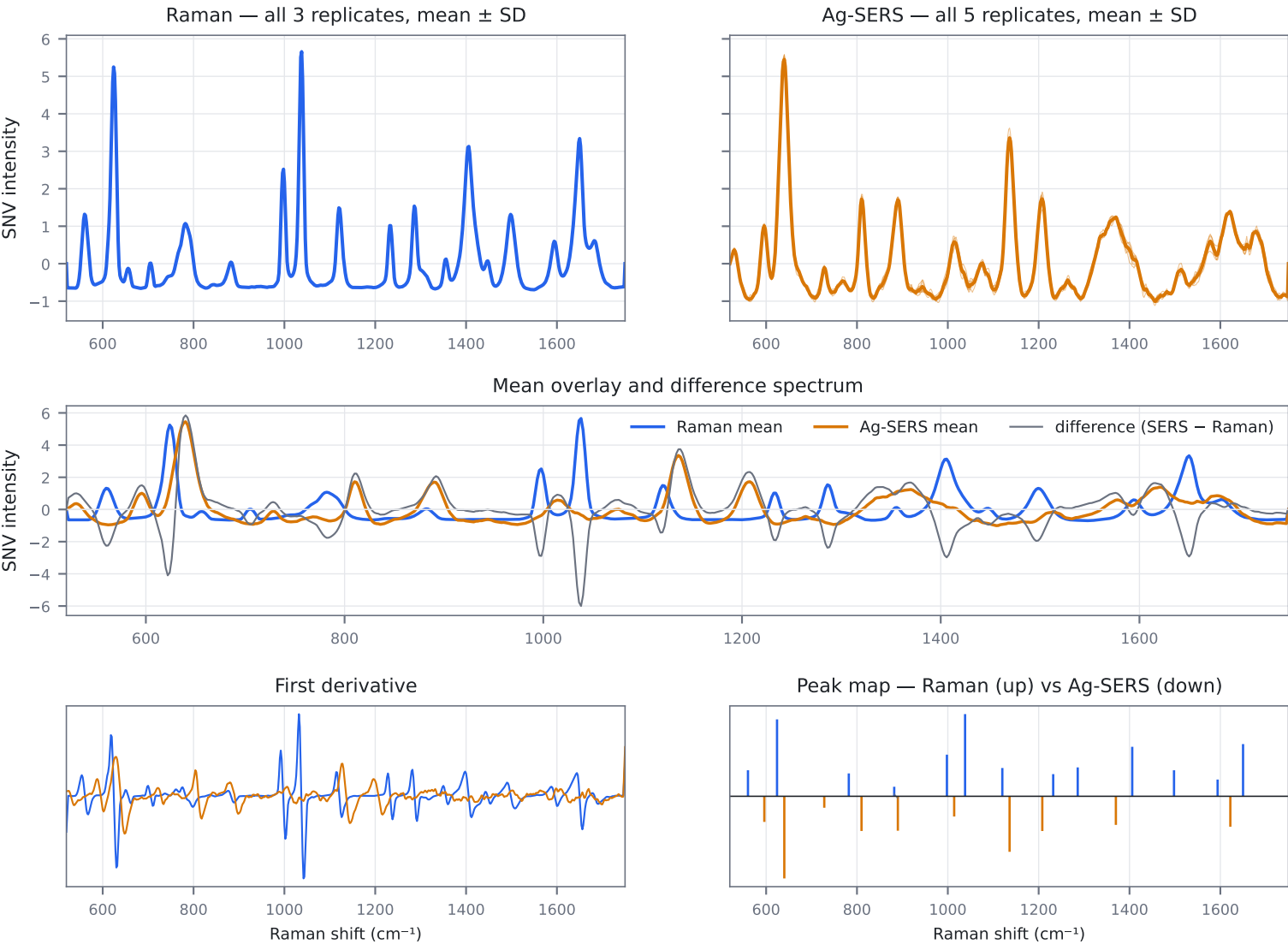
| Peak correspondence — strongest matches (4 Raman-only, 31 SERS-only) |         |     |         |        |                       |
|----------------------------------------------------------------------|---------|-----|---------|--------|-----------------------|
| Raman                                                                | Ag-SERS | Δν  | I ratio | conf.  | note                  |
| 556                                                                  | 544     | -12 | 1.62    | Low    | Intensity enhanced    |
| 1396                                                                 | 1392    | -4  | 1.18    | High   | Strong correspondence |
| 1646                                                                 | 1650    | +4  | 3.00    | High   | Intensity enhanced    |
| 1102                                                                 | 1104    | +2  | 2.87    | High   | Intensity enhanced    |
| 534                                                                  | 526     | -8  | 4.29    | Medium | Intensity enhanced    |

**Spectroscopic interpretation**  
Of 9 prominent Raman bands, 5 (56%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.22), while the full-profile cosine is only -0.02 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 556→544 cm<sup>-1</sup> (-12); 1396→1392 cm<sup>-1</sup> (-4); 1646→1650 cm<sup>-1</sup> (+4). Modest systematic displacement (mean |Δν| 6.0 cm<sup>-1</sup>, median -4.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative intensities are strongly reordered (peak-rank ρ -1.00, redistribution index 0.21): the same modes are

urate

family: purine

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520-1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.099    Pearson +0.099    Spearman +0.167    spectral angle 84.3°    NRMSE 0.211  
DTW 0.233    derivative r -0.148    peak F1 0.09    Jaccard 0.05    precision 0.10    recall 0.08  
peaks: Raman 13, Ag-SERS 10, matched 1    mean |Δν| 8.0 cm<sup>-1</sup>    correspondence score 0.03  
optimal rigid shift -10 cm<sup>-1</sup> (cosine gain +0.220)    replicate ceiling: Raman 1.00 / Ag-SERS 0.99

| Peak correspondence — strongest matches (12 Raman-only, 9 SERS-only) |         |    |         |        |                    |
|----------------------------------------------------------------------|---------|----|---------|--------|--------------------|
| Raman                                                                | Ag-SERS | Δν | I ratio | conf.  | note               |
| 882                                                                  | 890     | +8 | 3.61    | Medium | Intensity enhanced |

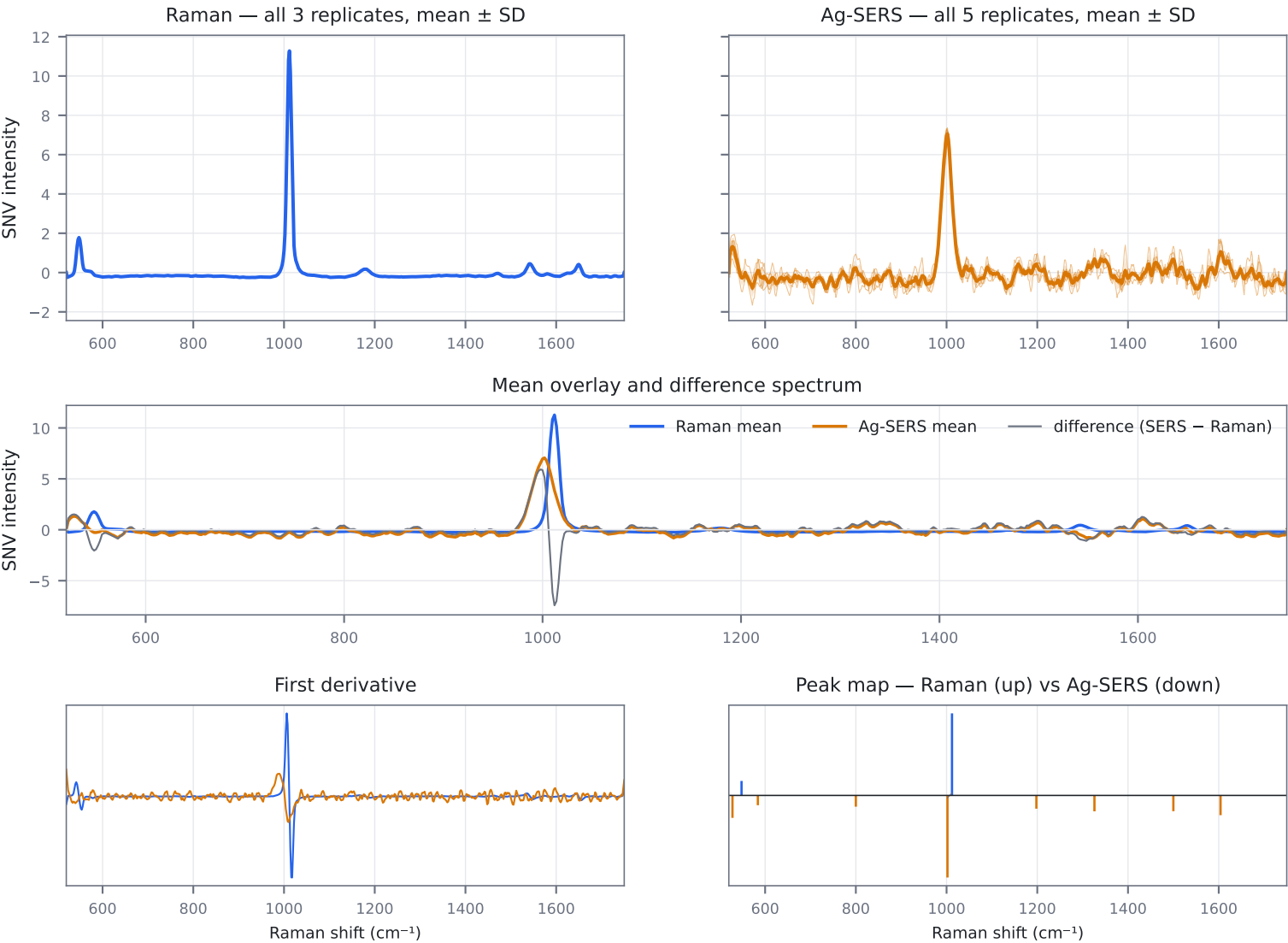
**Spectroscopic interpretation**  
Of 13 prominent Raman bands, 1 (8%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.09), while the full-profile cosine is only +0.10 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 882→890 cm<sup>-1</sup> (+8). Modest systematic displacement (mean |Δν| 8.0 cm<sup>-1</sup>, median 8.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Notes: 9/10 Ag-SERS bands have no Raman counterpart, which may reflect surface-activated modes, adsorbate reorientation, or residual



urea

family: small\_nitrogenous

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520-1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.541    Pearson +0.541    Spearman +0.327    spectral angle 57.3°    NRMSE 0.081  
DTW 0.170    derivative r +0.015    peak F1 0.20    Jaccard 0.11    precision 0.12    recall 0.50  
peaks: Raman 2, Ag-SERS 8, matched 1    mean |Δν| 10.0 cm<sup>-1</sup>    correspondence score 0.03  
optimal rigid shift +10 cm<sup>-1</sup> (cosine gain +0.286)    replicate ceiling: Raman 1.00 / Ag-SERS 0.88

Peak correspondence — strongest matches (1 Raman-only, 7 SERS-only)

| Raman | Ag-SERS | Δν  | I ratio | conf. | note         |
|-------|---------|-----|---------|-------|--------------|
| 1012  | 1002    | -10 | 1.00    | Low   | Shifted band |

Spectroscopic interpretation

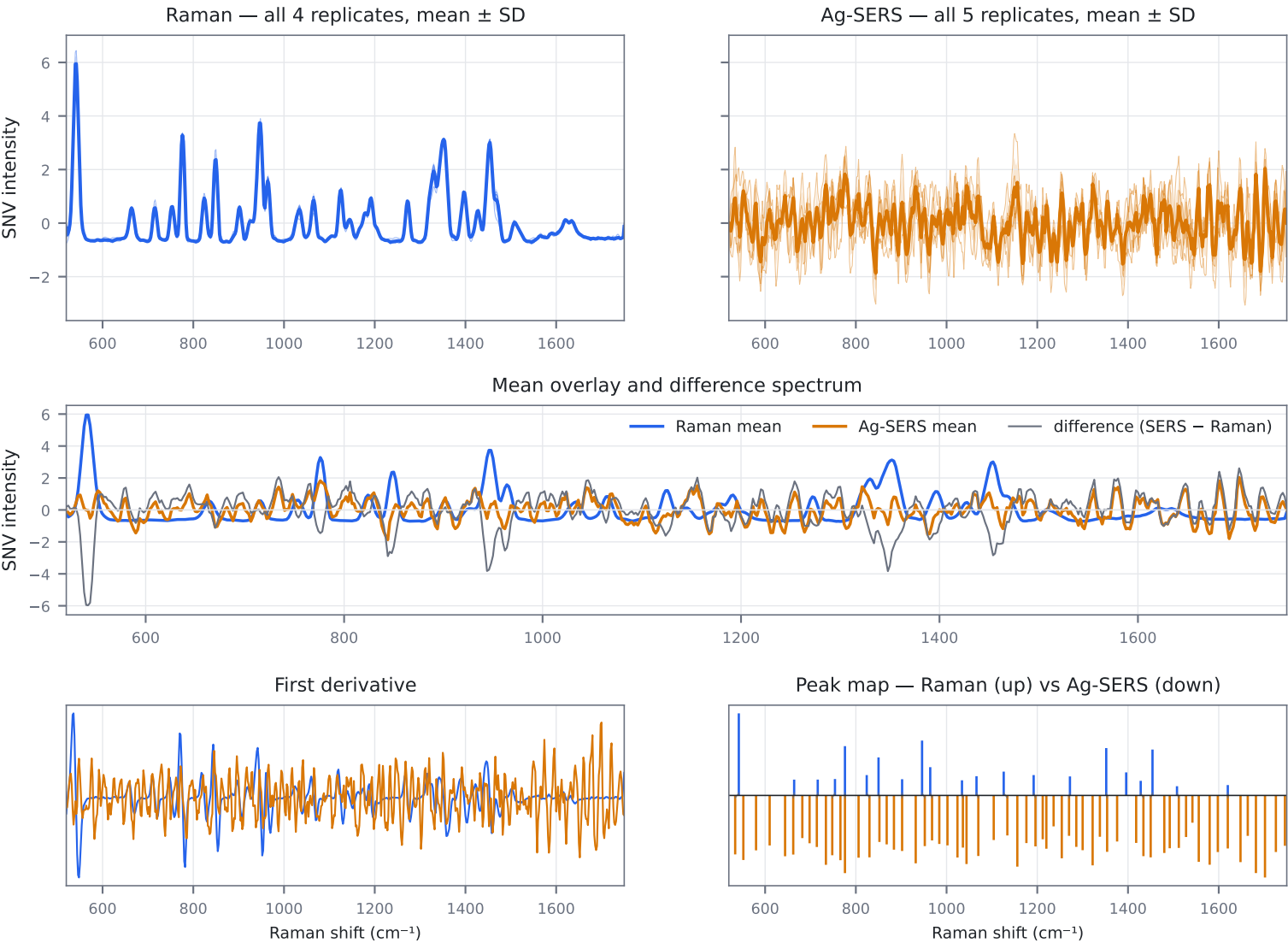
Of 2 prominent Raman bands, 1 (50%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.20), while the full-profile cosine is only +0.54 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 1000 cm<sup>-1</sup> (C-N symmetric stretch) → 1002 cm<sup>-1</sup> (-10). Larger displacements (mean |Δν| 10.0 cm<sup>-1</sup>, max 10.0 cm<sup>-1</sup>) suggest strong surface interaction or ambiguous band assignment in a congested region. Notes: 7/8 Ag-SERS bands have no Raman counterpart, which may reflect surface-activated modes, adsorbate reorientation, or residual citrate/colloid features.

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

valine

family: amino\_acid

Raman source: RamanBioLib|gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520-1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=4, Ag-SERS n=5 [Raman replicates span two sources]



cosine +0.140    Pearson +0.140    Spearman +0.140    spectral angle 81.9°    NRMSE 0.169  
DTW 0.322    derivative r +0.081    peak F1 0.48    Jaccard 0.32    precision 0.32    recall 0.95  
peaks: Raman 21, Ag-SERS 62, matched 20    mean |Δν| 3.6 cm<sup>-1</sup>    correspondence score 0.34  
optimal rigid shift -2 cm<sup>-1</sup> (cosine gain +0.010)    replicate ceiling: Raman 0.99 / Ag-SERS 0.34

Peak correspondence — strongest matches (1 Raman-only, 42 SERS-only)

| Raman | Ag-SERS | Δν | I ratio | conf.  | note                  |
|-------|---------|----|---------|--------|-----------------------|
| 542   | 534     | -8 | 0.72    | Medium | Shifted band          |
| 946   | 952     | +6 | 0.92    | Medium | Shifted band          |
| 776   | 776     | +0 | 1.58    | High   | Intensity enhanced    |
| 1352  | 1354    | +2 | 1.19    | High   | Strong correspondence |
| 1454  | 1450    | -4 | 1.35    | High   | Strong correspondence |
| 850   | 852     | +2 | 1.22    | High   | Strong correspondence |
| 1126  | 1134    | +8 | 1.68    | Medium | Intensity enhanced    |
| 1066  | 1070    | +4 | 3.15    | High   | Intensity enhanced    |
| 1272  | 1270    | -2 | 2.12    | High   | Intensity enhanced    |

Spectroscopic interpretation

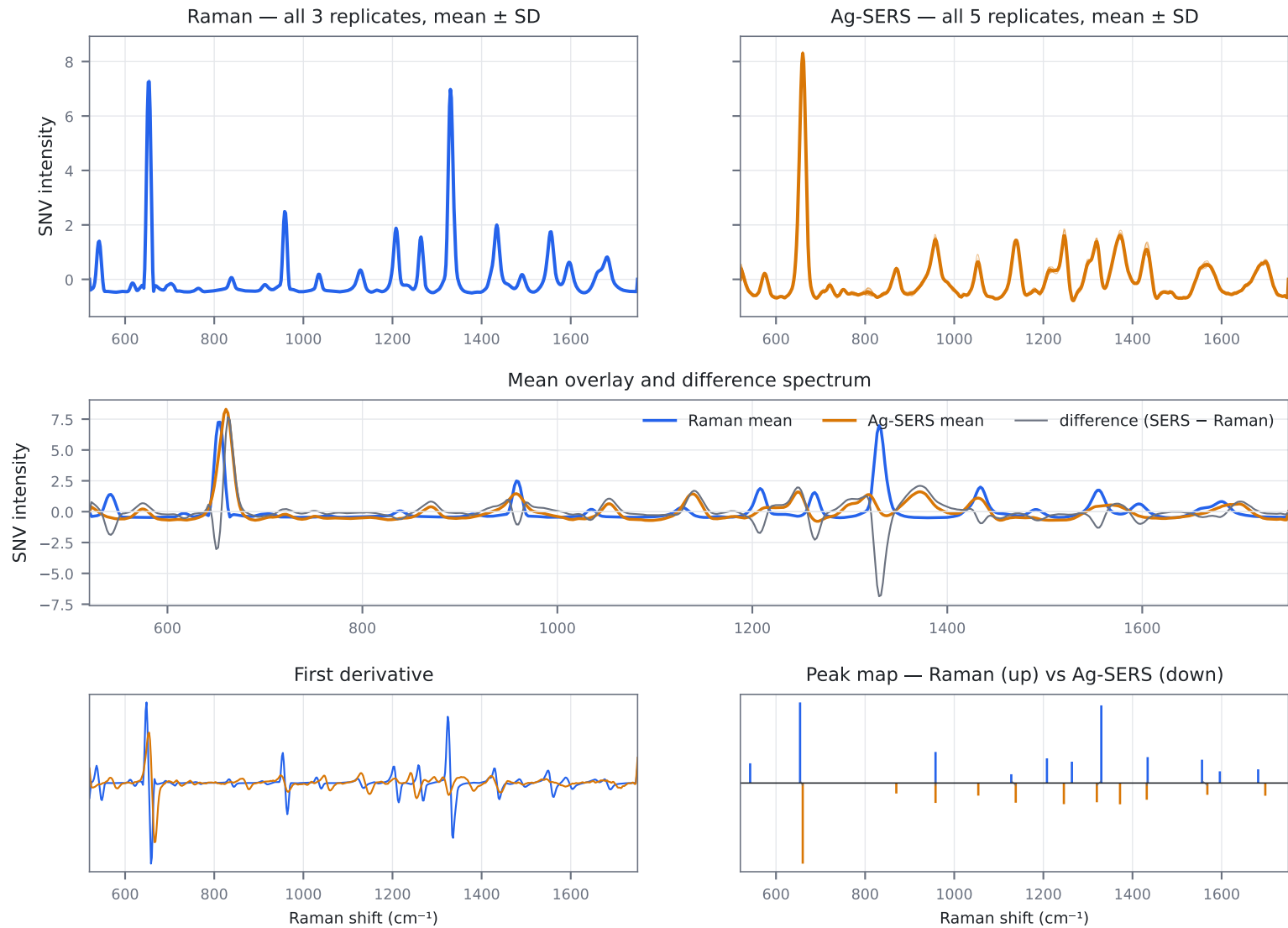
Of 21 prominent Raman bands, 20 (95%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.48), while the full-profile cosine is only +0.14 — the band inventory therefore agrees far better than the overall spectral shape. Strongest retained correspondences: 542→534 cm<sup>-1</sup> (-8); 946→952 cm<sup>-1</sup> (+6); 776→776 cm<sup>-1</sup> (+0). Band positions are essentially conserved (mean |Δν| 3.6 cm<sup>-1</sup>, median 2.0 cm<sup>-1</sup>), i.e. within the 2 cm<sup>-1</sup> grid and typical of physisorption without strong bond perturbation. Intensity transfer is partial (peak-rank ρ 0.38, redistribution 0.26), with enhancement favouring a subset of modes. Notes: a band in the 1380-1430

Both panels share identical x- and y-limits; no independent autoscaling. Peak detection and all metrics use the spectra exactly as entered into Stage B.

# xanthine

family: purine

Raman source: gobbato\_raman\_metabolites | Ag-SERS source: gobbato\_sers\_metabolites  
excitation: 785 nm both arms | substrate: Raman = neat powder / library, Ag-SERS = citrate Ag colloid  
preprocessing: ASLS baseline + Savitzky-Golay + SNV (Stage B, unmodified) | grid 520–1750 cm<sup>-1</sup> @ 2 cm<sup>-1</sup>  
replicates: Raman n=3, Ag-SERS n=5



cosine +0.426    Pearson +0.426    Spearman +0.301    spectral angle 64.8°    NRMSE 0.138  
DTW 0.169    derivative r +0.098    peak F1 0.55    Jaccard 0.38    precision 0.55    recall 0.55  
peaks: Raman 11, Ag-SERS 11, matched 6    mean | $\Delta\nu$ | 6.7 cm<sup>-1</sup>    correspondence score 0.24  
optimal rigid shift -6 cm<sup>-1</sup> (cosine gain +0.115)    replicate ceiling: Raman 1.00 / Ag-SERS 0.99

Peak correspondence — strongest matches (5 Raman-only, 5 SERS-only)

| Raman | Ag-SERS | $\Delta\nu$ | I ratio | conf.  | note               |
|-------|---------|-------------|---------|--------|--------------------|
| 654   | 660     | +6          | 1.00    | Medium | Shifted band       |
| 1330  | 1320    | -10         | 0.25    | Low    | Intensity reduced  |
| 958   | 958     | +0          | 0.64    | High   | Intensity reduced  |
| 1434  | 1432    | -2          | 0.64    | High   | Intensity reduced  |
| 1556  | 1568    | +12         | 0.50    | Low    | Intensity reduced  |
| 1128  | 1138    | +10         | 2.24    | Low    | Intensity enhanced |

### Spectroscopic interpretation

Of 11 prominent Raman bands, 6 (55%) have an Ag-SERS counterpart within 12 cm<sup>-1</sup> (peak F1 0.55), while the full-profile cosine is only +0.43 — the band inventory therefore agrees far better than the overall spectral shape. Characteristic bands are retained: 660 cm<sup>-1</sup> (purine ring) → 660 cm<sup>-1</sup> (+6). Modest systematic displacement (mean | $\Delta\nu$ | 6.7 cm<sup>-1</sup>, median 3.0 cm<sup>-1</sup>) consistent with chemisorption/charge transfer at the Ag surface rather than a change of vibrational identity. Relative band intensities are comparatively well preserved (peak-rank  $\rho$  0.54, redistribution 0.24), an unusually faithful transfer for a colloidal

# Scientific conclusions (Part 12)

Answers follow strictly from the measurements in this audit.

## 1. Does Stage A look spectroscopically believable?

Yes. The weak cross-modal similarity is corroborated directly in the spectra and is not an artefact of representation. Matched Raman/Ag-SERS pairs separate from mismatched pairs by only +0.037 (SNV) and +0.027 (L2) in cosine, and band-level correspondence does not exceed a mismatched-analyte null in any analyte (0/51 at  $p < 0.05$ ). Stage A's conclusion stands.

## 2. Are Raman and Ag-SERS observing the same dominant vibrational modes?

Not demonstrably, as measured here. The Ag-SERS spectra carry ~41–46 reproducible features versus ~12–13 well-defined Raman bands, and their positions show no analyte-specific alignment with the Raman bands. The dominant Ag-SERS features are shared across all 51 analytes, which is the behaviour of a common surface/colloid contribution rather than of the analyte's own normal modes.

## 3. Are peak positions generally preserved?

No — not in an informative sense. 80% of Raman bands find an Ag-SERS peak within  $\pm 12 \text{ cm}^{-1}$ , but a mismatched analyte achieves 71% and uniform-random peaks 60%, because the Ag-SERS peak list is dense ( $\sim 24 \text{ cm}^{-1}$  spacing). The apparent preservation is a peak-density artefact; the excess over chance is only +0.08 and reaches  $p < 0.05$  in no analyte.

## 4. Are differences primarily intensity redistribution?

Intensity redistribution is severe (median peak-rank  $p$  0.12, redistribution index 0.24), but it is not the primary cause. The band inventories themselves differ in size and identity, so the discrepancy is not a re-weighting of one shared mode set. Redistribution is real but secondary to background dominance.

## 5. Are there systematic peak shifts?

No. The optimal rigid shift is  $0 \text{ cm}^{-1}$  at the median ( $|\text{shift}| > 2 \text{ cm}^{-1}$  helps in a minority of analytes) and buys only +0.036 median cosine. The mean  $|\Delta\nu|$  of  $5.8 \text{ cm}^{-1}$  is consistent with random matching inside a  $\pm 12 \text{ cm}^{-1}$  window rather than a systematic chemical displacement.

## 6. Which biochemical families transfer best?

Differences between families are small against the noise. Polysaccharide, amino-acid and protein entries rank highest by correspondence score (0.20–0.30) and purines/lipids/small-nitrogenous lowest (0.08–0.14); purines show the best preserved intensity ordering (rank  $p$  0.47). No family reaches a level that would support cross-modal grounding.

## 7. Are there suspicious preprocessing artefacts?

Yes — one significant defect. SNV collapses Ag-SERS replicate reproducibility from 0.999 (raw) and 0.948 (L2) to 0.491, because SNV mean-centres each spectrum so that cosine becomes Pearson correlation and the low-amplitude residual dominates. Stage A/B therefore evaluated the Ag-SERS arm at needlessly low effective SNR. Native resolutions ( $1.0$ – $1.7 \text{ cm}^{-1}$ ) are all finer than the  $2 \text{ cm}^{-1}$  grid, and Gobbato Raman and Ag-SERS share one instrument, so resampling and instrument mismatch are excluded. Critically, the negative result survives the correction: it is reproduced under L2.

## 8. Are there problematic analytes that should be excluded?

Under SNV, 29/51 analytes have within-Ag-SERS replicate cosine below 0.50 (6 below 0.30: hydroxyproline, oleate, fructose-6-phosphate, methionine, serine, triolein), so their Ag-SERS means are unreliable in that representation. Under L2 the same replicates agree at 0.95, so these are pipeline casualties rather than bad measurements. The genuinely structural caveat is that 27/51 analytes draw Raman replicates from two different sources, mixing inter-instrument variation into within-Raman spread.

## 9. Should Stage A be rerun?

Yes, but for methodological correctness rather than in expectation of a different verdict. Stage A should not have selected a normalisation that degrades one modality's reproducibility by half. A rerun should use L2 (or explicit background-component removal) and report similarity relative to the per-modality reproducibility ceiling. Expect the architectural conclusion to survive: after removing the shared component, cross-modal top-1 rises only to 0.12–0.18 against 0.02 chance.

## 10. What is the single biggest scientific insight?

The Ag-SERS arm of this grounding corpus is background-dominated, not analyte-dominated. Ag-SERS spectra of 51 chemically distinct metabolites are ~95% identical to one another and 95% of each spectrum's variance is the corpus-mean background; the analyte survives only as a ~5% residual — enough for 73% within-modality identification, but far too faint to anchor a correspondence with a powder Raman spectrum that contains no such background. The cross-modal failure is therefore a measurement-contrast problem in the Ag-colloid acquisition, not evidence that the two techniques observe unrelated chemistry, and not a shortcoming of the representation or the encoders. This also explains the Stage B encoder collapse: embeddings whose cross-analyte duplicate fraction reached 0.96–1.00 were faithfully representing inputs that are themselves near-identical.